

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J17 MLB

LAST_MODIFIED=Fri May 31 13:43:58 2013

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
A	0002056619	PRODUCTION RELEASED	2013-05-30

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DRAWING
TITLE=J17
ABBREV=DRAWING
LAST_MODIFIED=Fri May 31 13:43:58 2013

DRAWING TITLE		SCHEM, MLB, J17	
	Apple Inc.		DRAWING NUMBER 051-9884
			SIZE D
			REVISION A.0.0
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BOM VARIANTS

BOM NUMBER	BOM NAME	BOM OPTIONS
639-4206	PCBA, MLB, GOOD, HYNIX, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_HYNIX, FBA, FBB, SSD:Y, EEEE:F8YL
639-4418	PCBA, MLB, GOOD, ELPIDA, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_ELPIDA, FBA, FBB, SSD:Y, EEEE:FCTC
985-0047	PCBA, MLB, DEV, J17	DEVELOPMENT, J17_DEVEL

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754532	1	H5W, SR14E, PRQ, C0, 3.20, 84W, +4*2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD
33754505	1	H5W, QE70, Q5, C0, 3.26, 84W, +4*2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD_TDP

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET,MOLEX, LGA1150, CPU-HF	U0500	CRITICAL	

BOM Groups

BOM GROUP	BOM OPTIONS
J17_COMMON	COMMON, ALTERNATE, J17_COMMON1, J17_PROGPARTS
J17_COMMON1	XDP, SPEAKERID, TBTHV:P12V, CPUVCC:4PHASE, EXT_GPU:YES, VDDQ:P1V5
J17_PROGPARTS	SMC:PROG, BOOTROM:PROG, TBTRM:PROG, CIVROM:PROG, CAMROM:PROG, BLCMCU:PROG
J17_DEVEL	XDP_CONN, LPCPLUS, DDRVREF_DAC
DEVEL_SENSORS	TEMPSNSDEV, AP_ISNS:Y
J17_PRODUCTION	AP_ISNS:N

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM_OPTION
33754541	1	1C, SR198, 287, LPT-D, C2, PRQ, 23X22MM, FCBG4708	U1100	CRITICAL	
33754562	1	1C, QEW, LPT-D, C2, SQ5, 23X22MM, FCBG4708	U1100	CRITICAL	PCH_TDP
33851113	1	1C, T87, CR-4C, B1, PRQ, 288 FCBG4, 12X12MM	U2800	CRITICAL	
33850616	1	1C, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3880	1	IC, PROGRMD, EFI ROM, V0095, J17	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3874	1	IC, PROG, SMC-A3, V2.14F9, PVT, J17	U5000	CRITICAL	SMC:PROG
338S1159	1	IC, SMC, LM4FS1BH5BBCIGR, SCPL	U5000	CRITICAL	SMC:BLANK
341S3860	1	IC, EEPROM, CR, V23.10, J17	U2890	CRITICAL	TBTROM:PROG
335S0865	1	IC, EEPROM, SERIAL, 8KB, MLP8	U2890	CRITICAL	TBTROM:BLANK
341S3735	1	IC, ENET 1MBIT, SPI, ROM, V1.13 J17	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC, CAMERA FLASH, V7230, J16/17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK
341S3674	1	IC, BLC, MCU, PROGRAMMED, V9204, DB	U8100	CRITICAL	BLCMCU:PROG
337S3978	1	IC, BLC MCU LCU2132FBD64/01, LQFP64	U8100	CRITICAL	BLCMCU:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL, MLB, 2D	EEEE_F8YL	CRITICAL	EEEE:F8YL
825-7896	1	LABEL, MLB, 2D	EEEE_FCTC	CRITICAL	EEEE:FCTC

J17 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-9884	1	SCH, MLB, J17	SCH1	CRITICAL	J17
820-3478	1	PCBF, MLB, J17	PCB1	CRITICAL	J17

J17 ALTERNATES

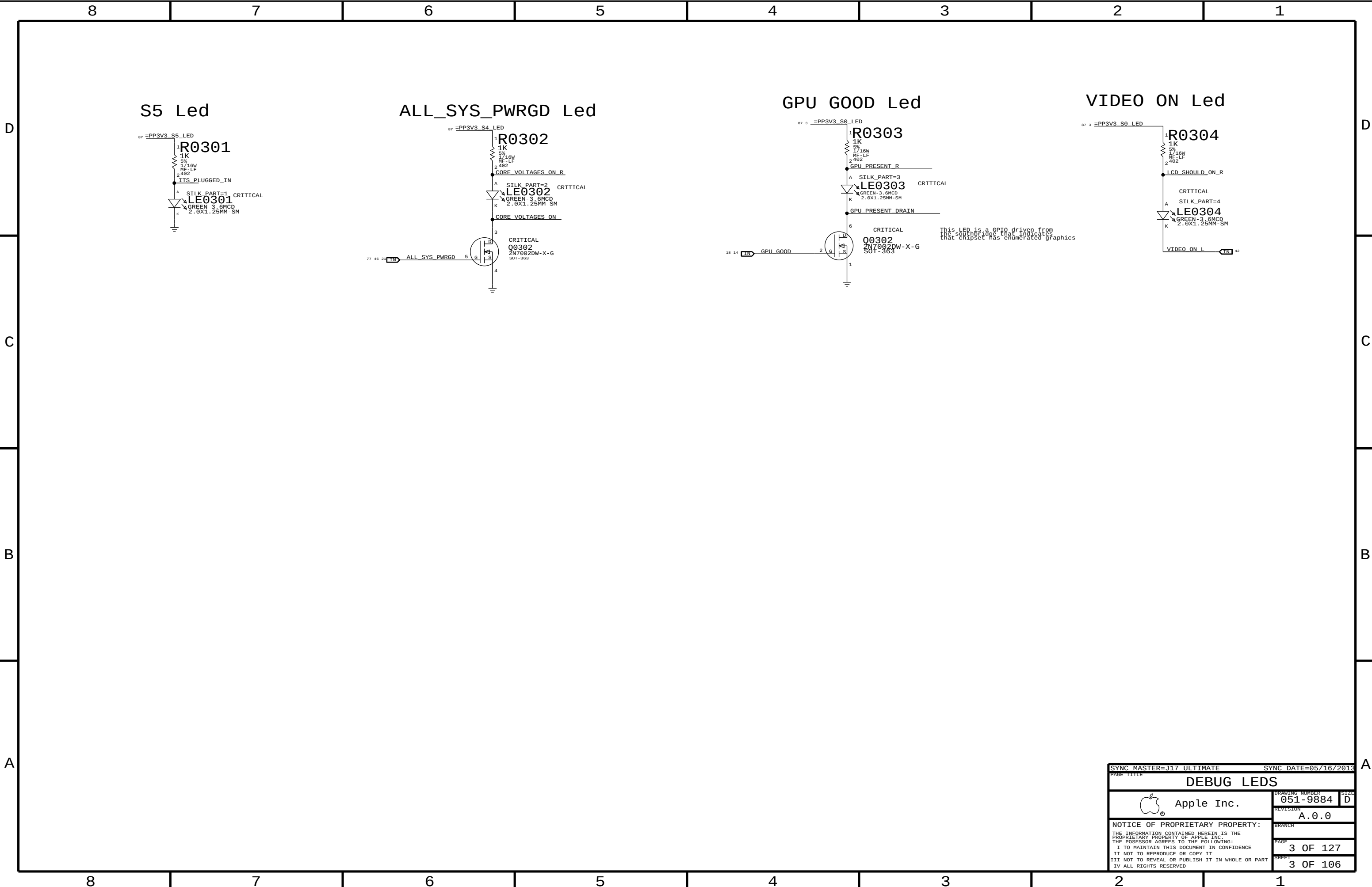
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diodes Array
377S0155	377S0104		ALL	USB Diode
157S0084	157S0058		ALL	Enet Magnetics
376S0975	376S1081		ALL	P/NCH DUAL FET
128S0365	128S0368		ALL	150UF CAPS BLK
138S0803	138S0804		ALL	2.2UF CAPS SOFT
126-0161	126-0160		ALL	56UF CAPS 10X12 THD
197S0481	197S0480		Y1950	25MHz PCH Xtal
197S0479	197S0478		ALL	12MHZ Cam Xtal
372S0186	372S0185		ALL	Alt temp diode
107S0251	107S0249		R5520	Alt 2mOhm sense
107S0254	107S0241		R5450, R5480	Alt 5mOhm sense
107S0255	107S0240		R5450, R5480	Alt 1mOhm sense
341S3747	341S3735		U3990	ENET ROM,ATMEL,V1.13
138S0681	138S0638		ALL	10uF Caps
376S1106	376S0969		ALL	N-Ch FET
138S0860	138S0775		ALL	Single-source 1uF
138S0859	138S0788		ALL	Single-source 10uF

Y4200 and Y8105

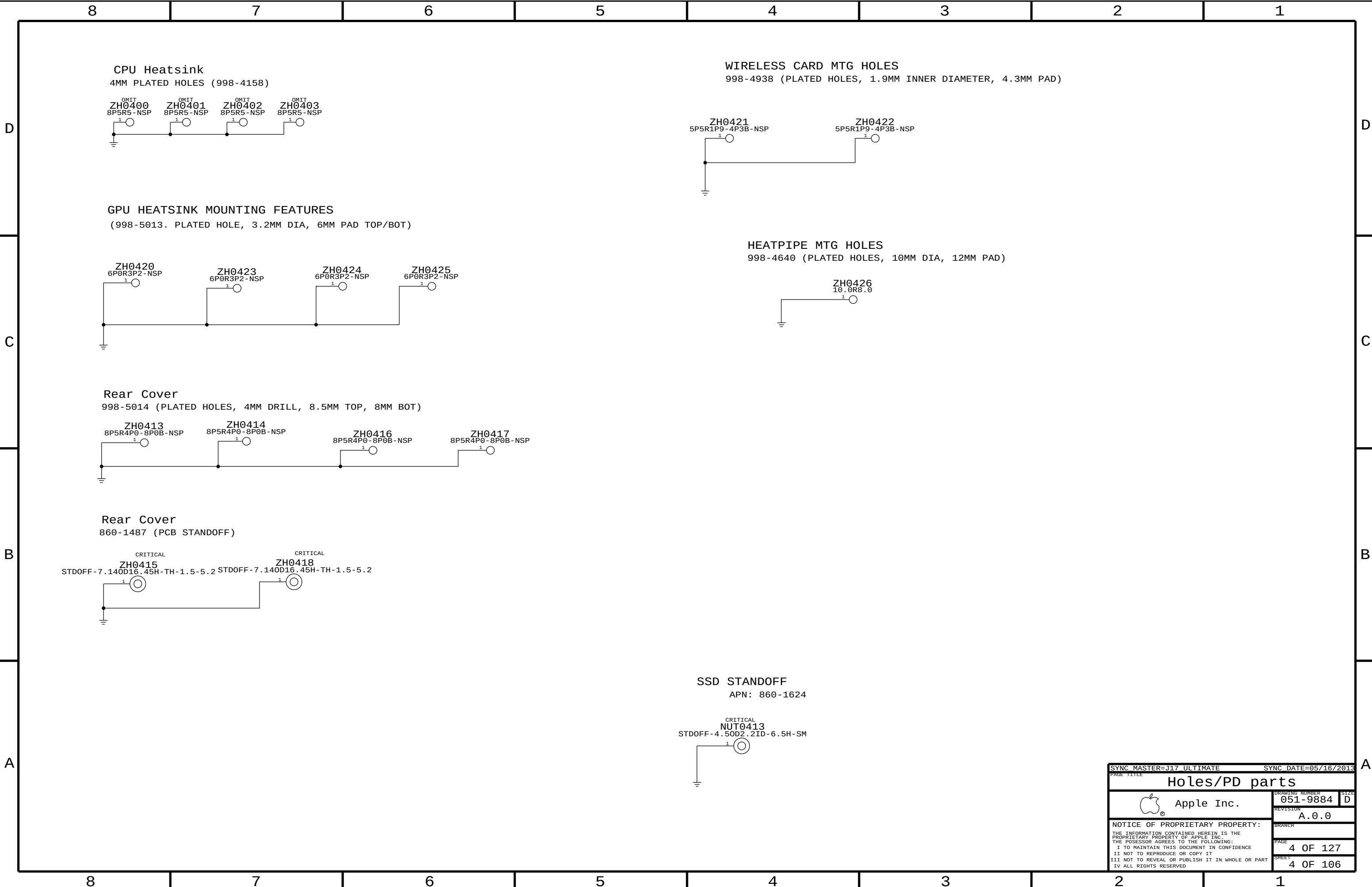
GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4438	1	IC, GPU, NV, GK107-EGX, PS-A2	U8700	CRITICAL	GPU: GK107EGX
333S0638	4	IC, GDDR5, 2GBIT1, 64KX32, GEMMA-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_HYNIX
333S0695	4	IC, GDDR5, 2GBIT1, 64KX32, B-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_ELPIDA

SYNCH MASTER=J17 DINI		SYNCH DATE=02/06/2013	
PAGE TITLE			
BOM Configuration			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-9884	D
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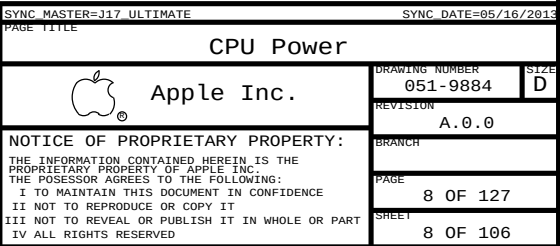


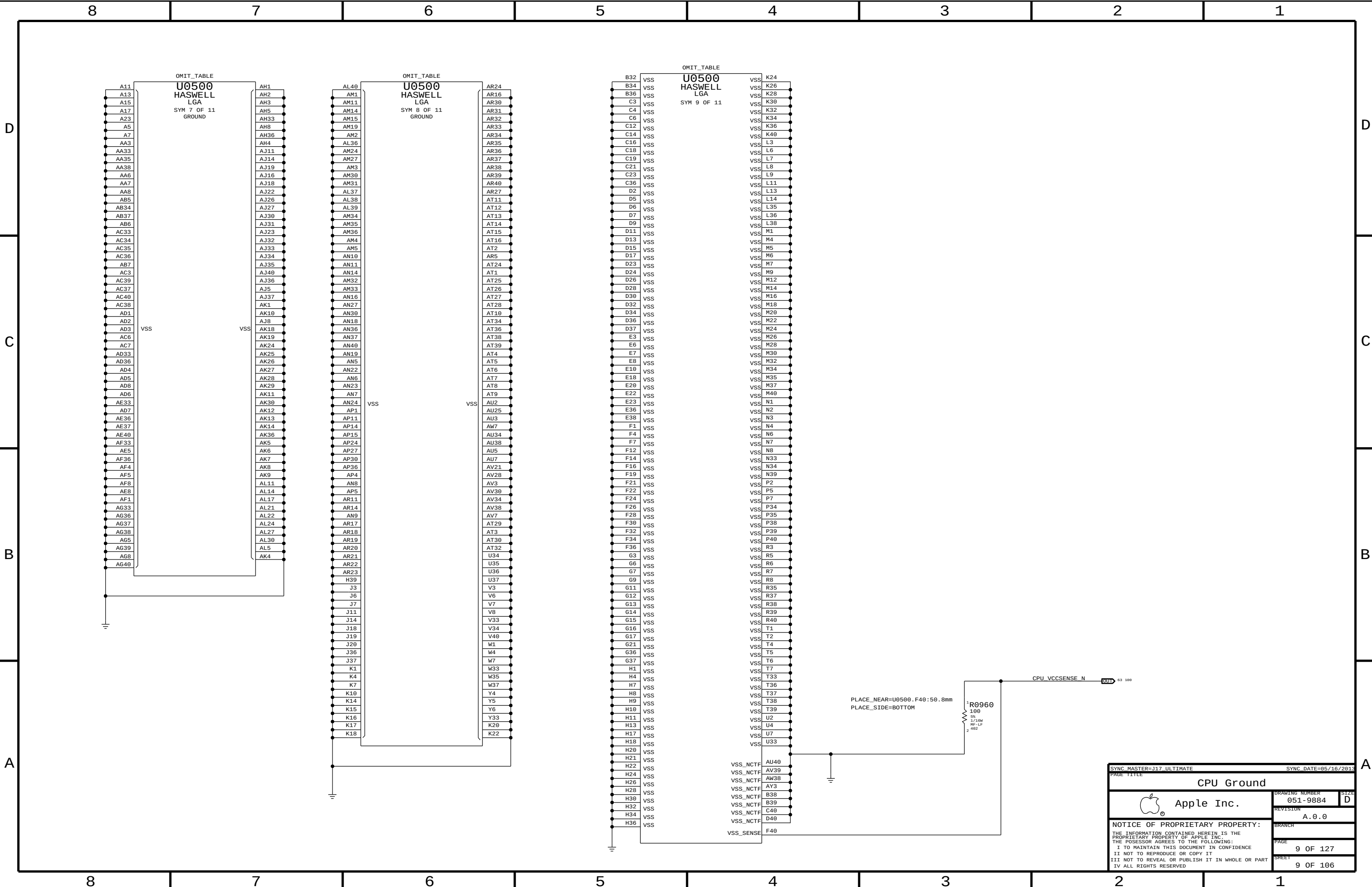
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Holes/PD parts			
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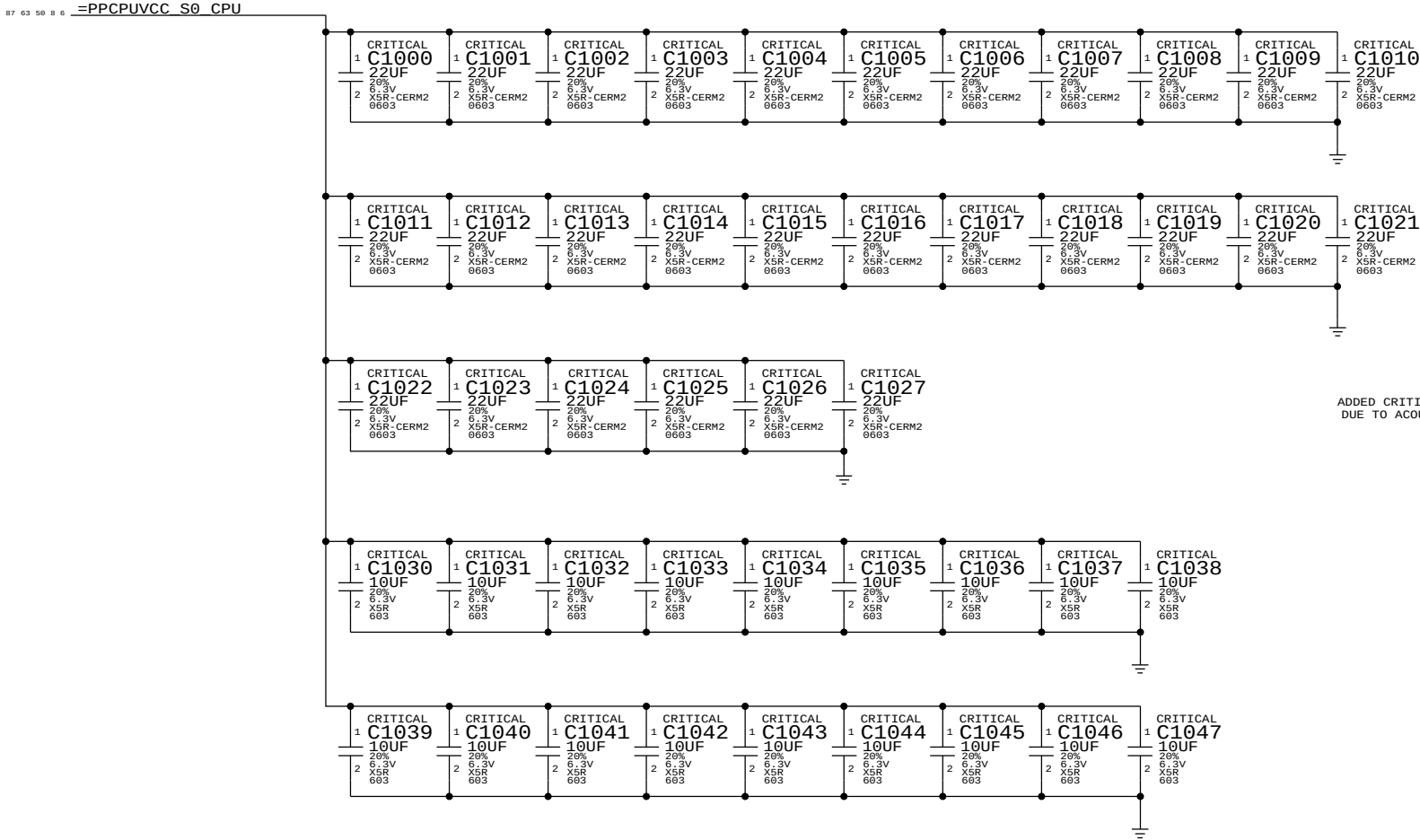




CPU VCORE DECOUPLING

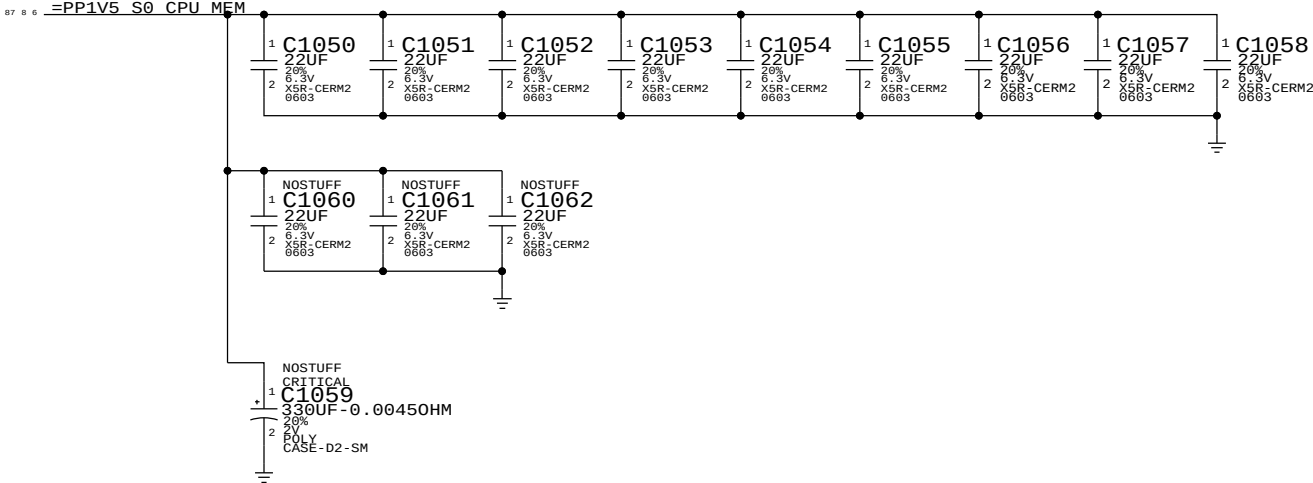
Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71

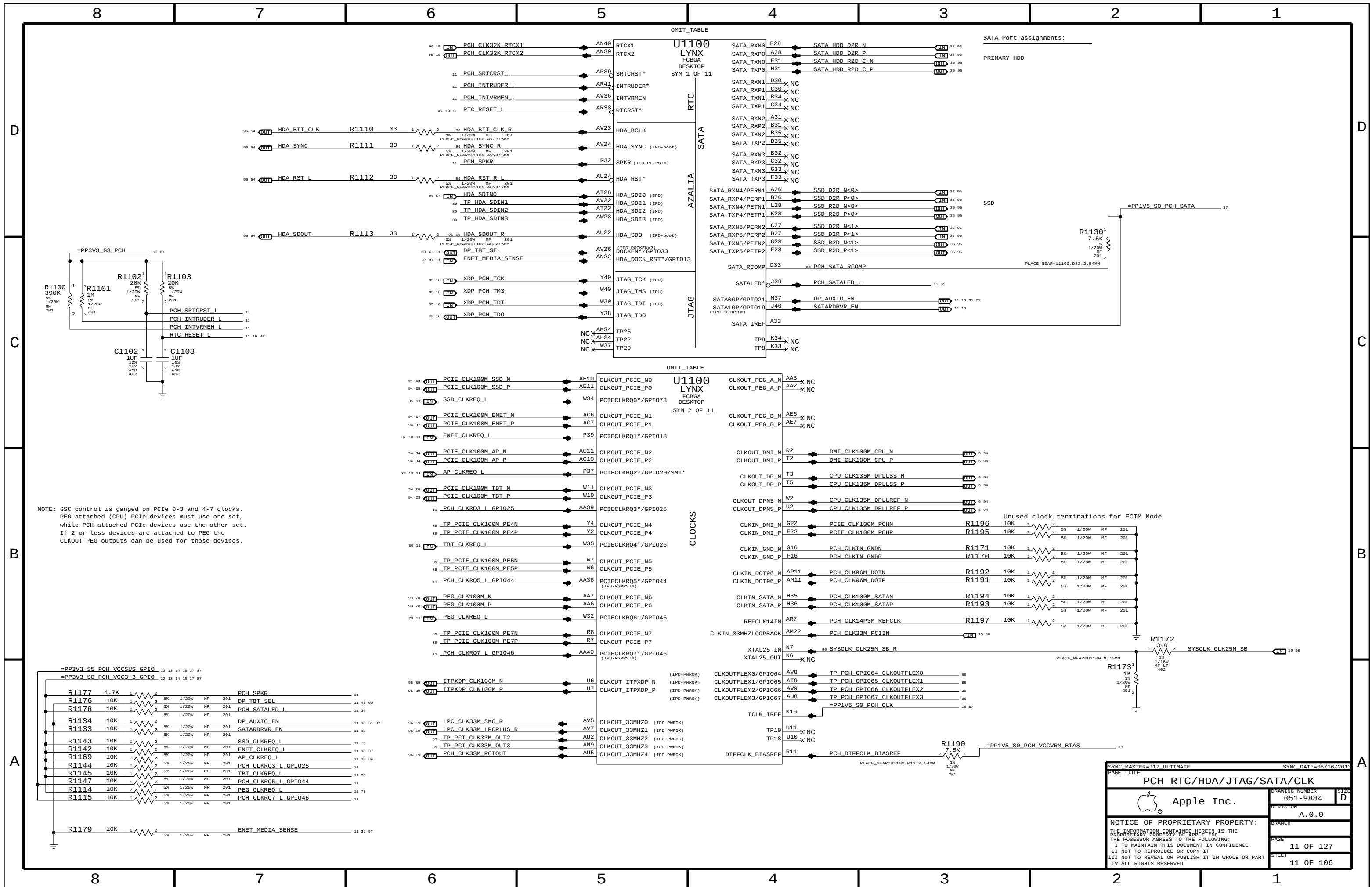


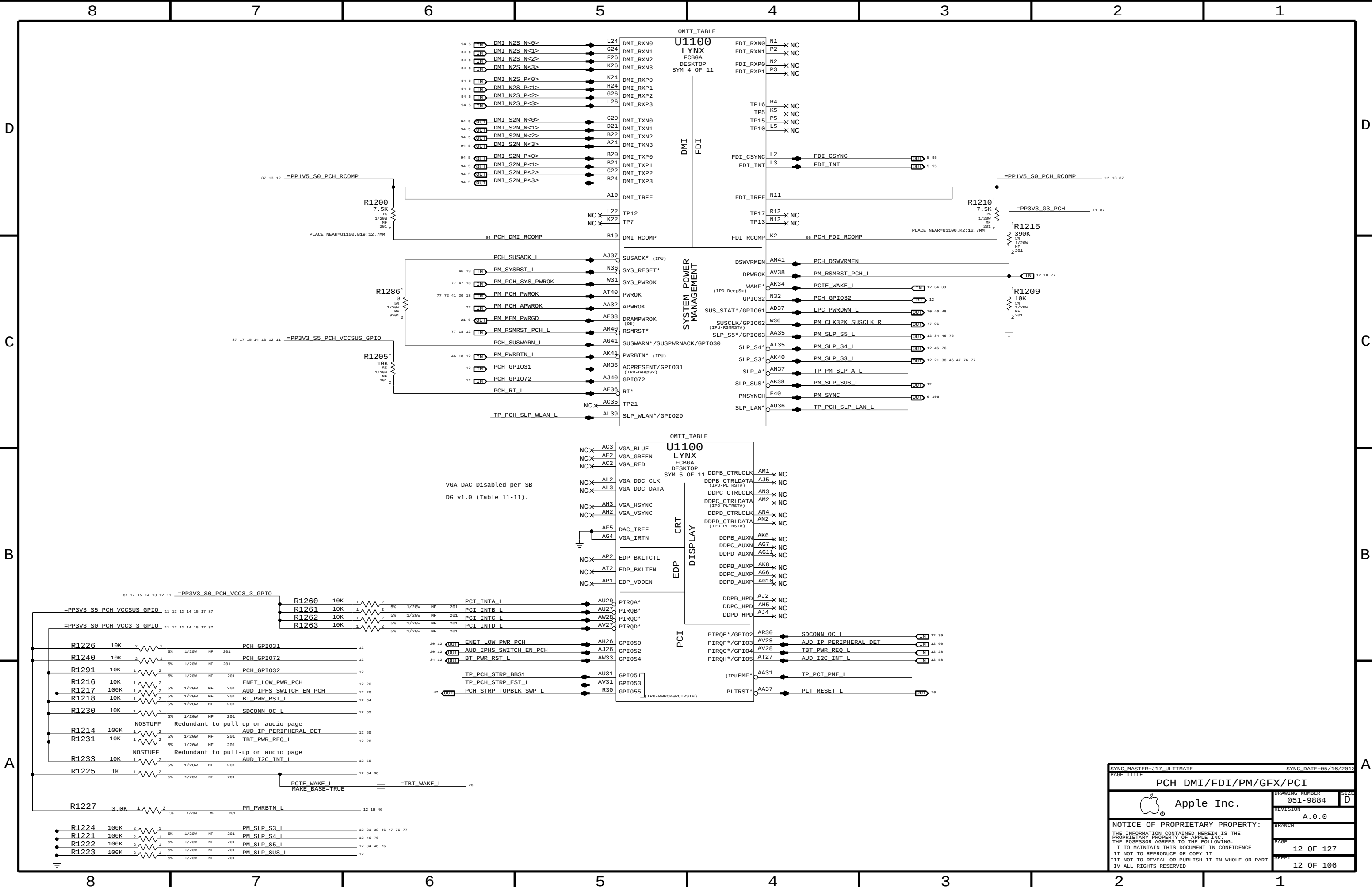
Memory (CPU VCCDDR) DECOUPLING

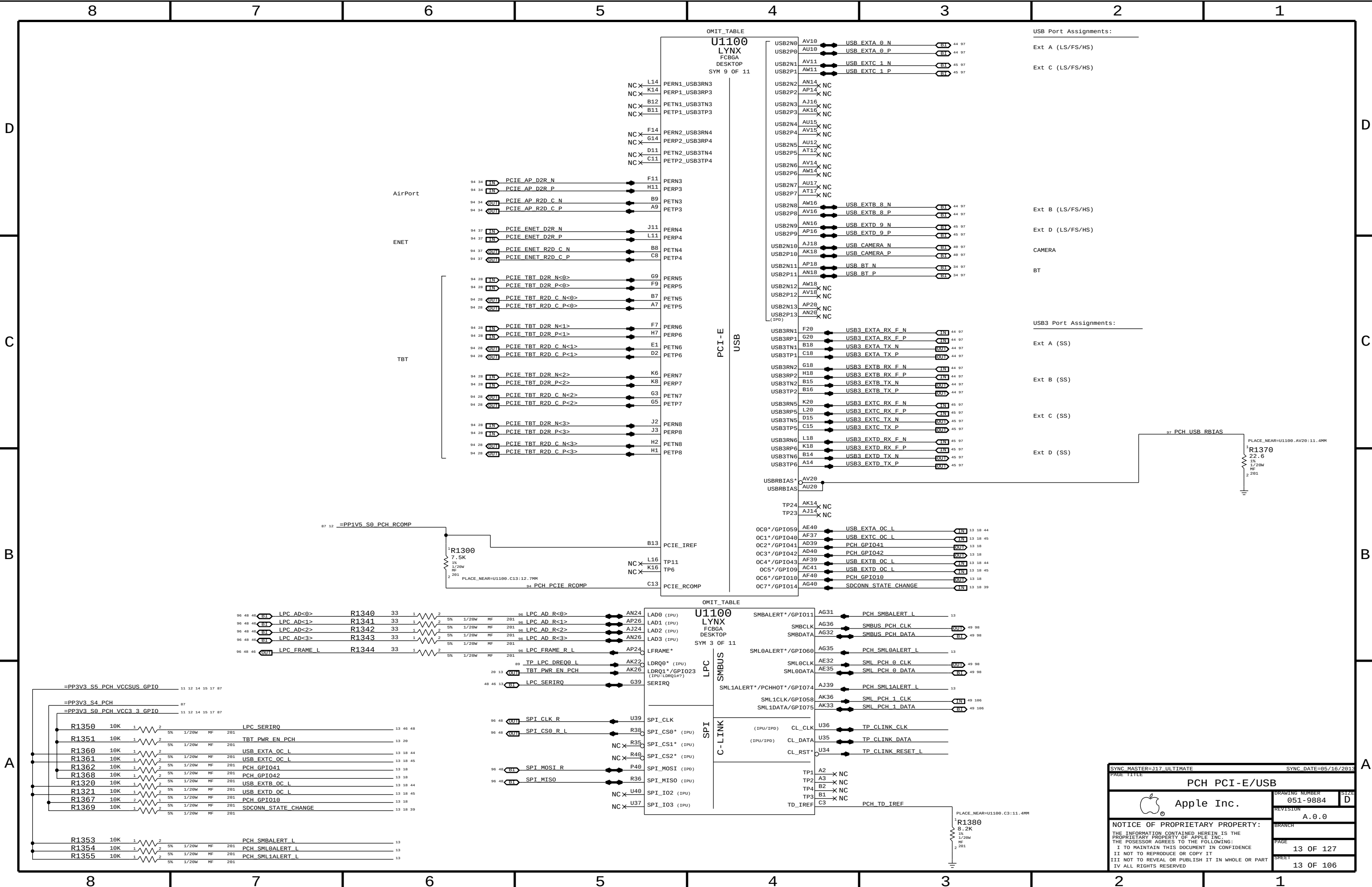
Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

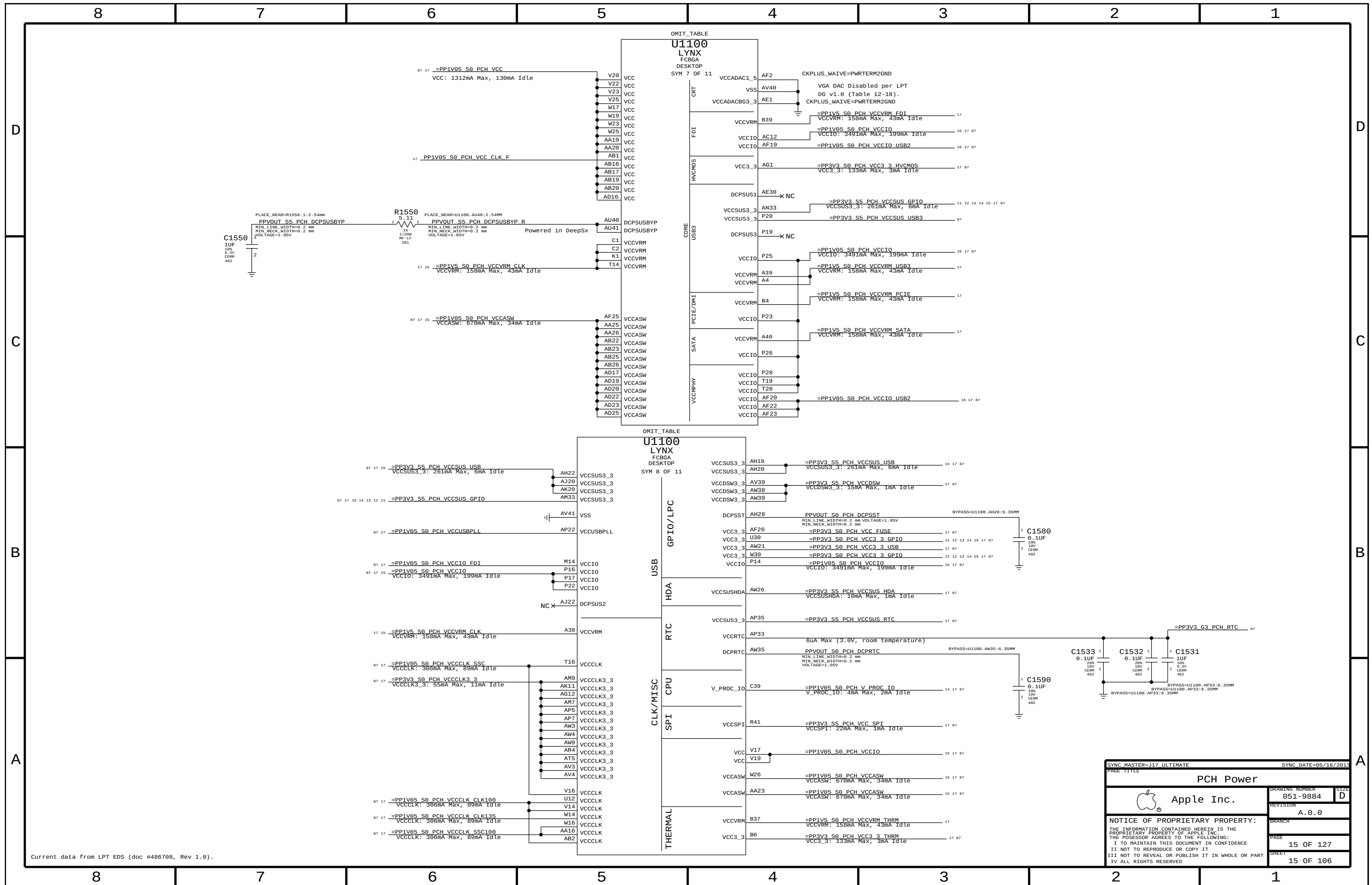


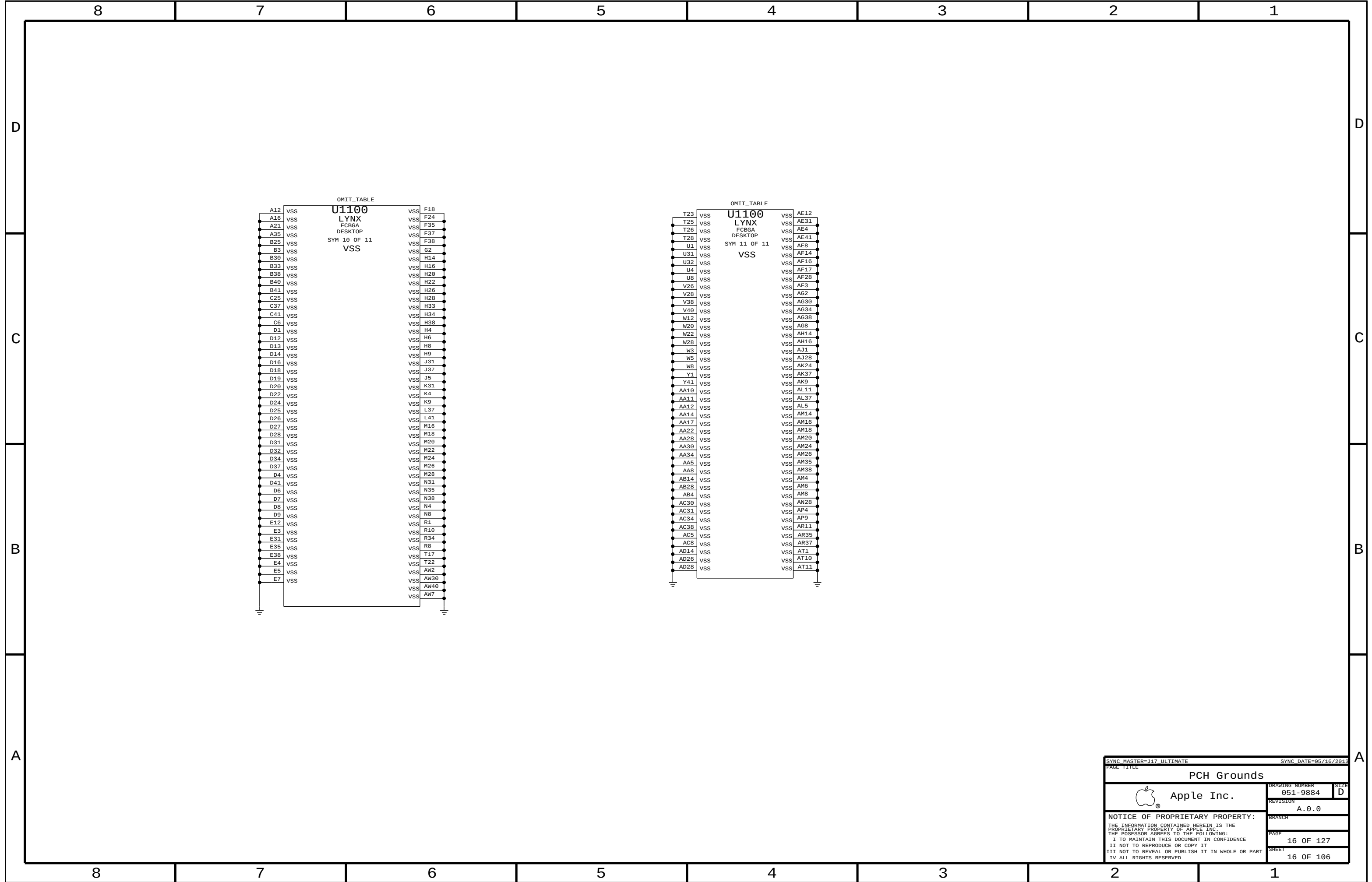
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		DRAWING NUMBER	051-9884
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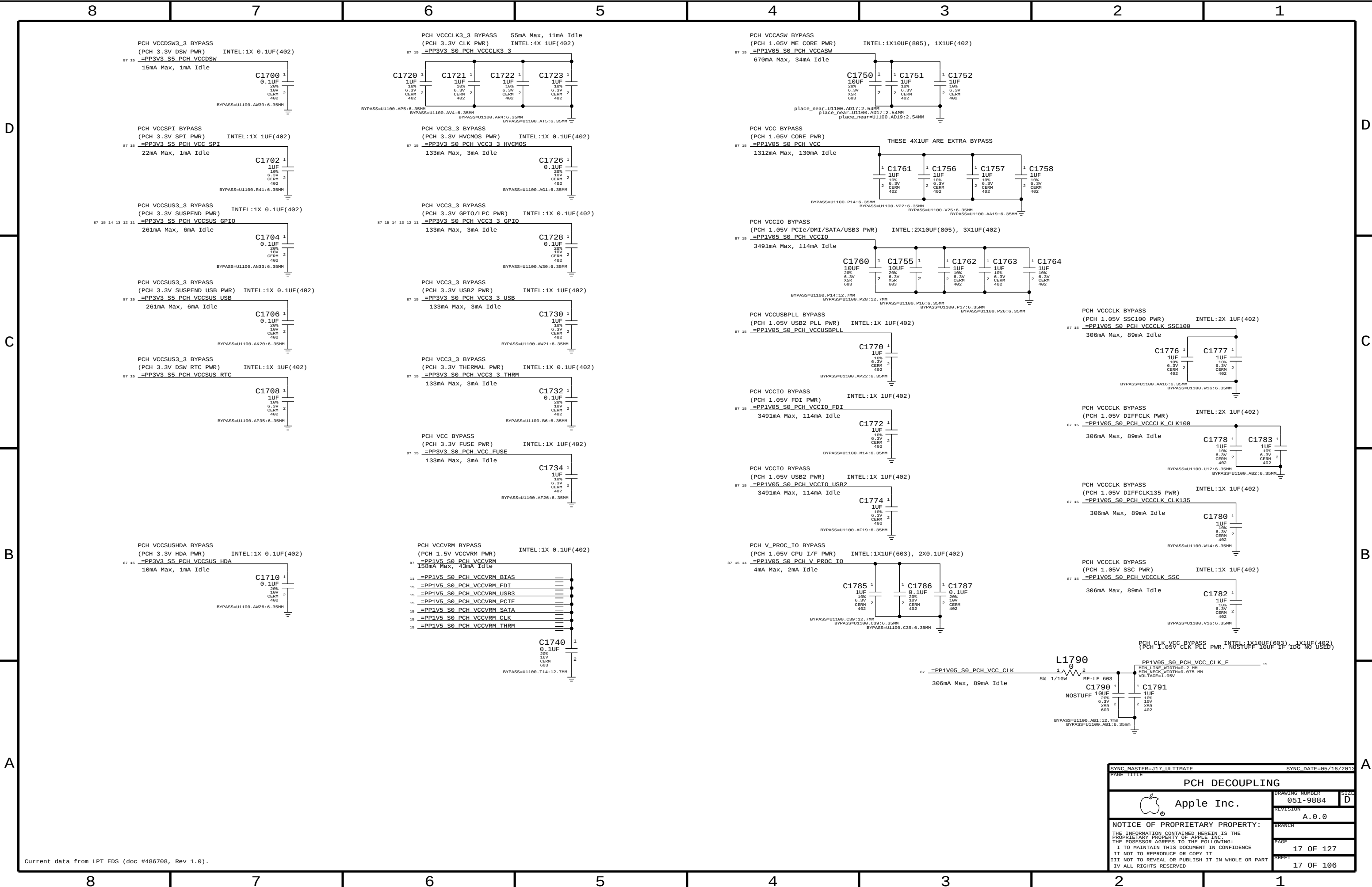






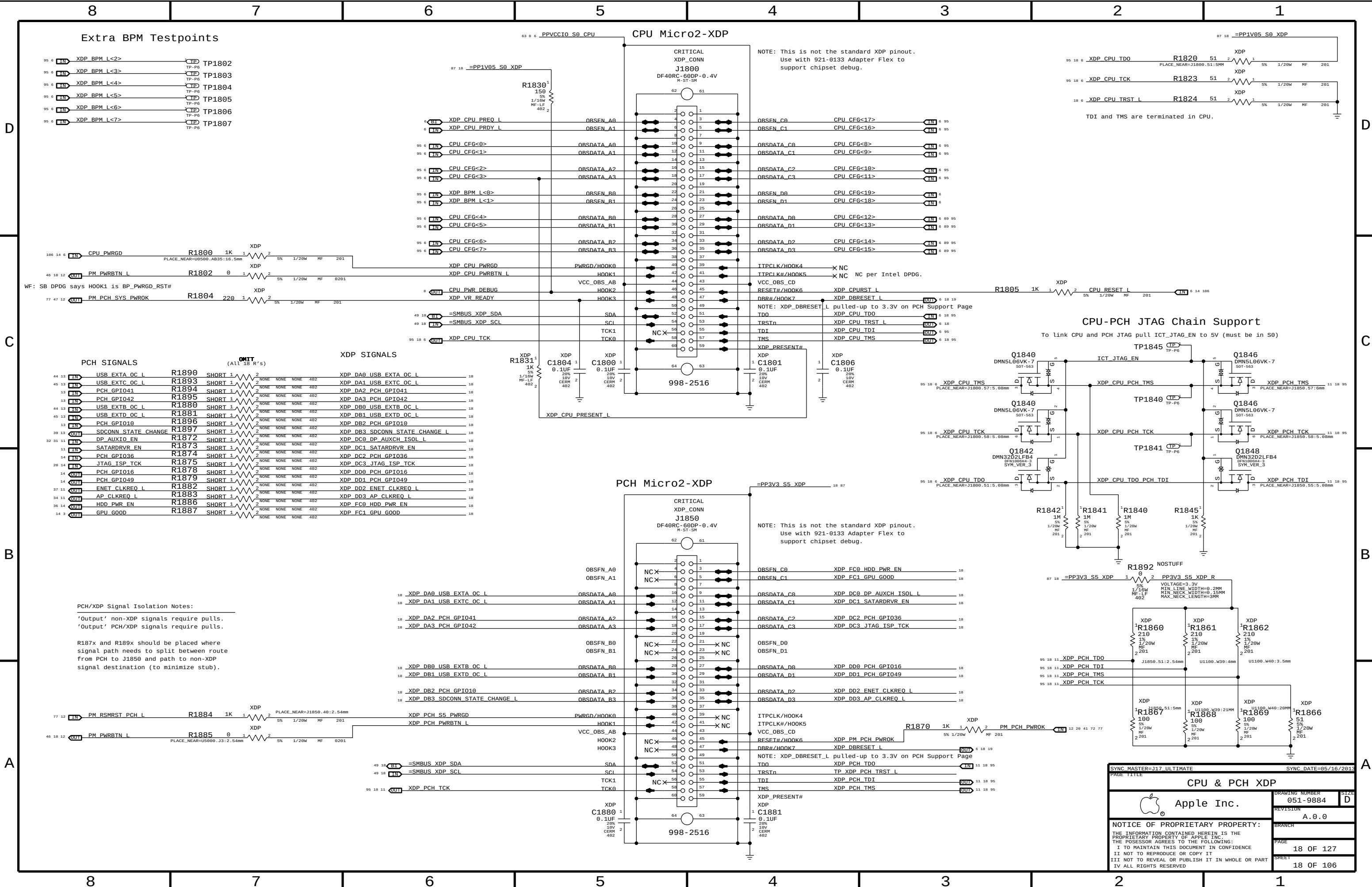




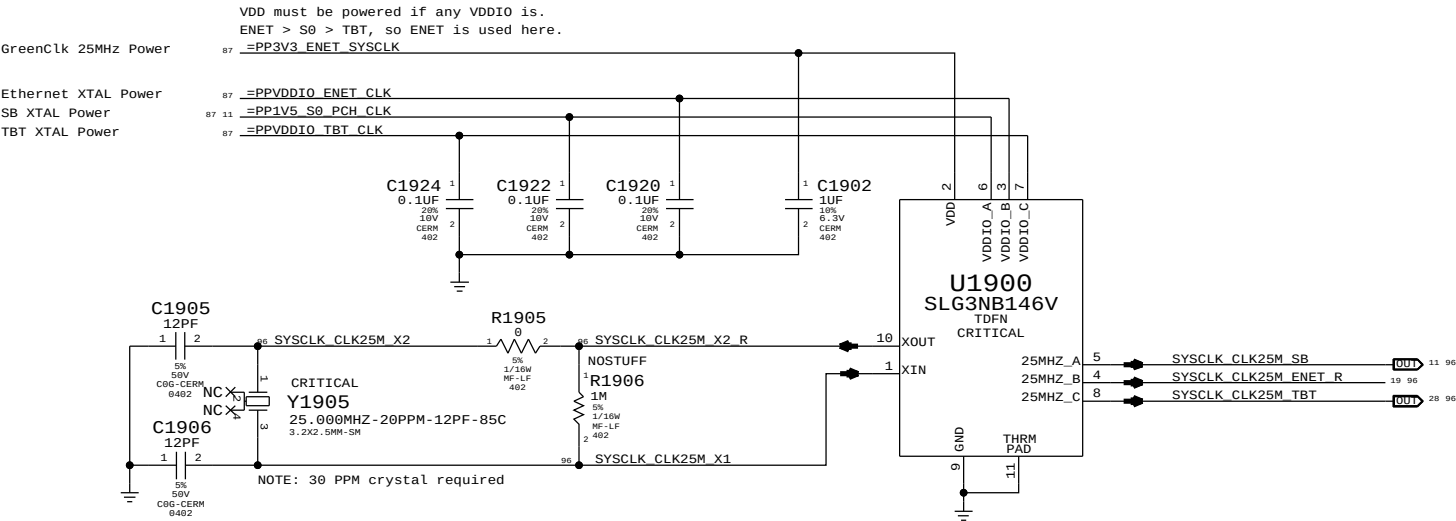


Current data from LPT EDS (doc #486708, Rev 1.0).

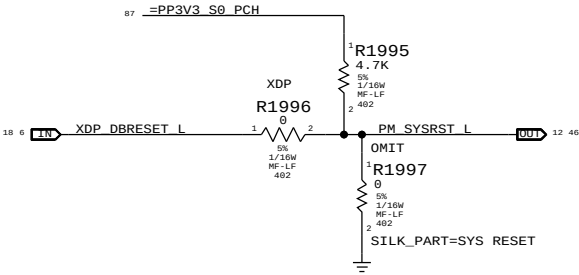
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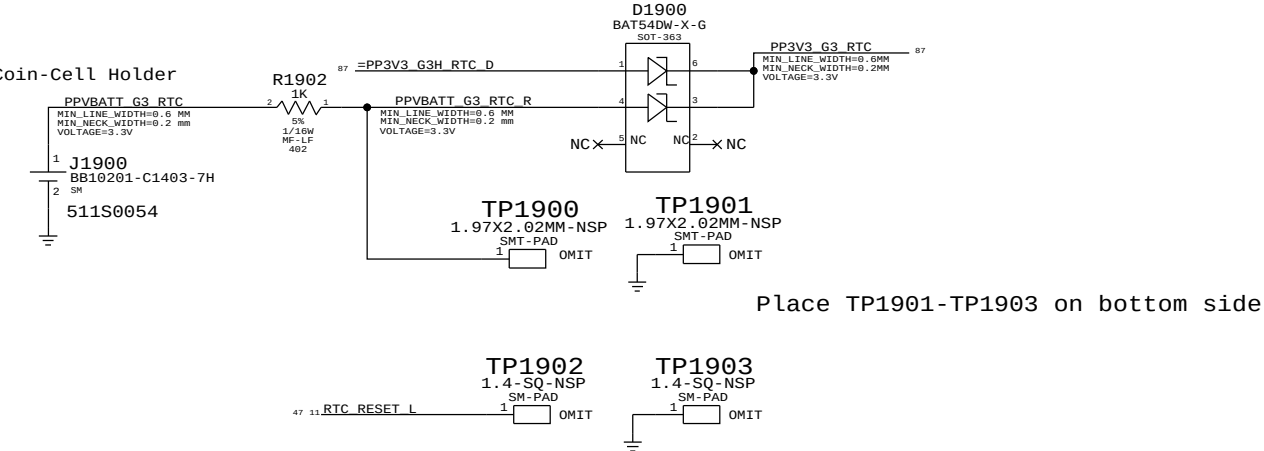
System 25MHz Clock Generator



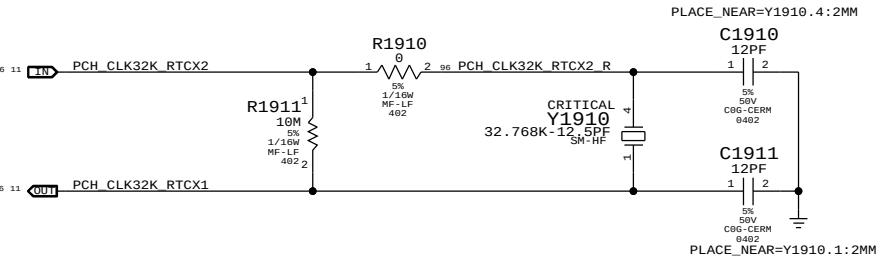
PCH Reset Button



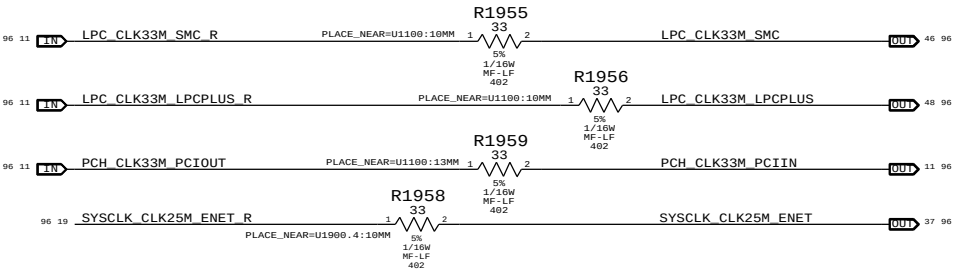
RTC Power Sources



PCH RTC Crystal

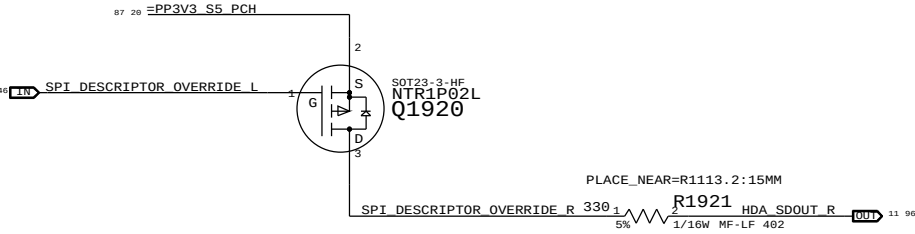


Clock series termination

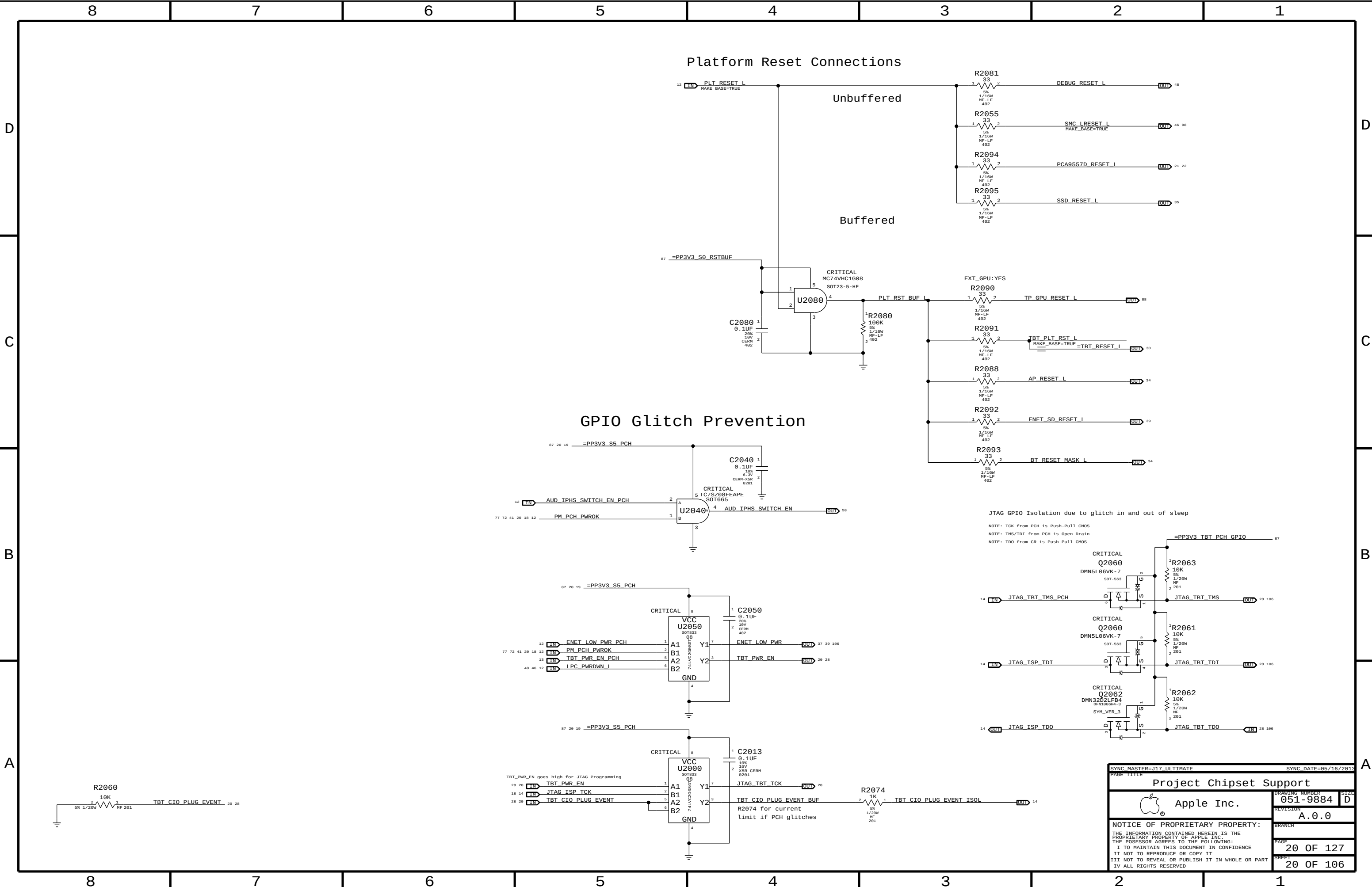


PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		Chipset Support	
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Project Chipset Support

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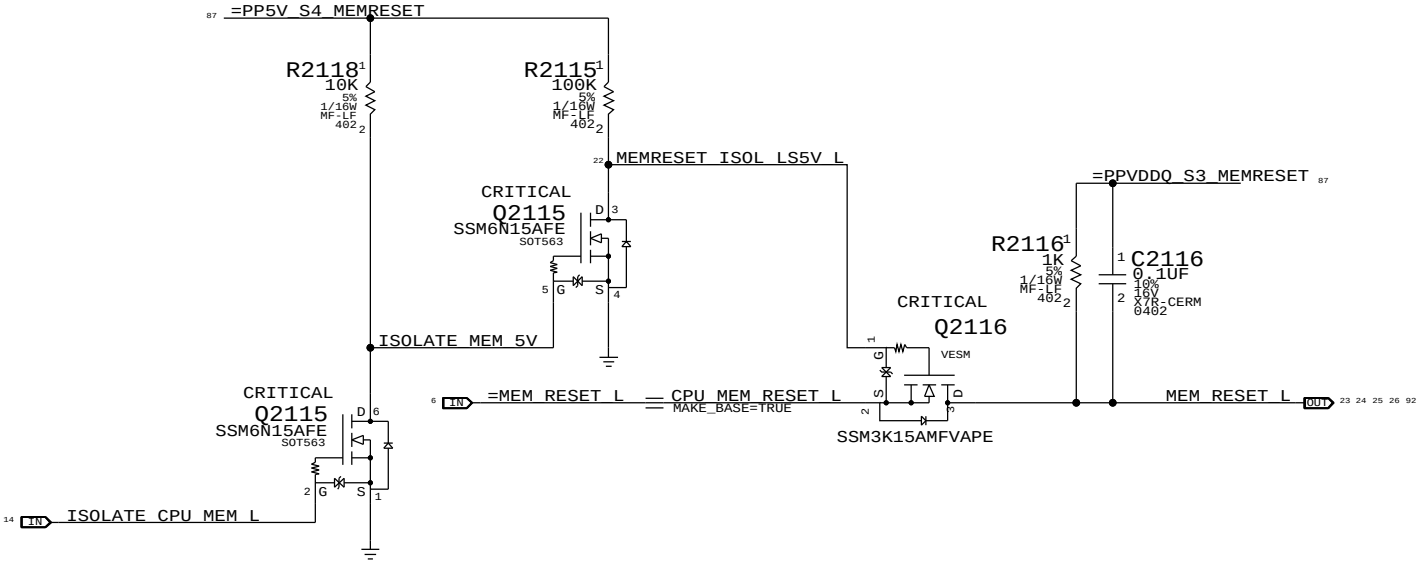
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

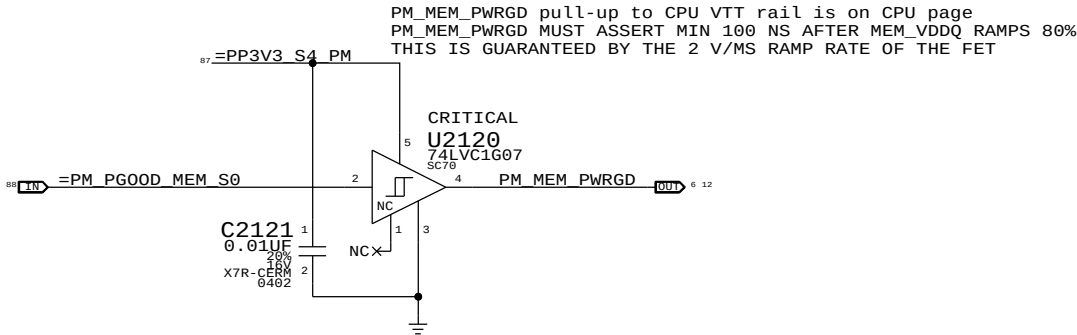
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

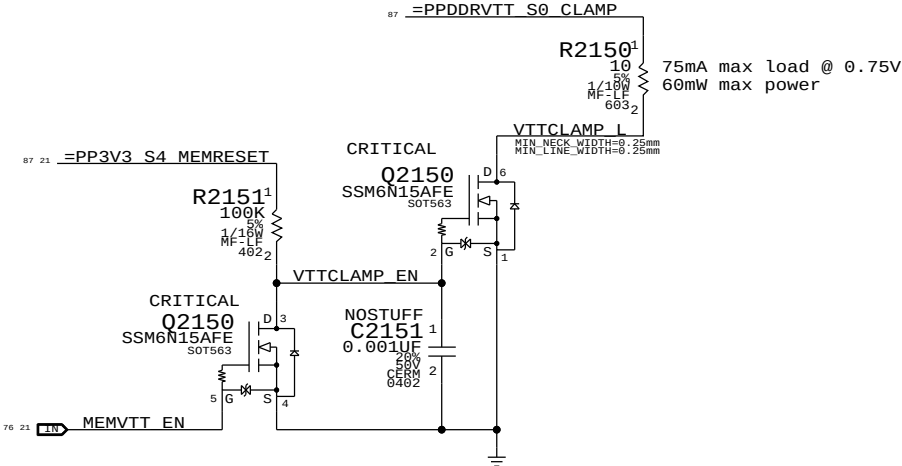
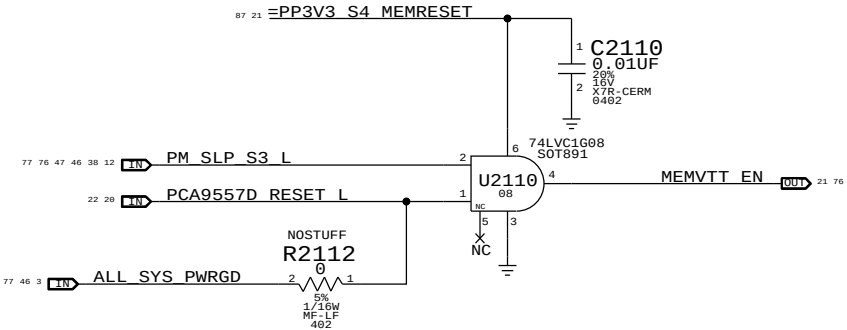


MEM S0 "PGOOD" FOR CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

CPU Memory S3 Support

Apple Inc.

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051-9884

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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

FETs for CPU isolation during S3

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure ISOLATE_CPU_MEM_L is low to remove short due to CPU.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.

OMIT
R2218
SHORT

PP3V3_S3_VREFMRGN

MIN_LINE_WIDTH=0.3 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=3.3V

DDR3 (1.5V)

DDR3L (1.35V)

=I2C_VREFDACS_SCL

=I2C_VREFDACS_SDA

Addr=0x98(WR)/0x99(RD)

Addr=0x30(WR)/0x31(RD)

=I2C_PCA9557D_SCL

=I2C_PCA9557D_SDA

RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PCA9557D_RESET_L

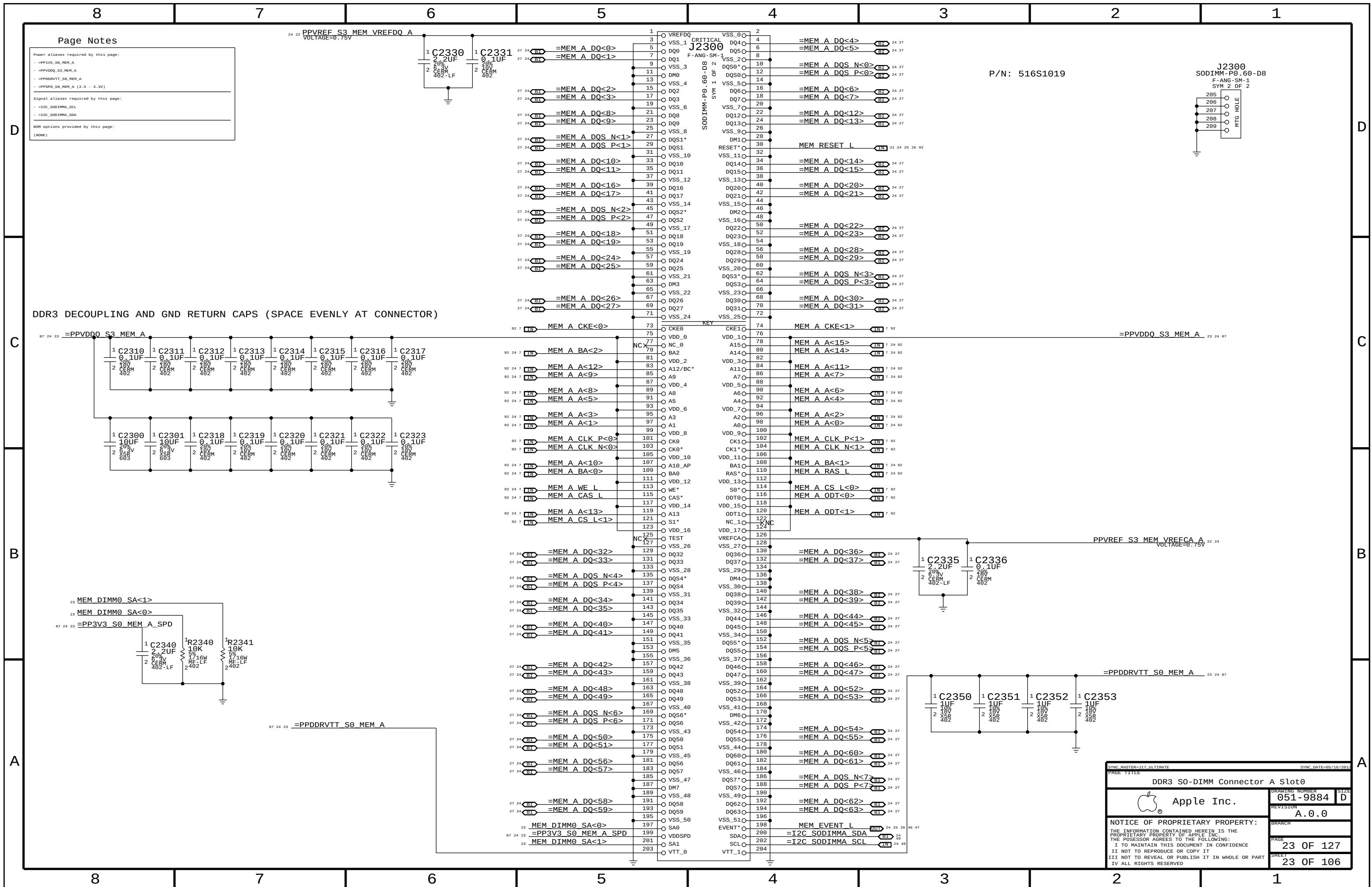
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	DDR3 (1.5V)		DDR3L (1.35V)		
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)
VRef current:	+901uA - 911uA (- = sourced)		+811uA - 816uA (- = sourced)		+36uA - 36uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output

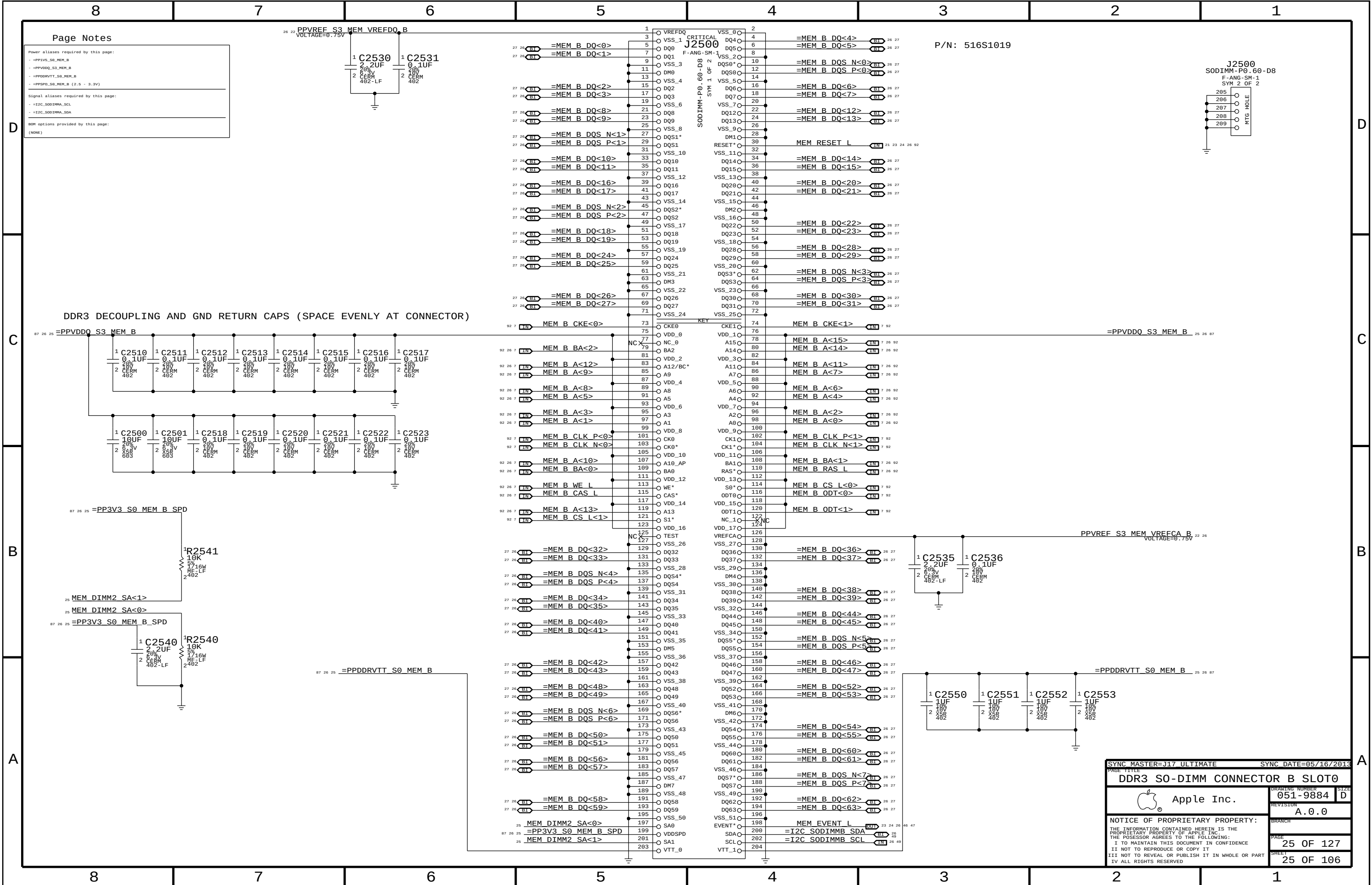
NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

VRef Dividers

Always used, regardless of margining option.

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		DDR3 VREF MARGINING	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

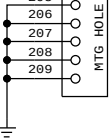
BOM options provided by this page:

(NONE)

P/N: 516S1019

J2500
SODIMM-P0.60-D8

F-ANG-SM-1
SYM 2 OF 2



DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

=PPVDDQ_S3_MEM_B

PPVREF_S3_MEM_VREFCA_B

=PPDDRVT_S0_MEM_B

SYNC MASTER=J17 ULTIMATE SYNC DATE=05/16/2013

PAGE TITLE: DDR3 SO-DIMM CONNECTOR B SLOT0



Apple Inc.

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SIZE: D

REVISION: A.0.0

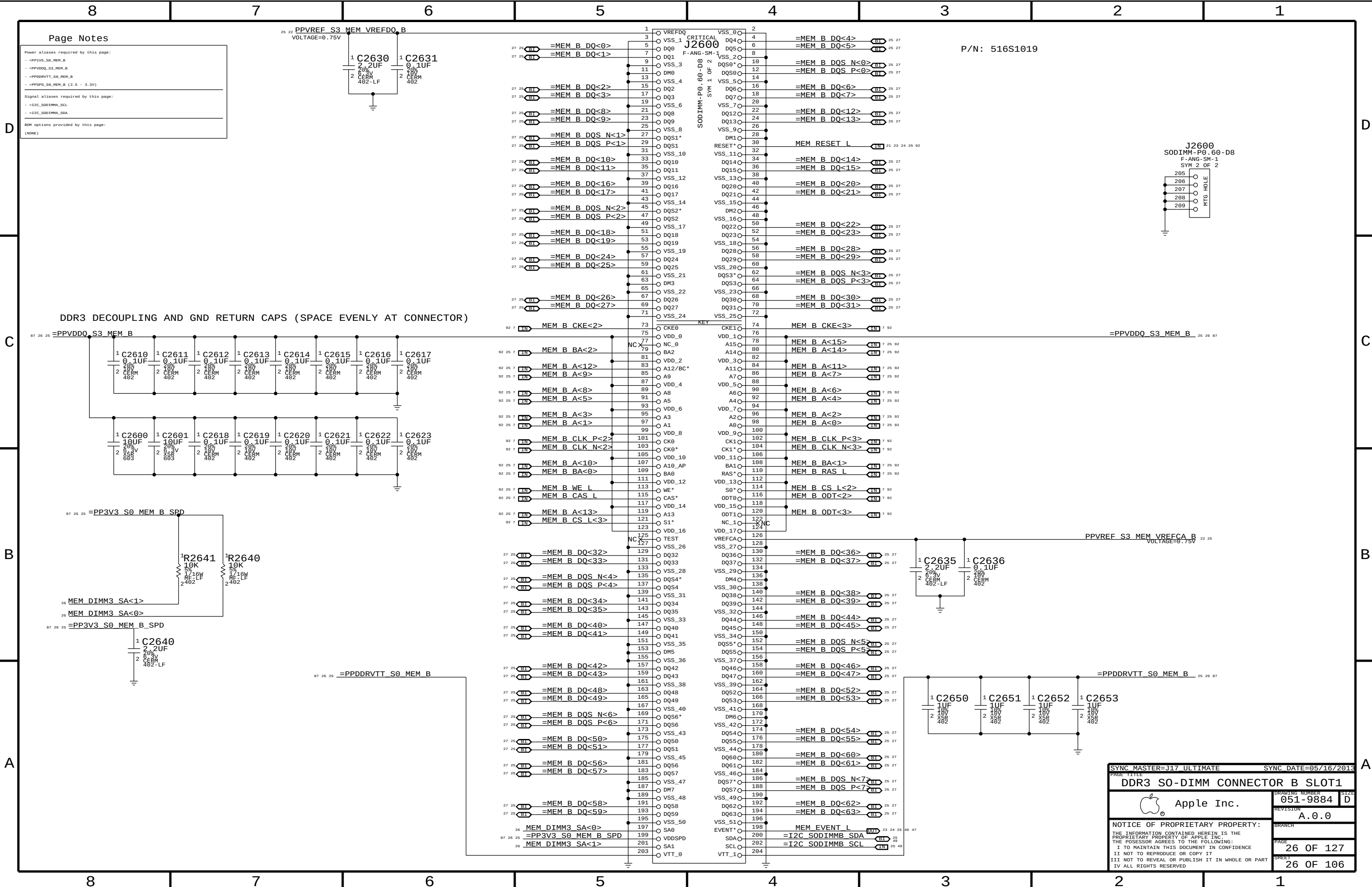
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

P/N: 516S1019

J2600
SODIMM-P0.60-D8
F-ANG-SM-1
SYM 2 OF 2

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		DDR3 SO-DIMM CONNECTOR B SLOT11	
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8		7		6		5		4		3		2		1	
THERE ARE NO PIN SWAPS															
D	92 7	MEM A DQS N<0>													
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			MAKE_BASE=TRUE												
	92 7	MEM A DQ<7>	MAKE_BASE=TRUE												
	92 7	MEM A DQ<6>	MAKE_BASE=TRUE												
	92 7	MEM A DQ<5>	MAKE_BASE=TRUE												
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	92 7	MEM A DQ<3>	MAKE_BASE=TRUE												
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SYNC DATE=05/16/2013

DDR3 ALIASES AND BITSWAPS

 Apple Inc.

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051-9884

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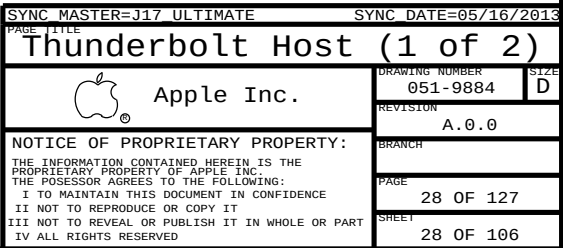
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SHEET
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BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

D ☐ =TBT_CLKREQ_L ☐ =TBT_RESET_L D



B B



R3011 VIN VOUT Max Current = 2A (85C) U3010
A2 A1 B2 B1
SENSE U3030 RESET*
VDD Pull-up: R2810
TBT_PWR_ON_PDC_RST_L
PP3V3_S0_PCH_GPIO

1.05V TBT "LC" Switch

1.05V TBT "CIO" Switch



R3016

100k

10k

1A/150W

Max Current = 2A (85°C)

U3015

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



Max Current = 4A (85C)

Part TPS22920 SYNC MASTER=J17 ULTIMATE SYNC DATE=05/16/2013

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BRANCH			
PAGE	30 OF 127		
SHEET	30 OF 106		

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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	TI Alternate
128S0398	128S0220		ALL	3.3V INPUT CAP



Thunderbolt Connector A

B

DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNCH MASTER=J17 ULTIMATE		SYNCH DATE=05/16/2013	
PAGE 1 OF 1			
Thunderbolt Connector A			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9884		D
REVISION		A.0.0	
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		SHEET 31 OF 106	

8	7	6	5	4	3	2	1
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	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)

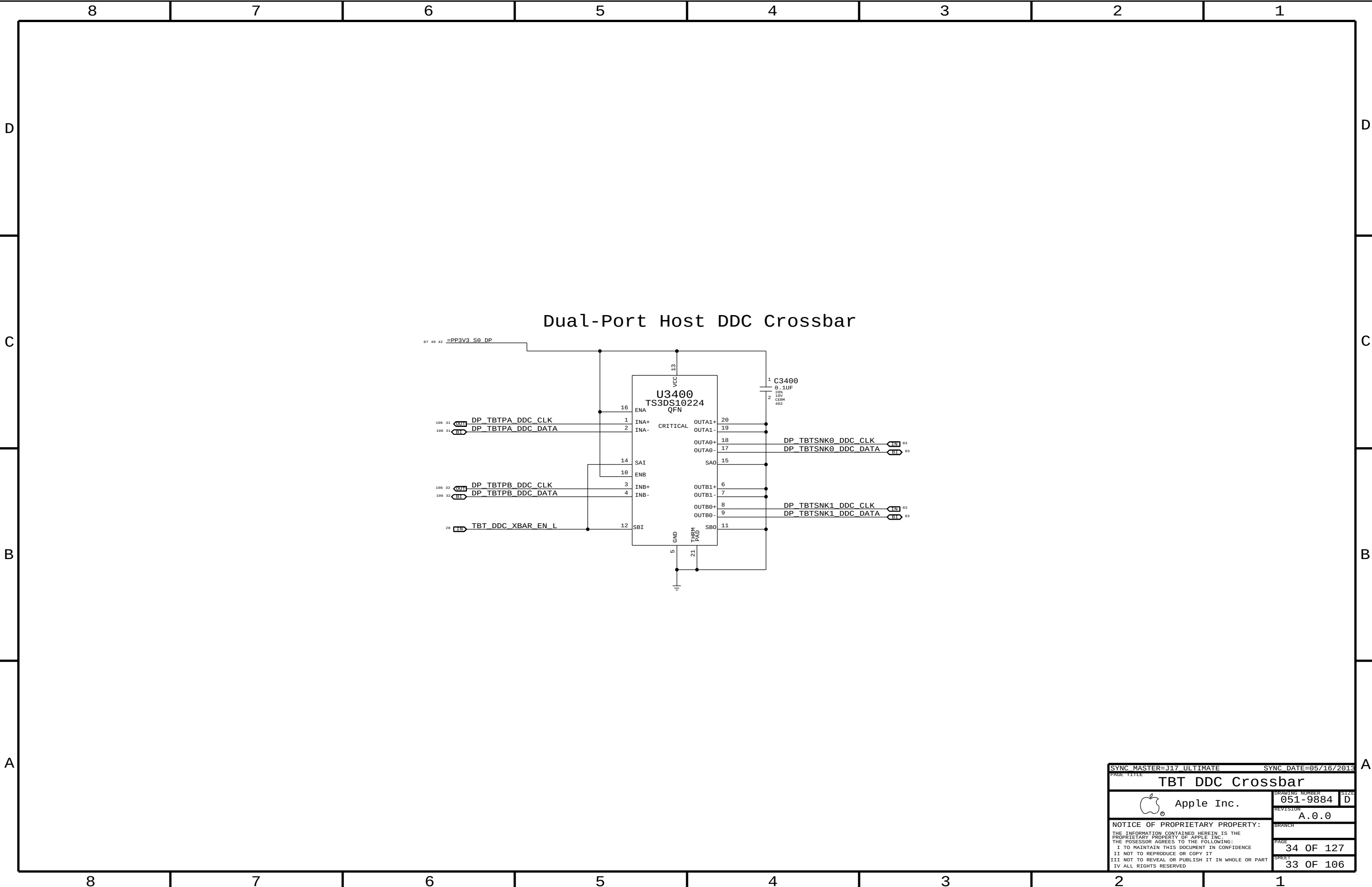


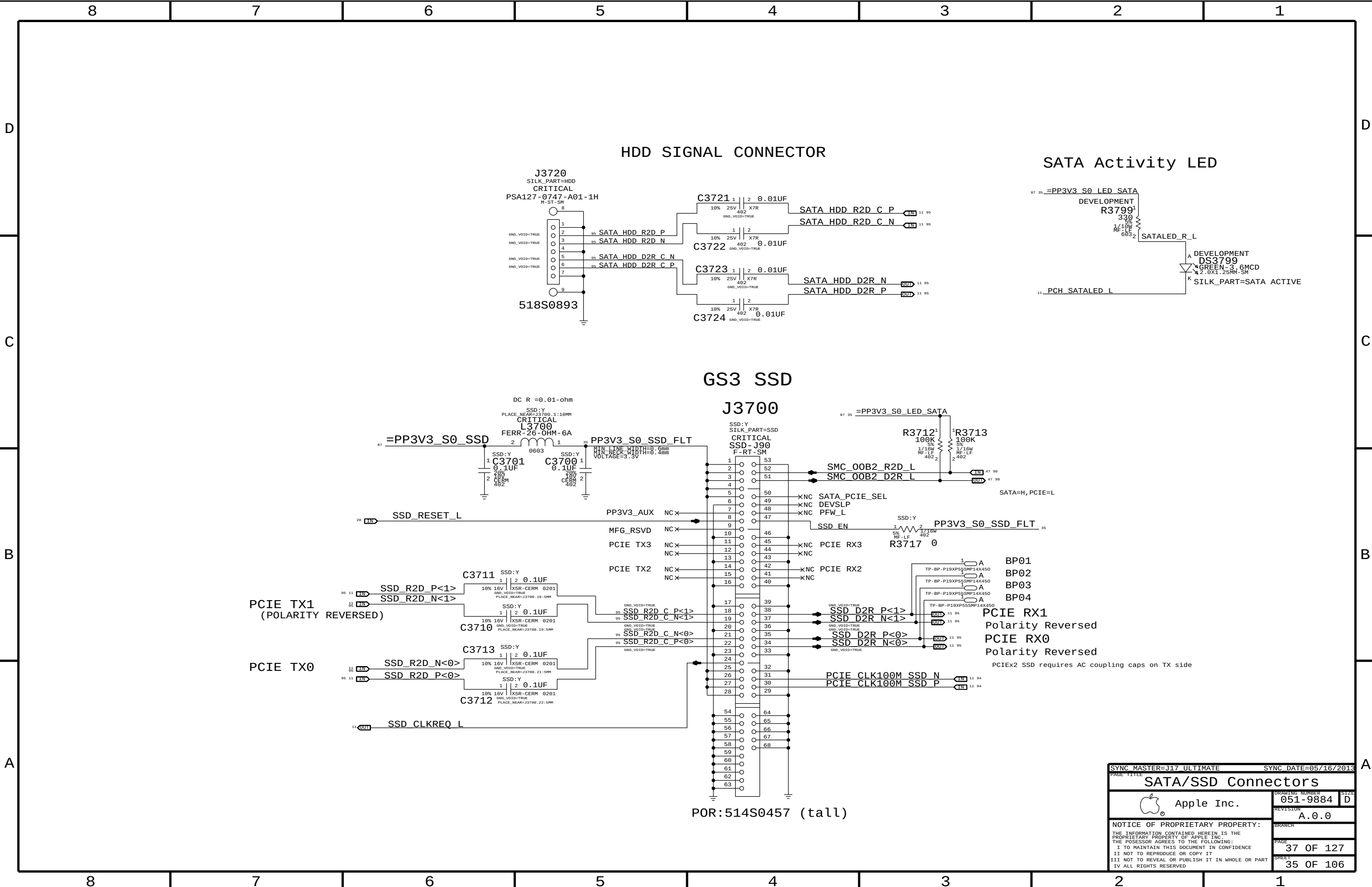
Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V



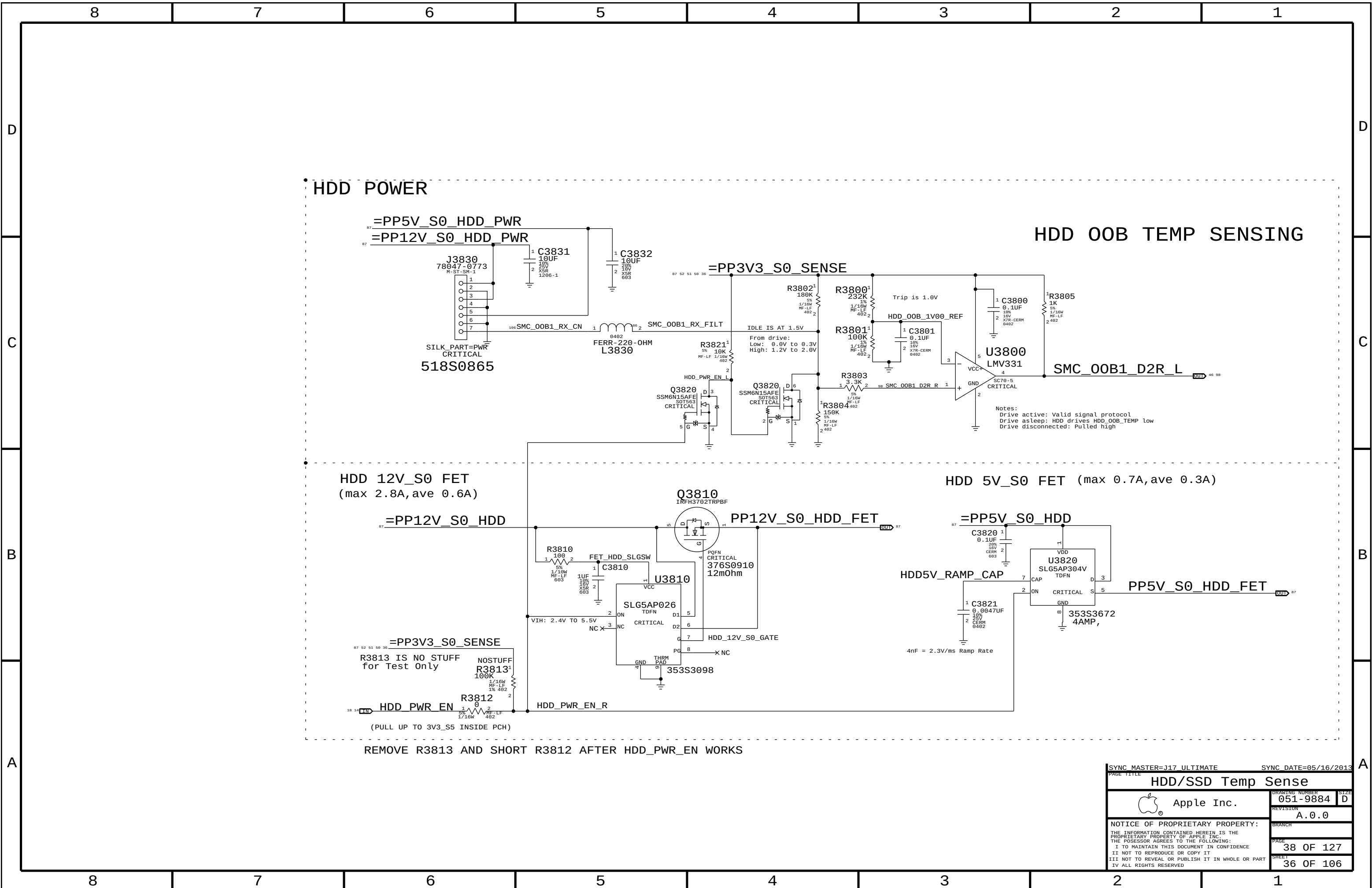
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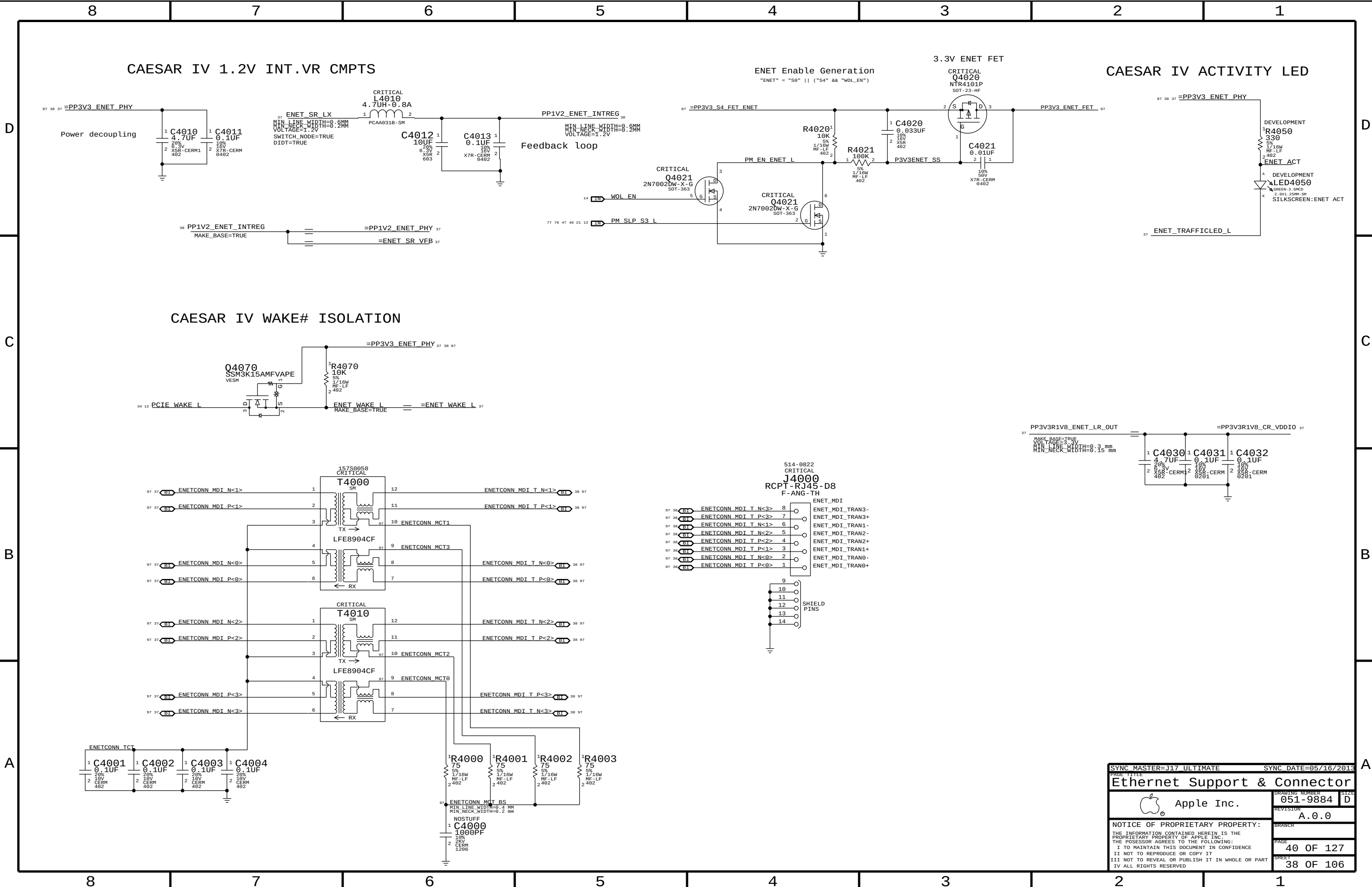
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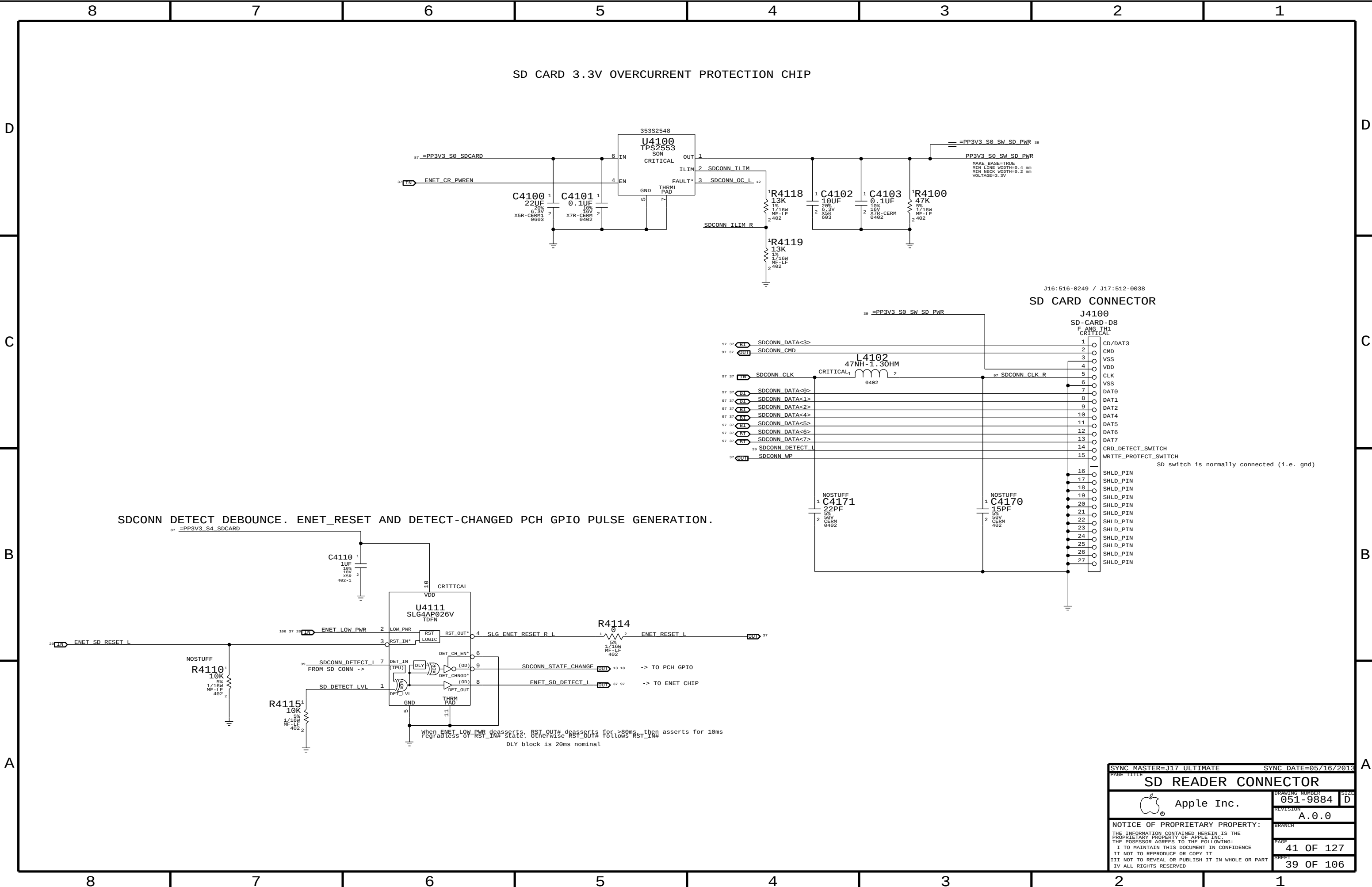




SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
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SATA/SSD Connectors			
 Apple Inc.		DRAWING NUMBER	051-9884
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		SHEET	35 OF 106





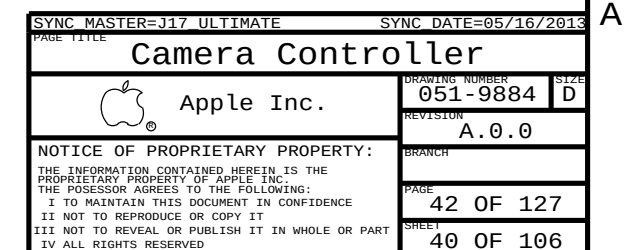


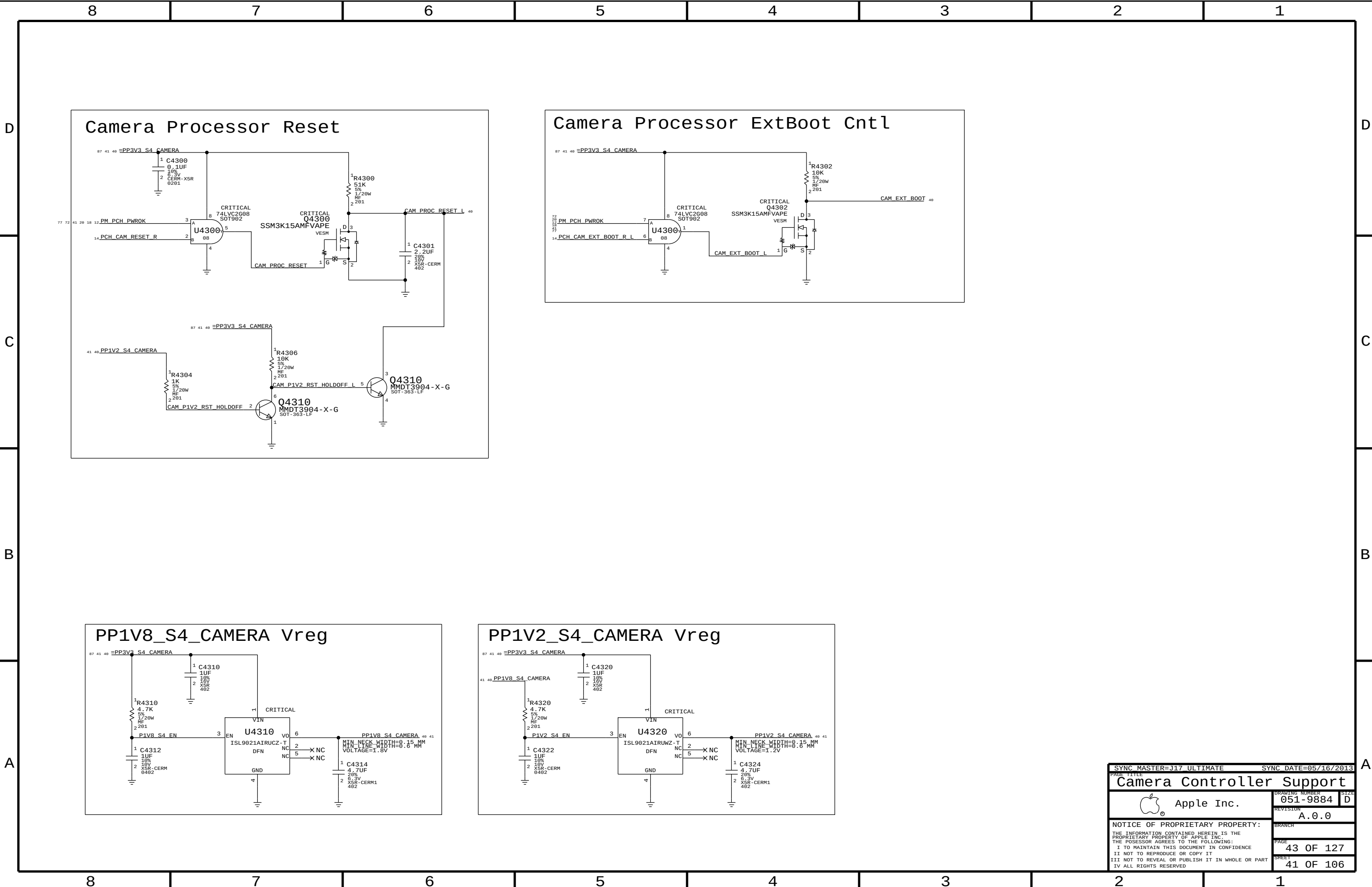
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Apple Inc.		DRAWING NUMBER	051-9884
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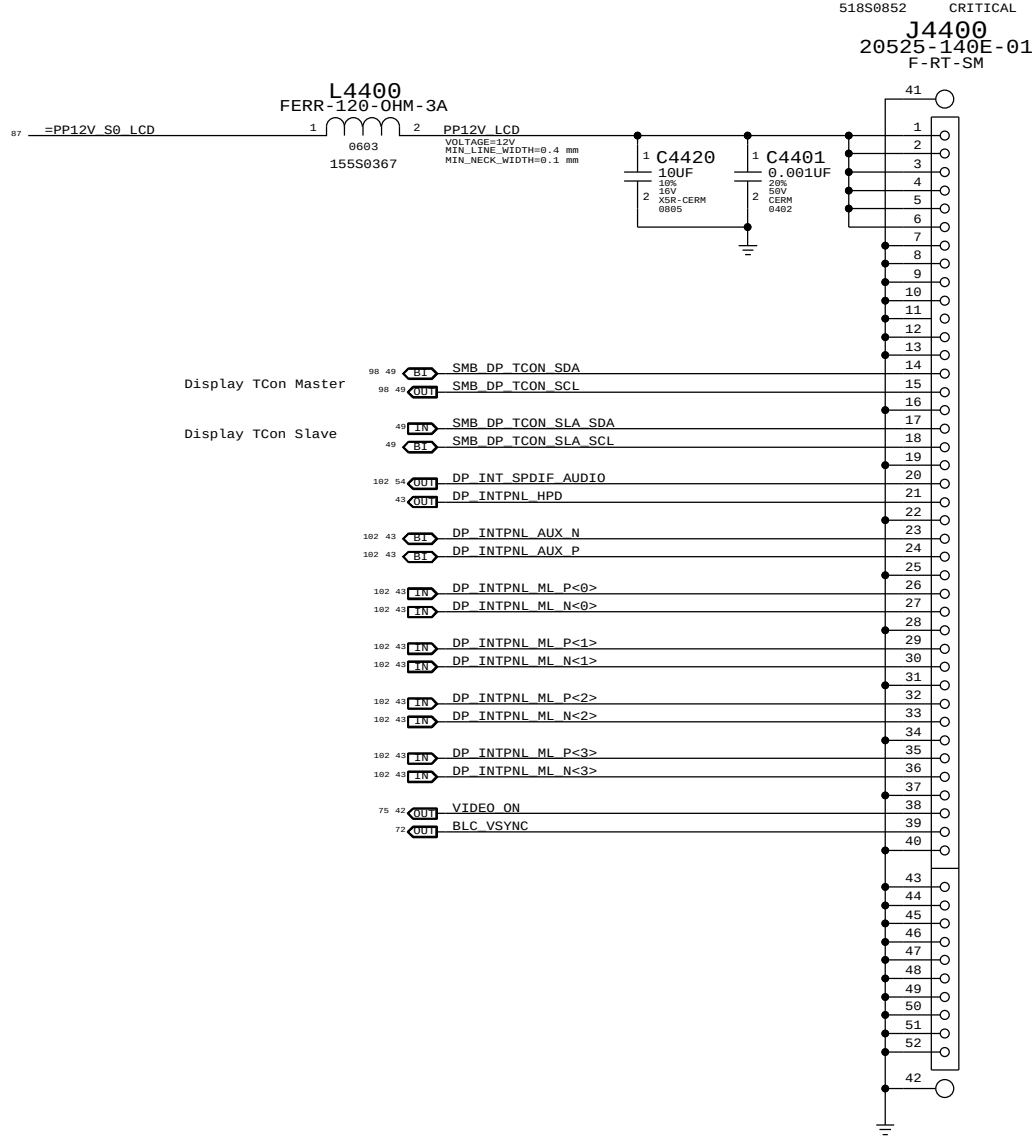


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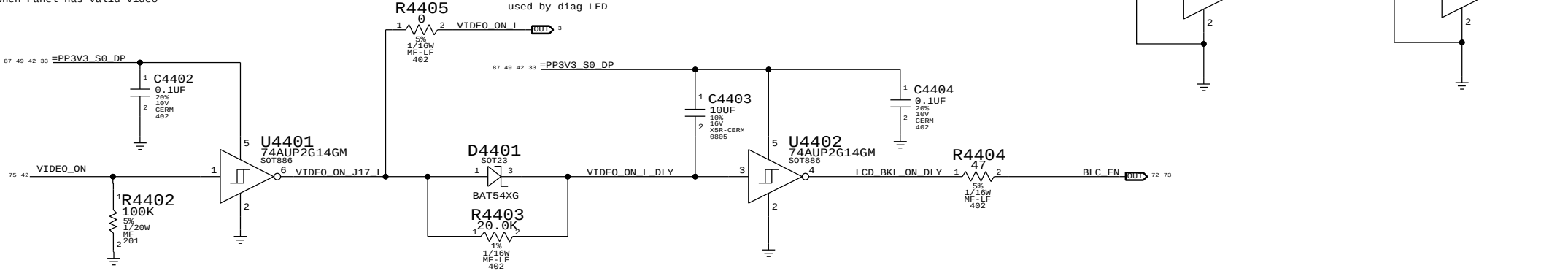


INTERNAL DP (STRAIGHT)

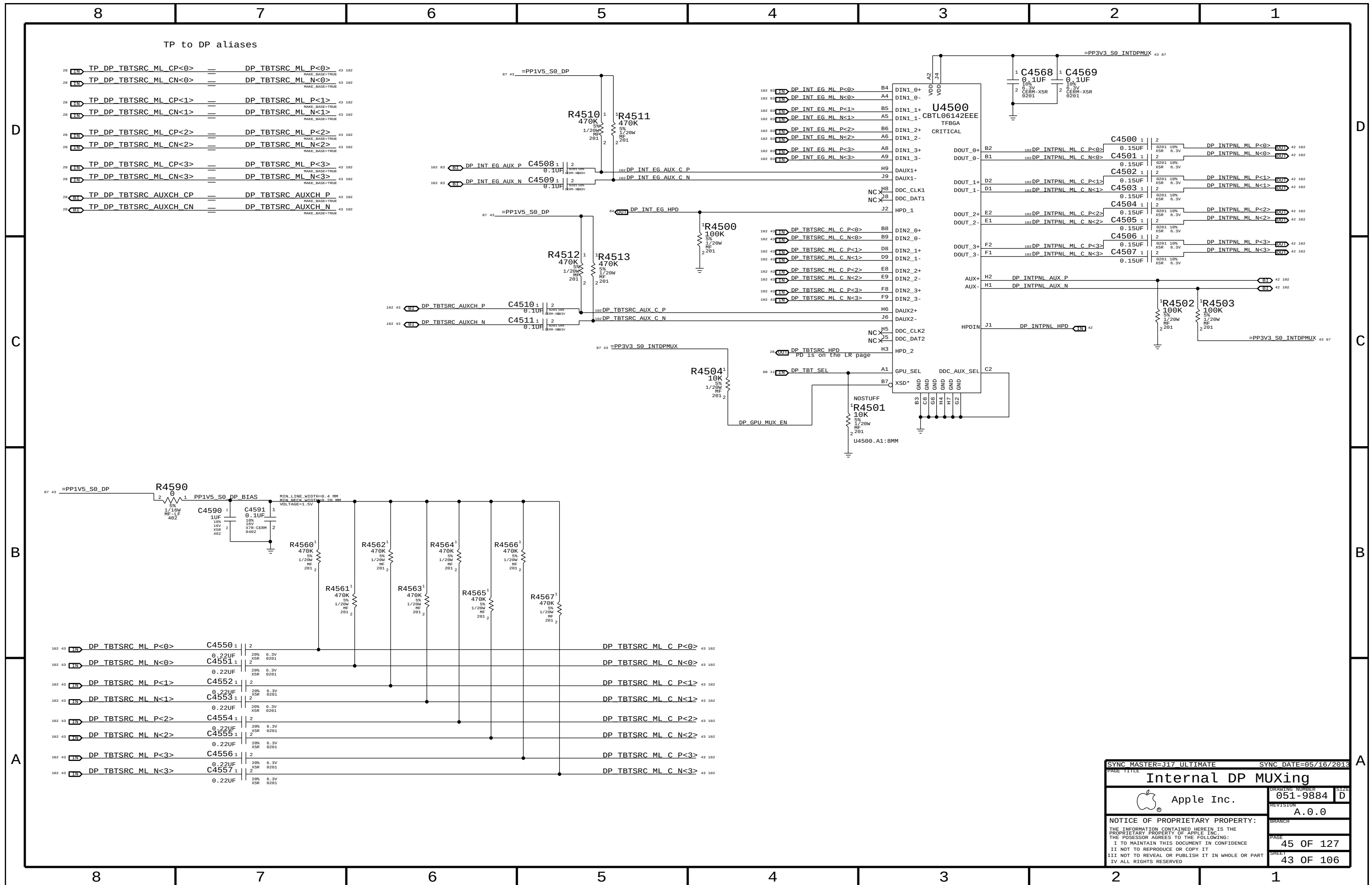


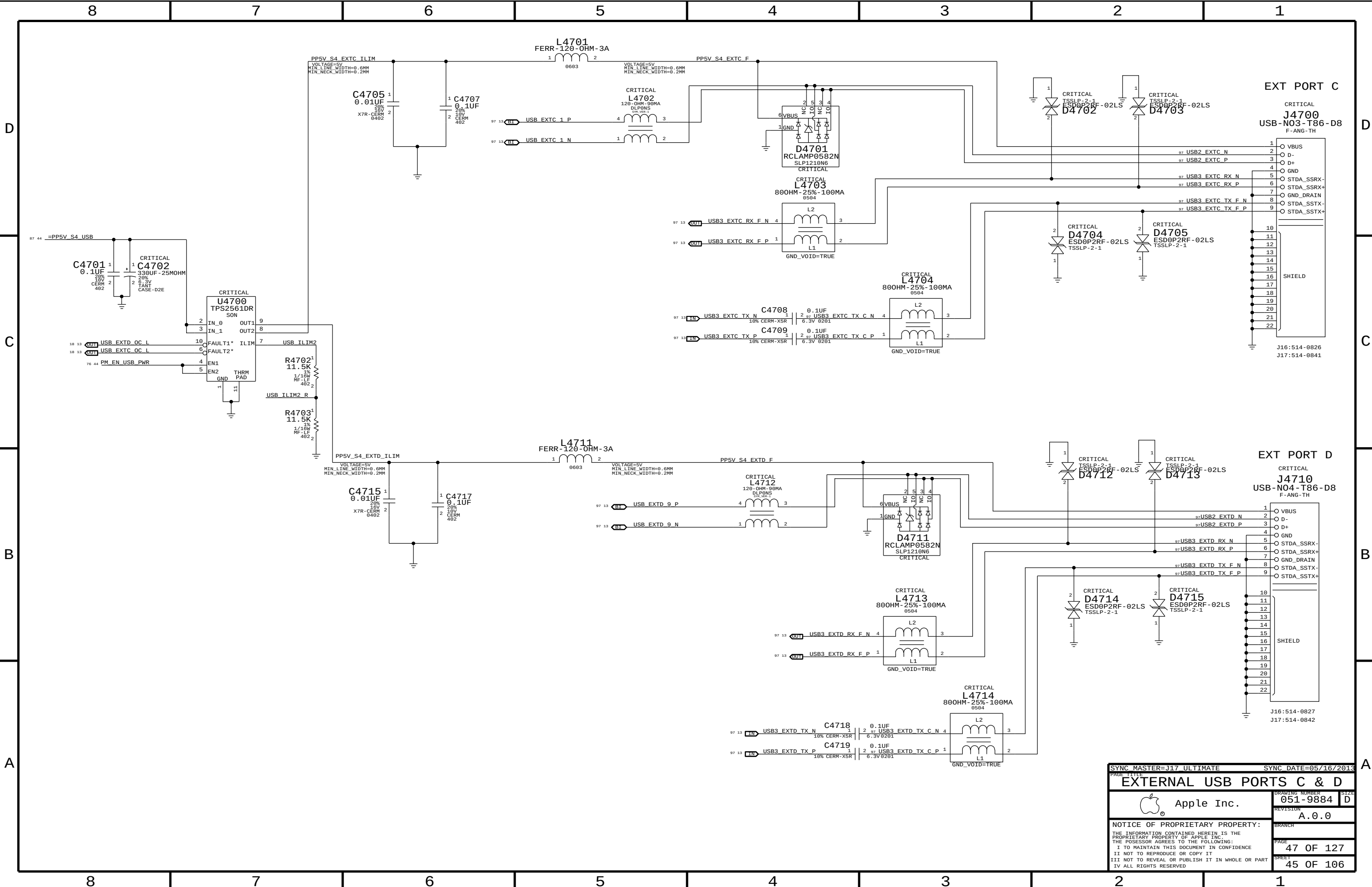
BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

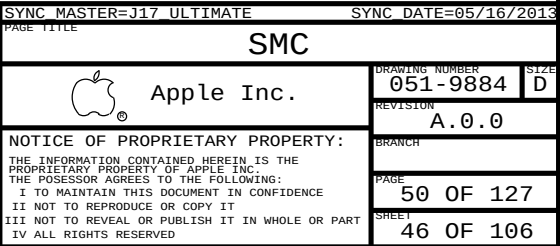


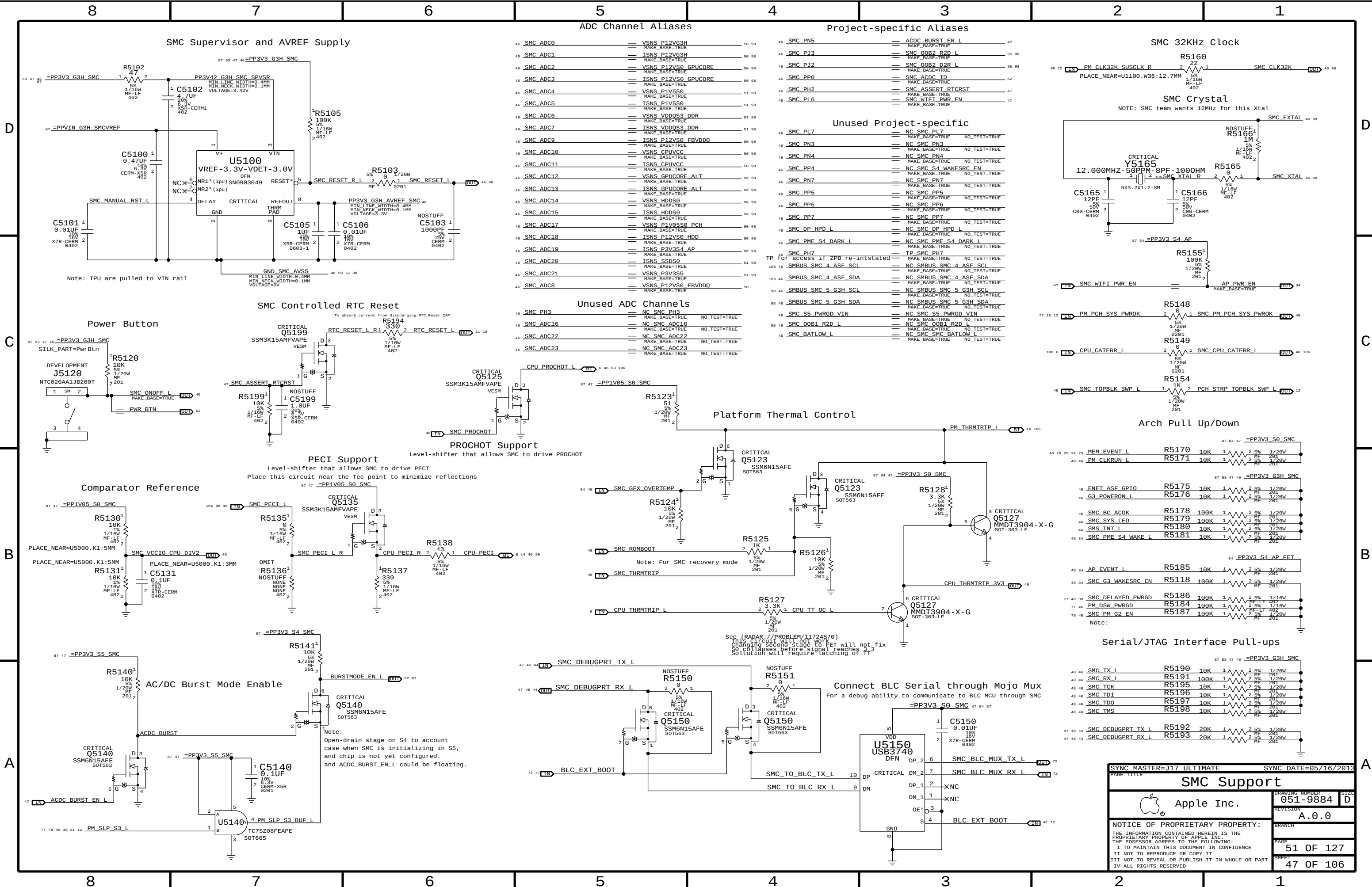
SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
Internal DP Support		DRAWING NUMBER	051-9884
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		PAGE	44 OF 127
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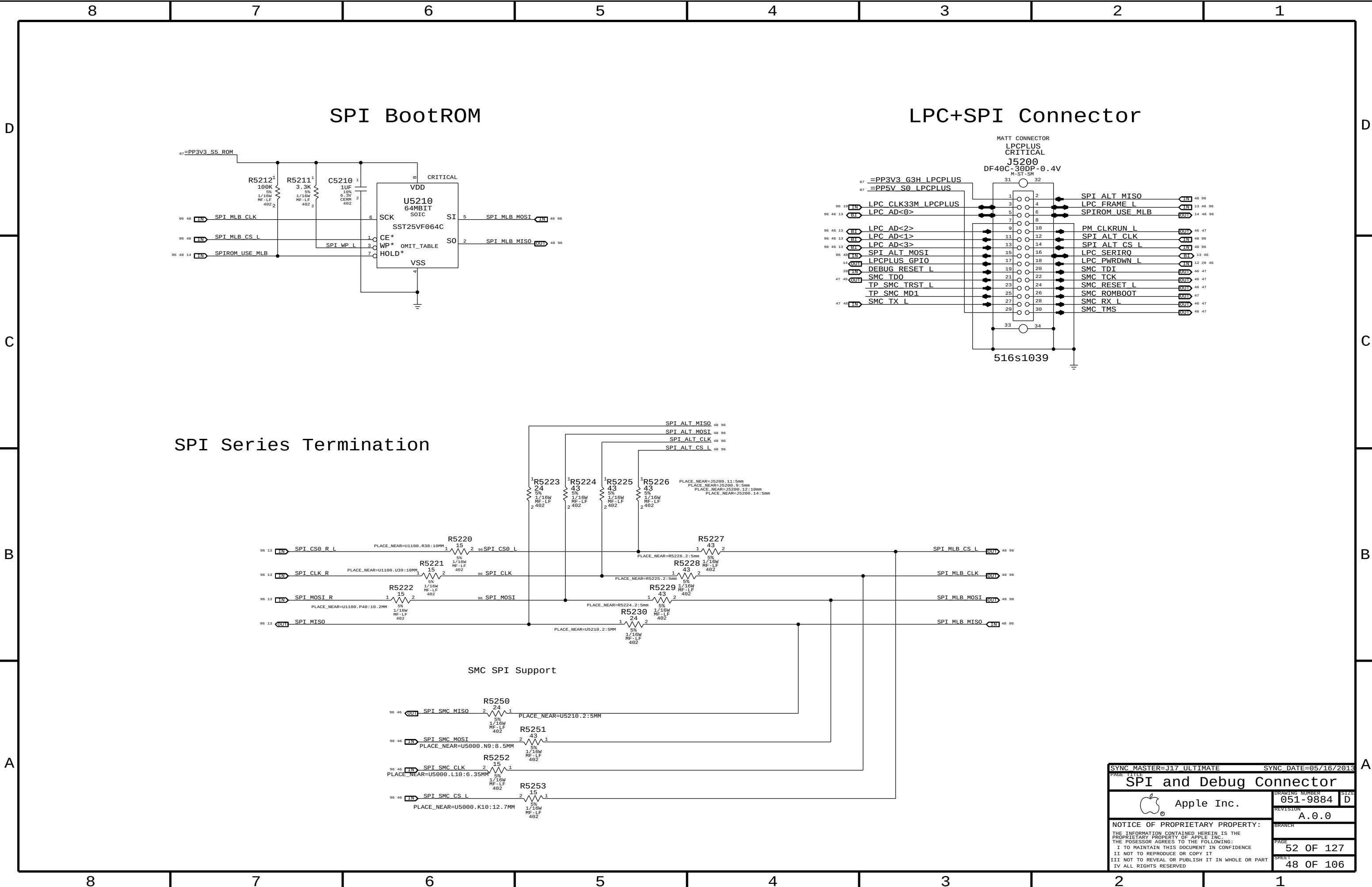




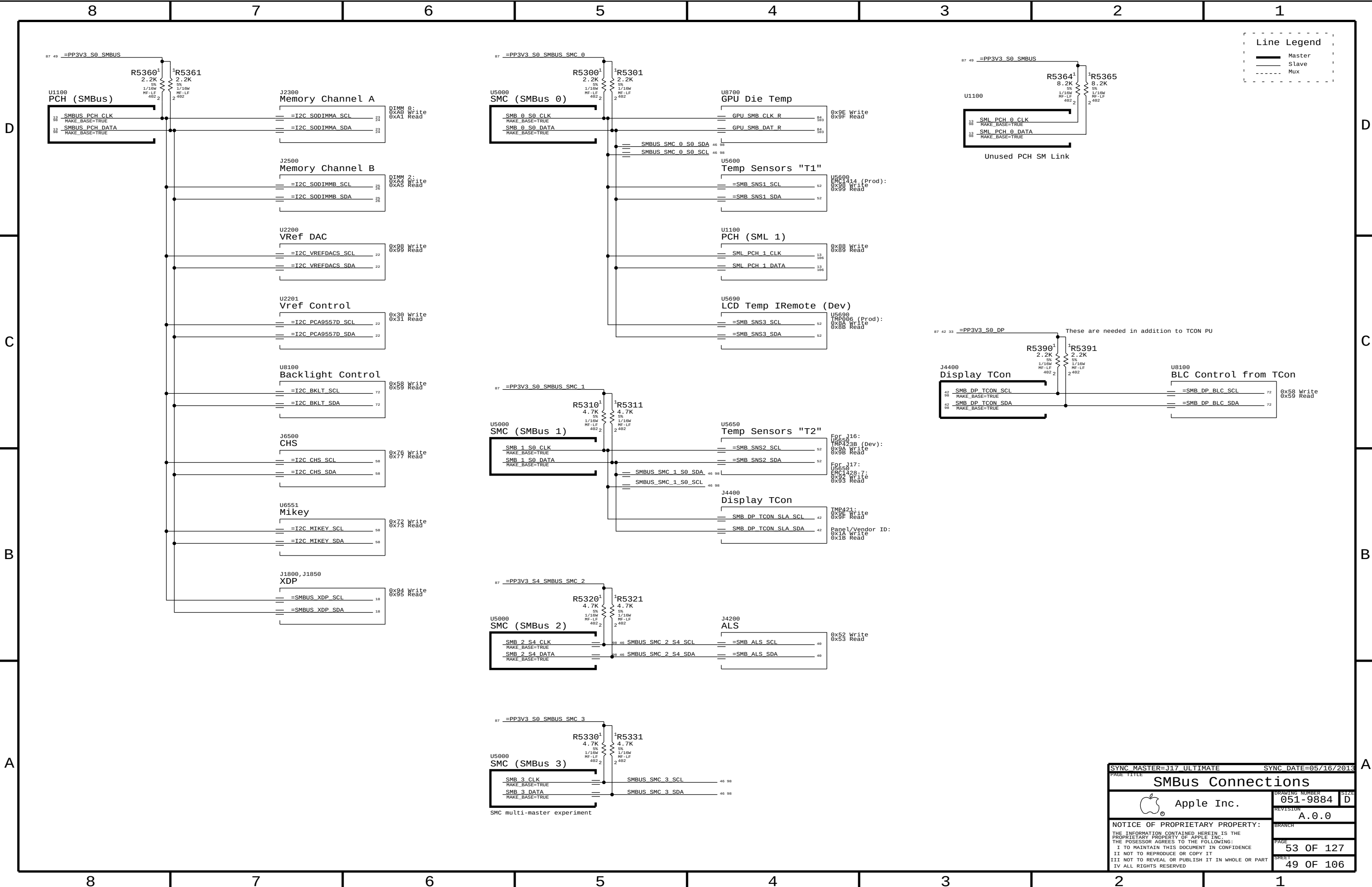
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PAGE TITLE		EXTERNAL USB PORTS C & D	
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		SHEET	45 OF 106

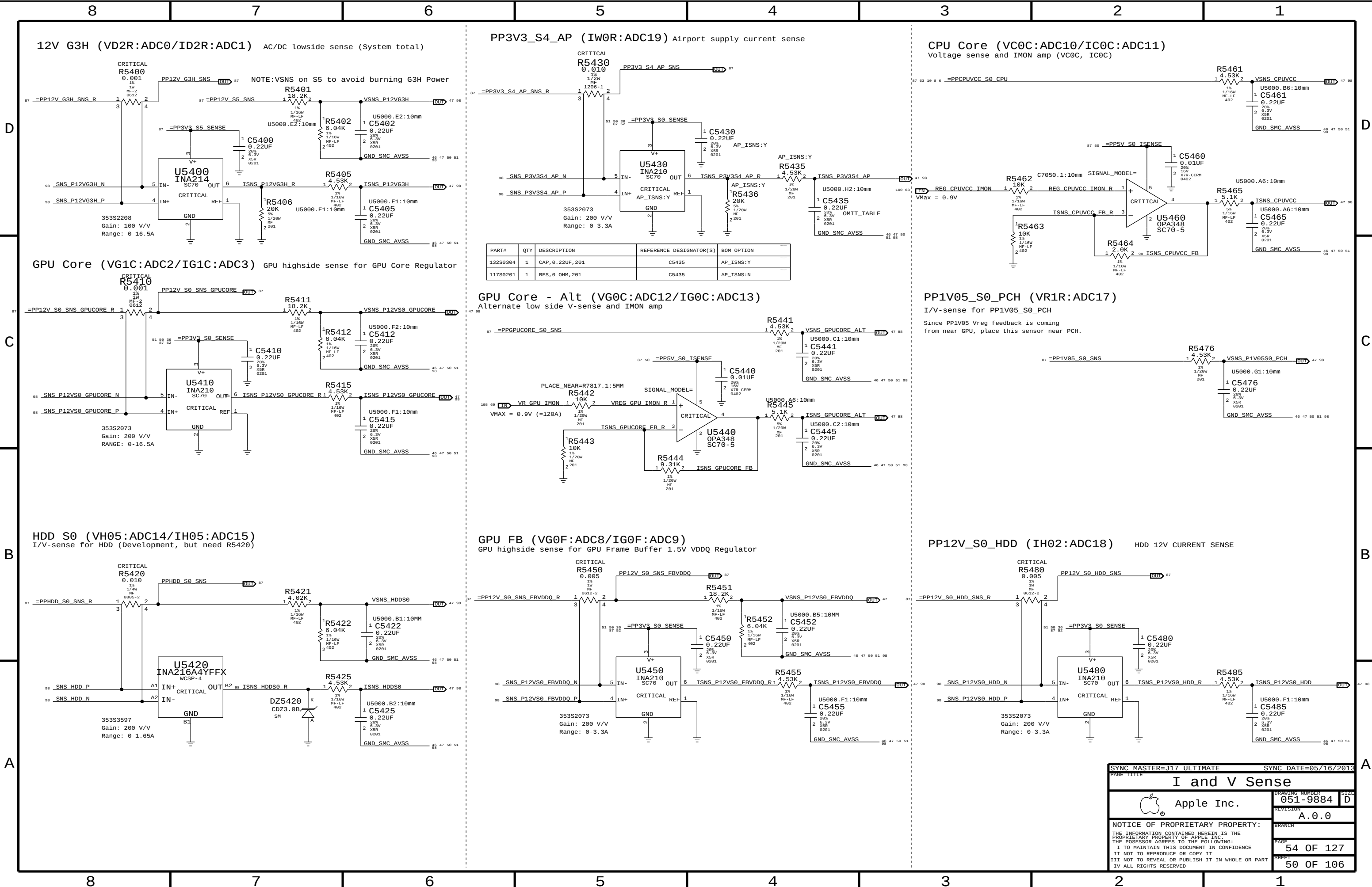


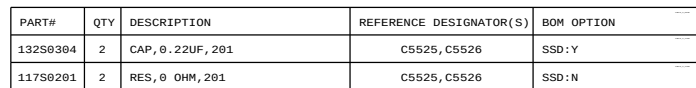
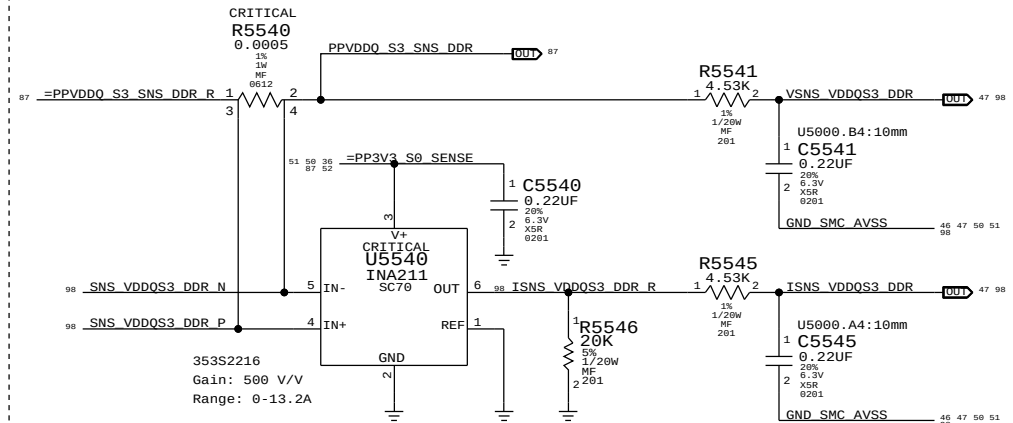
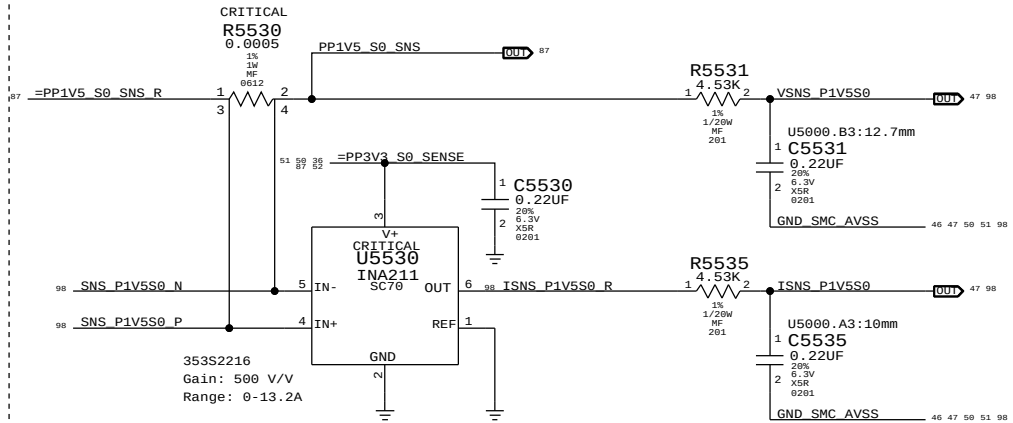




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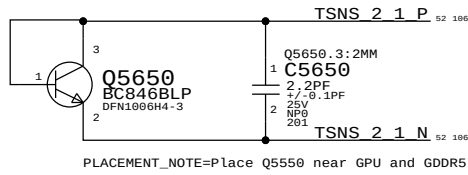


SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE I and V Sense(Continued)			
 Apple Inc.		DRAWING NUMBER 051-9884	SIZE D
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		SHEET 51 OF 106	

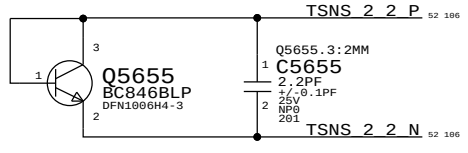
Temperature Sensor T2 EMC1428: Near GPU VR

SNS T2: TEMP SENSOR IC

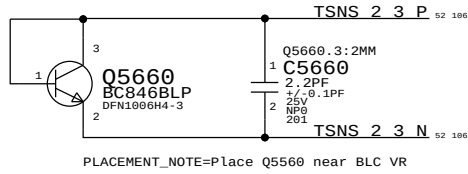
GPU Proximity



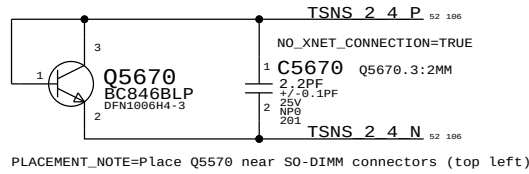
Ambient



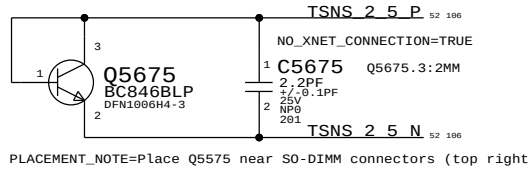
BLC Proximity



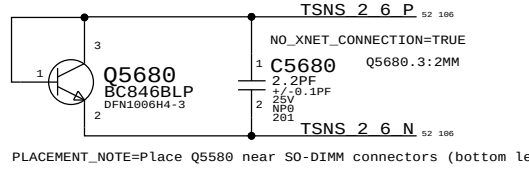
SO-DIMM Proximity 1



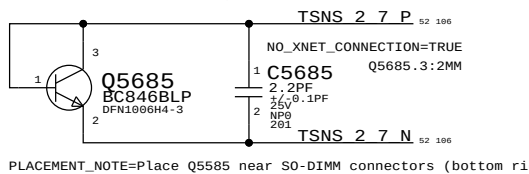
SO-DIMM Proximity 2



SO-DIMM Proximity 3



SO-DIMM Proximity 4



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

GPU Prox (TG0p)

Ambient (TA0p)

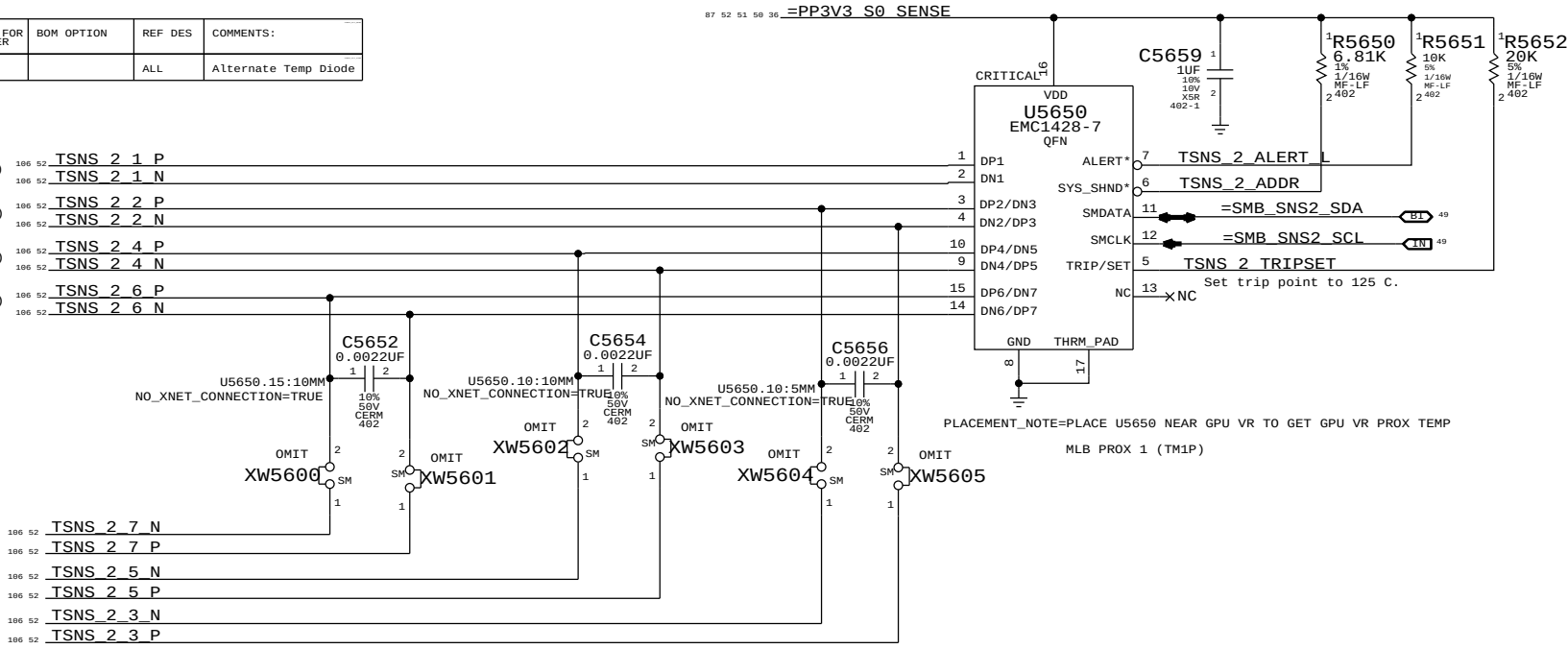
SO-DIMM Prox 1 (TM0p)

SO-DIMM Prox 3 (TM2p)

SoDIMM Prox 4 (TM3p)

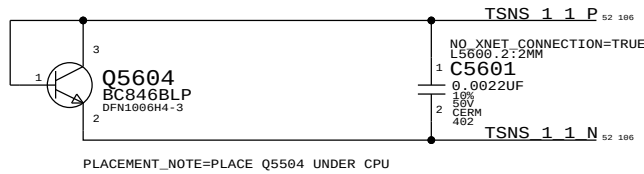
SoDIMM Prox 2 (TM1p)

BLC PROX (Tb0p)

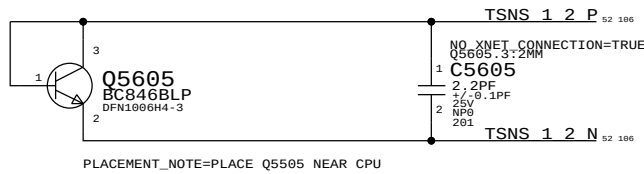


Temperature Sensor T1 EMC1414: Near PSU Conn

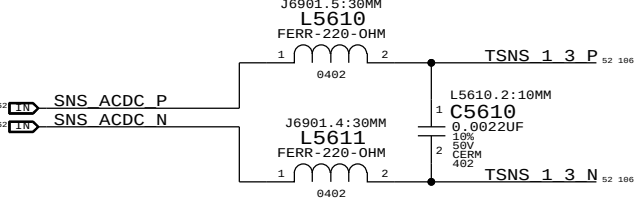
MLB Proximity



CPU Proximity



AC/DC Via connector to diode inside PSU



MLB PROX 2 (Tm2p)

CPU Prox (TC0p)

AC/DC (Tp2h)

TSNS_1_1_P

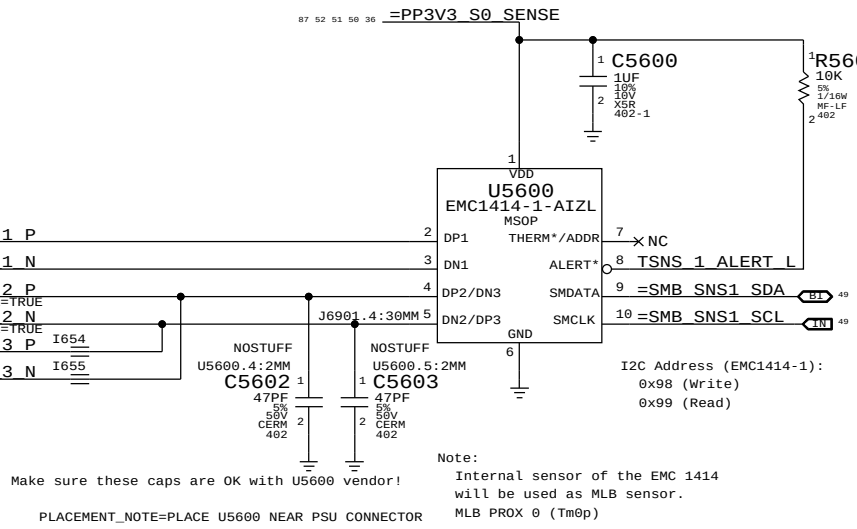
TSNS_1_1_N

TSNS_1_2_P

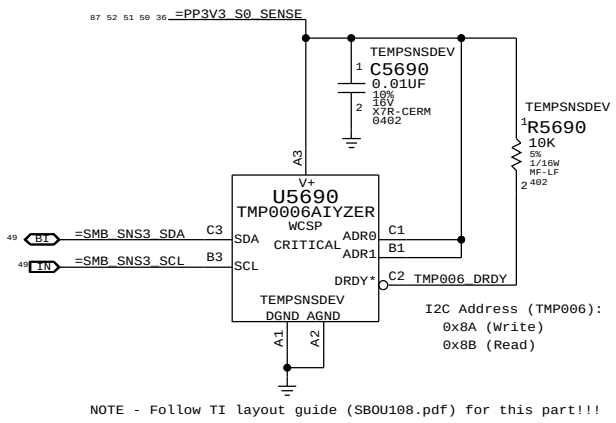
TSNS_1_2_N

TSNS_1_3_P

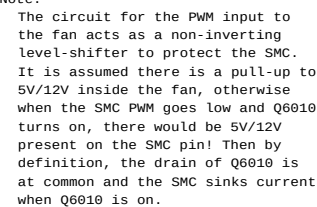
TSNS_1_3_N



Temperature Sensor T3: LCD Remote Sensor(Dev Only)



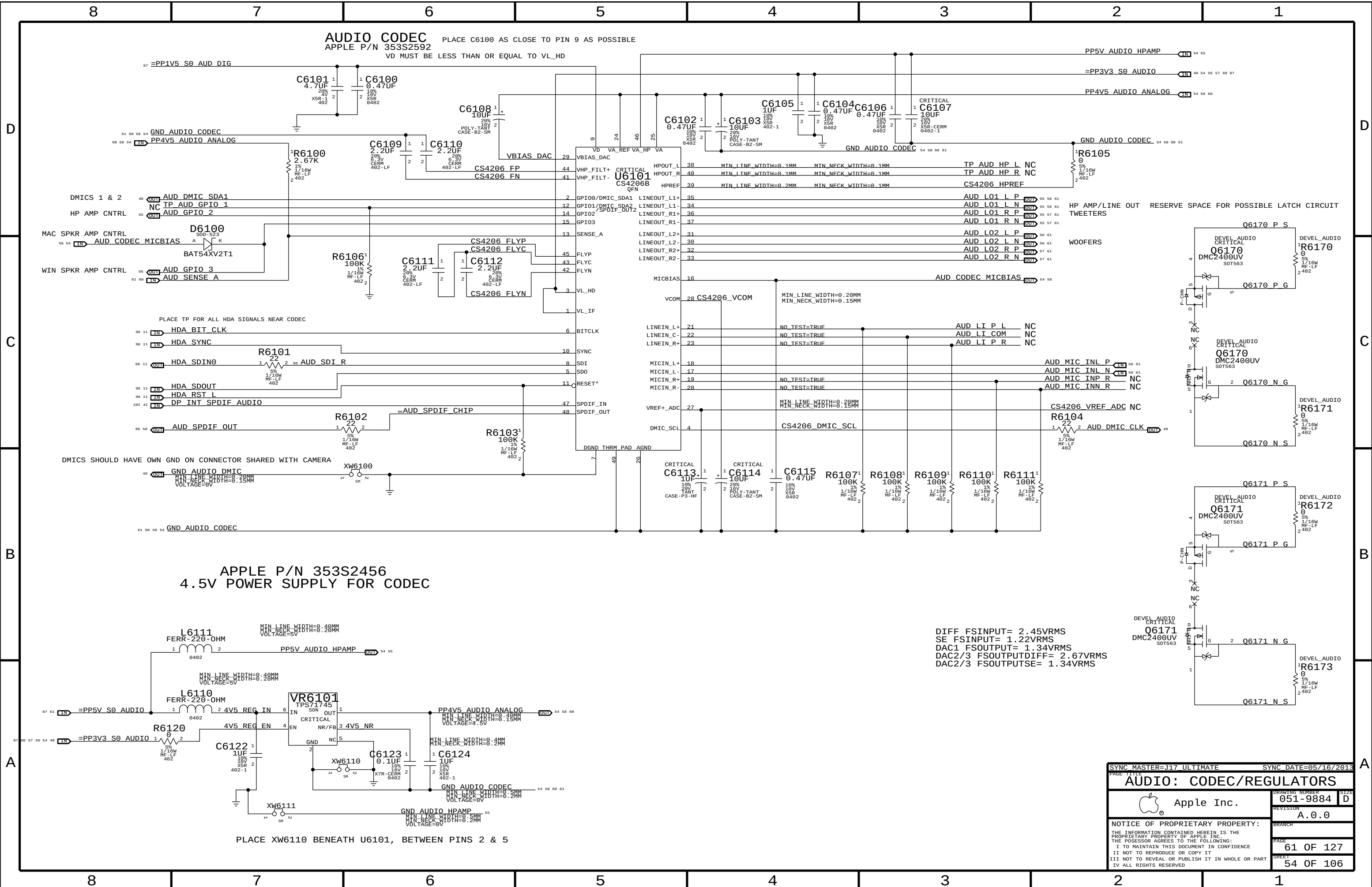
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Temperature Sensors	
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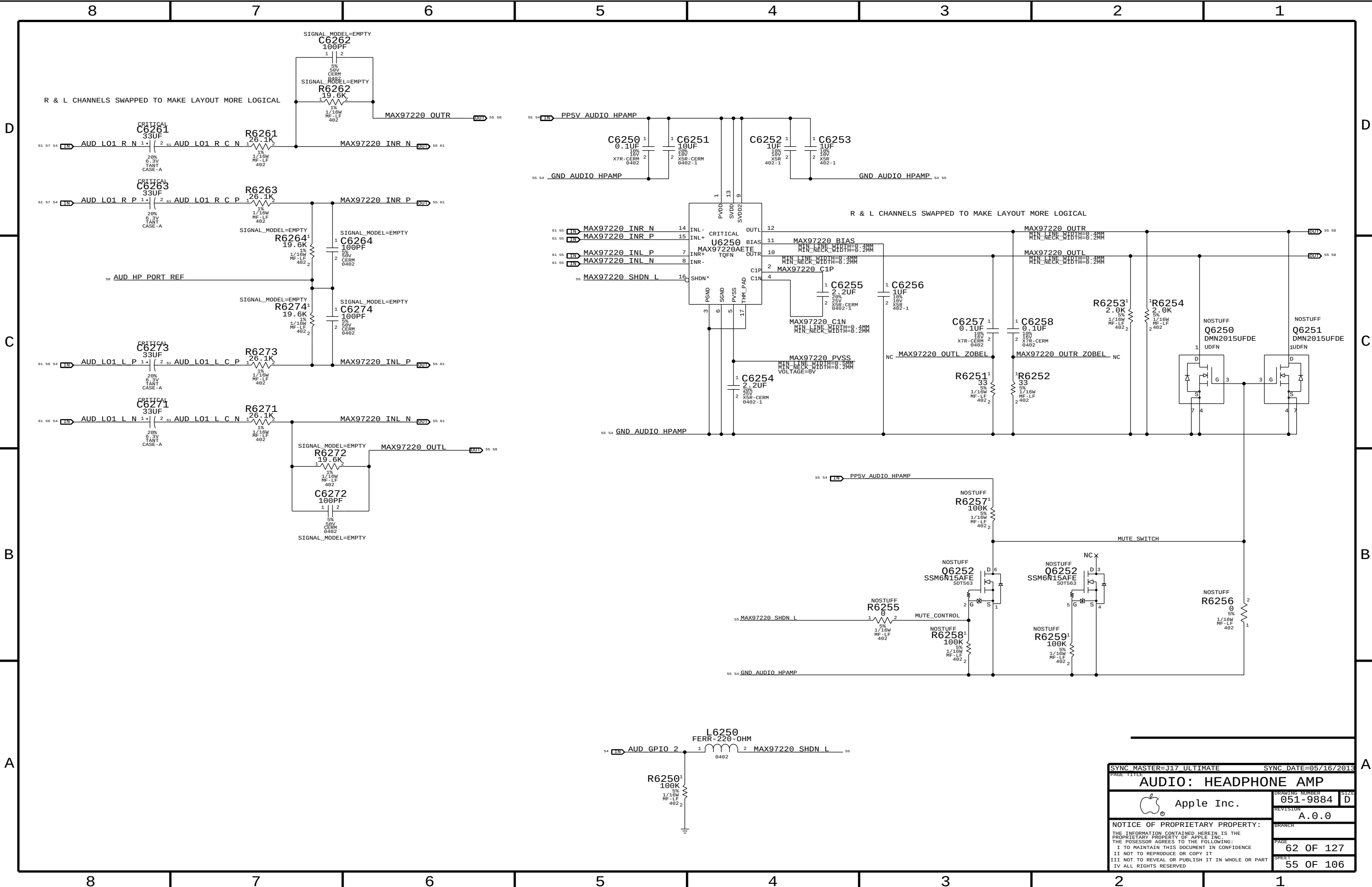
Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



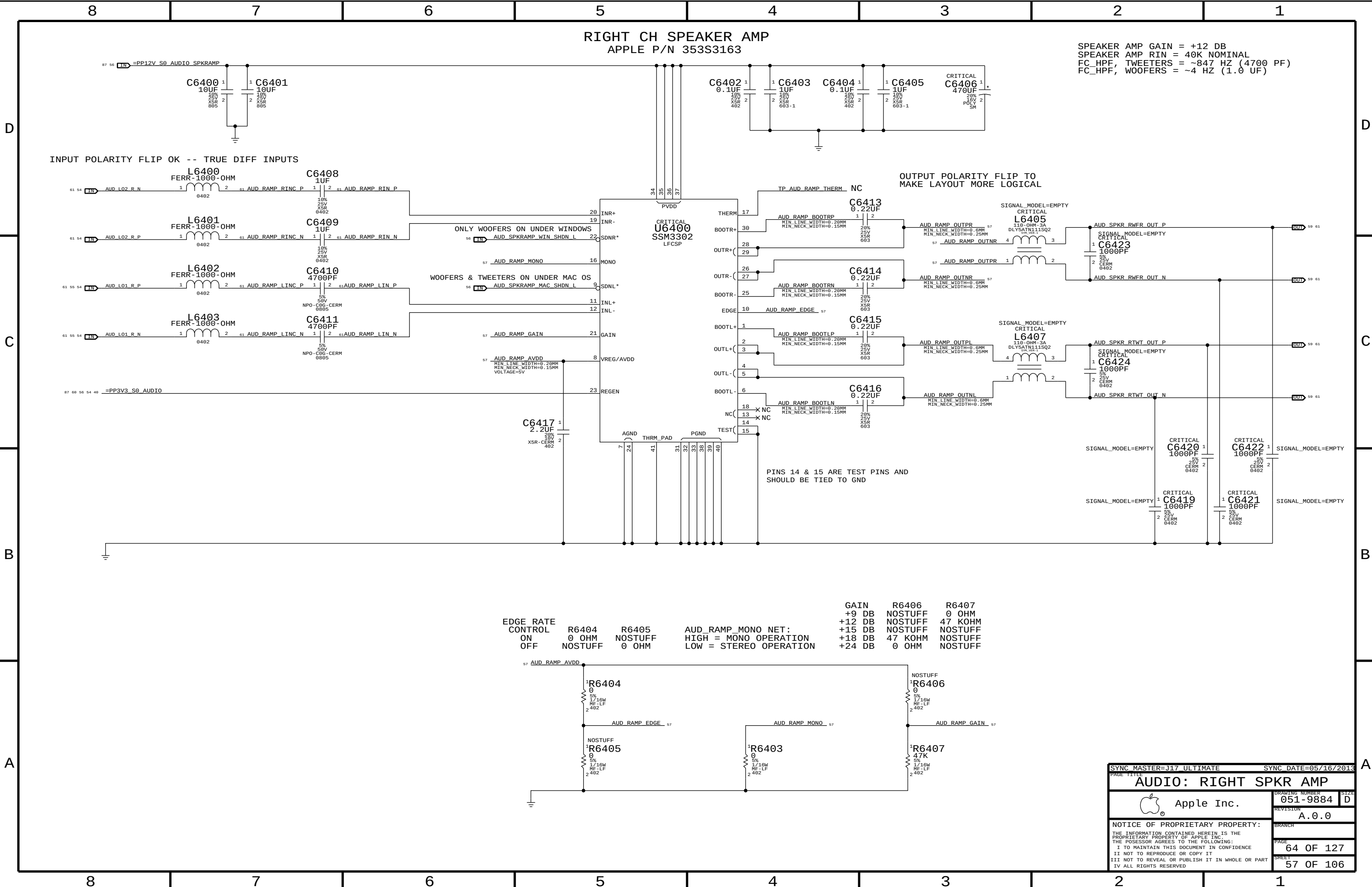
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PAGE TITLE			
System Fan			
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BRANCH		PAGE	
		60 OF 127	
		SHEET	
		53 OF 106	

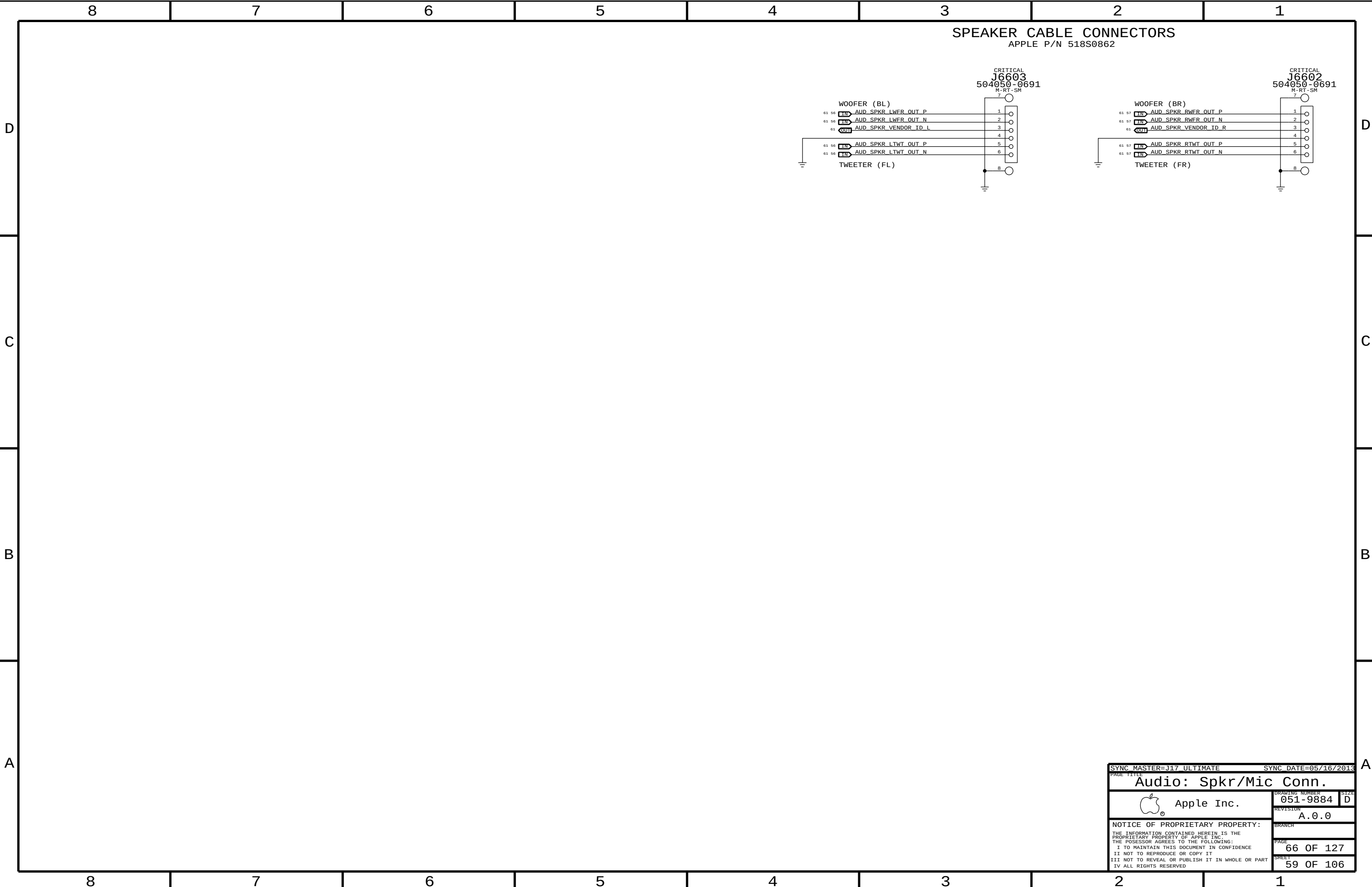


PAGE TITLE		SYNC DATE=05/16/2013	
AUDIO: CODEC/REGULATORS		DRAWING NUMBER	
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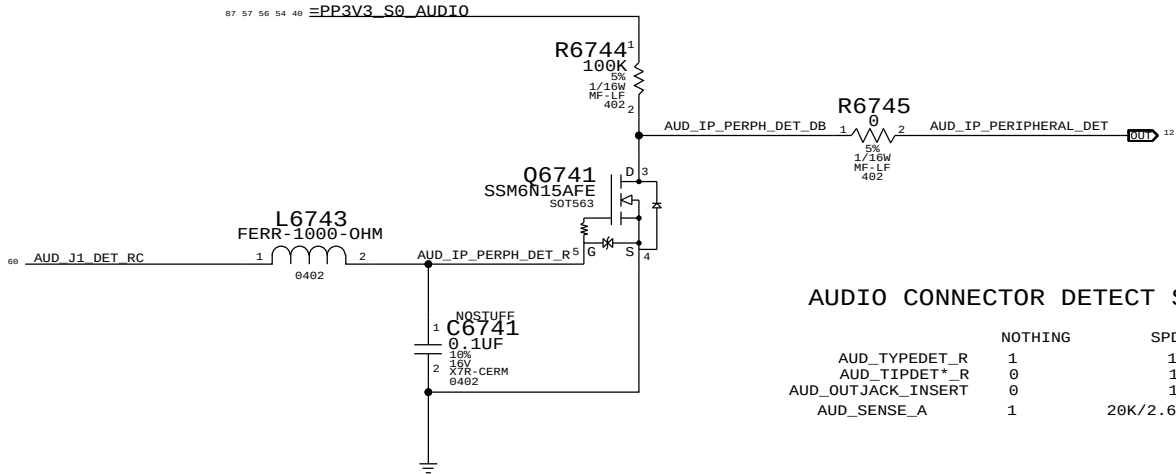


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PAGE TITLE		AUDIO: HEADPHONE AMP	
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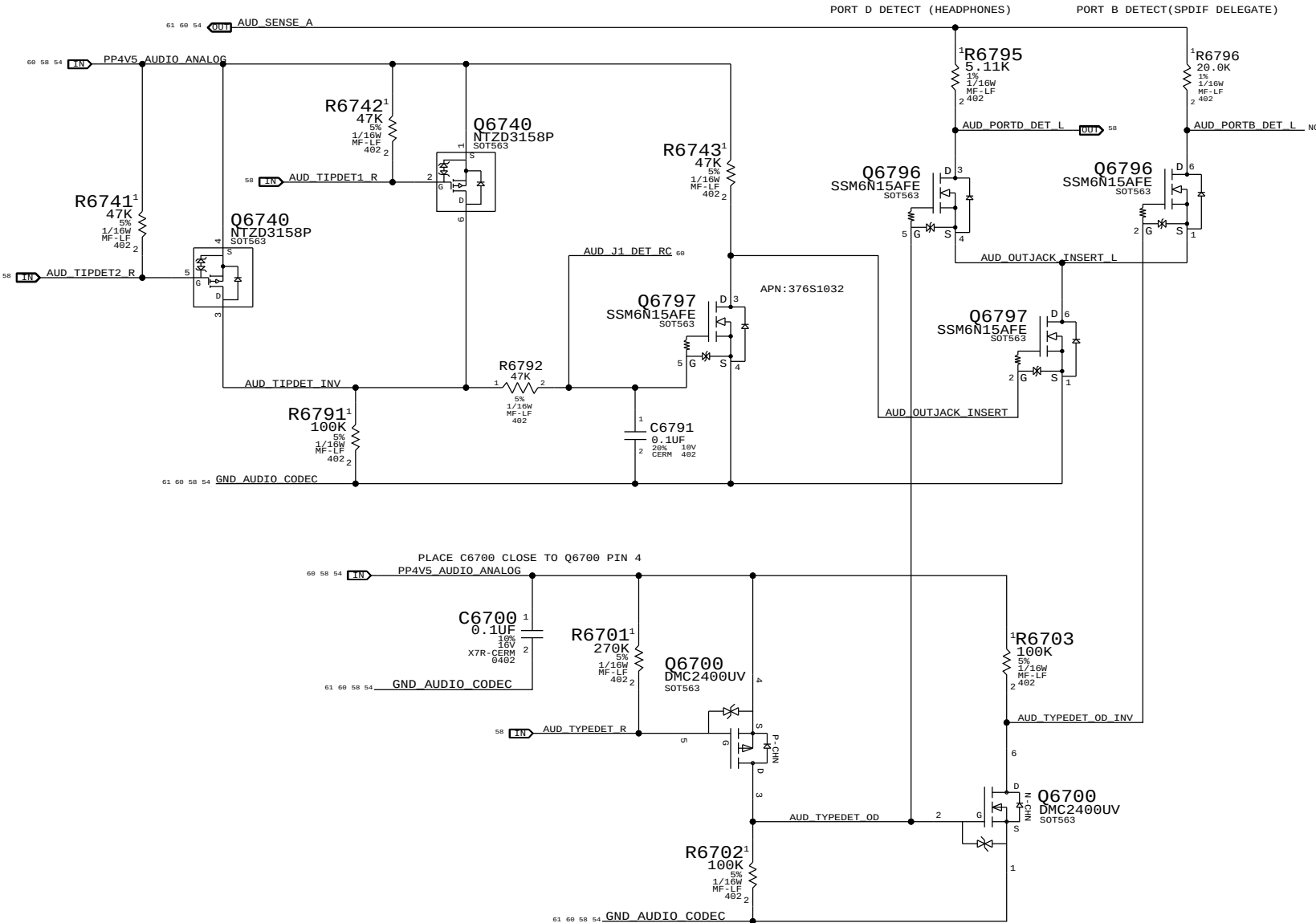
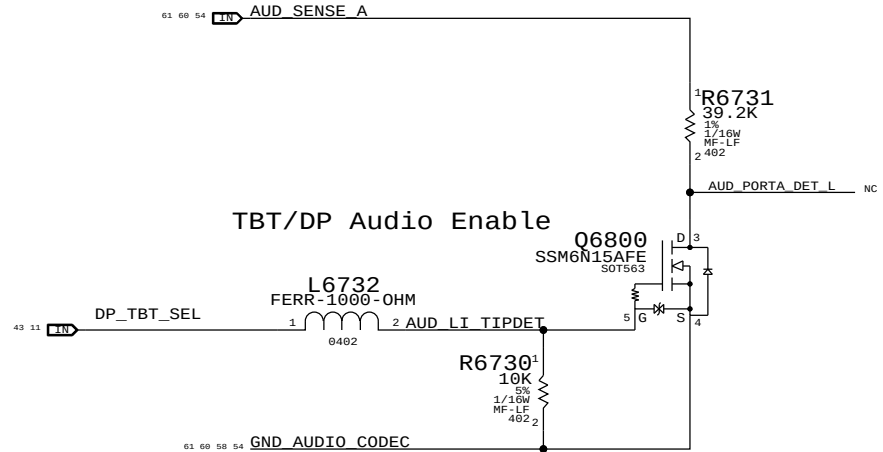


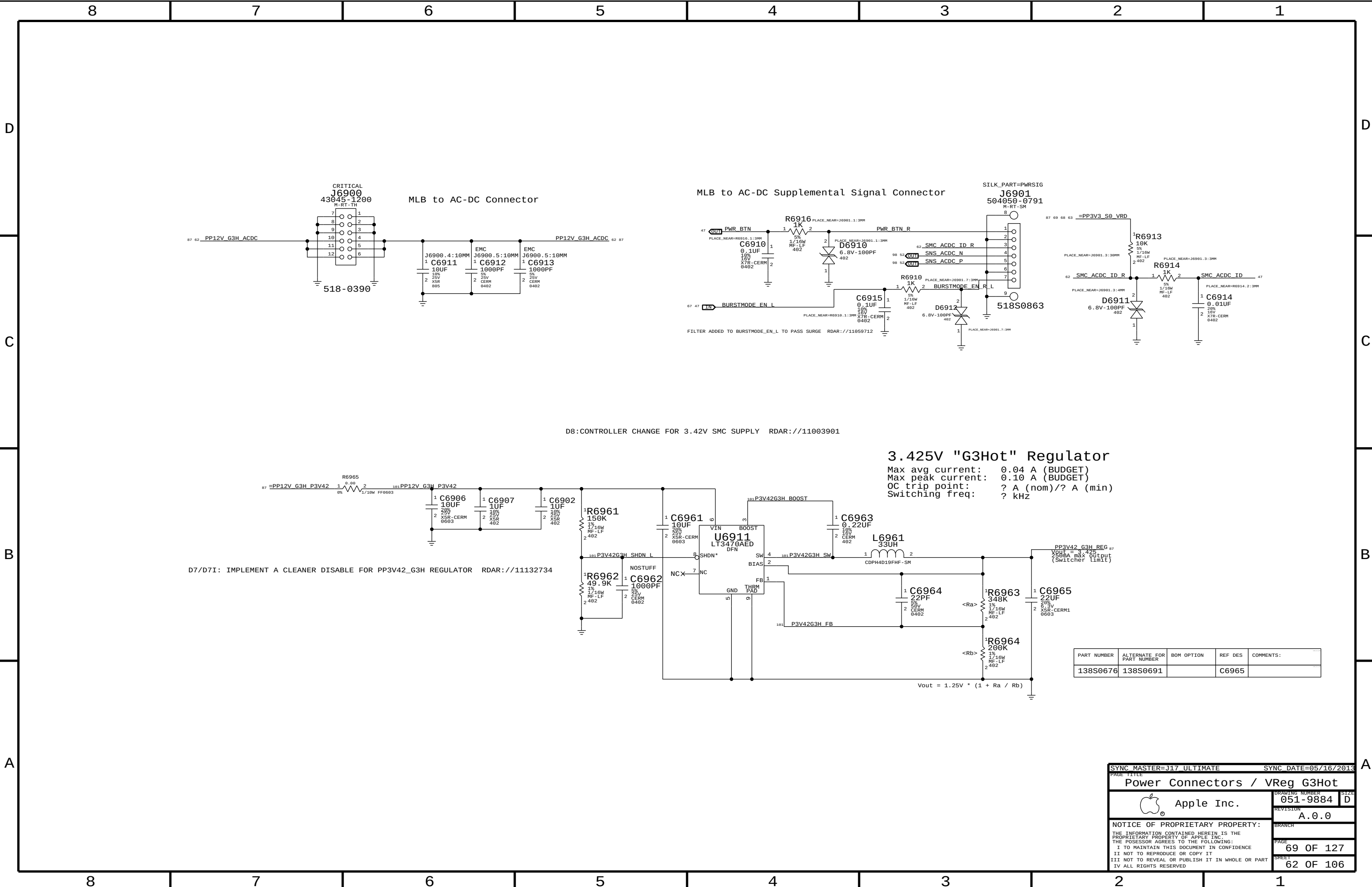


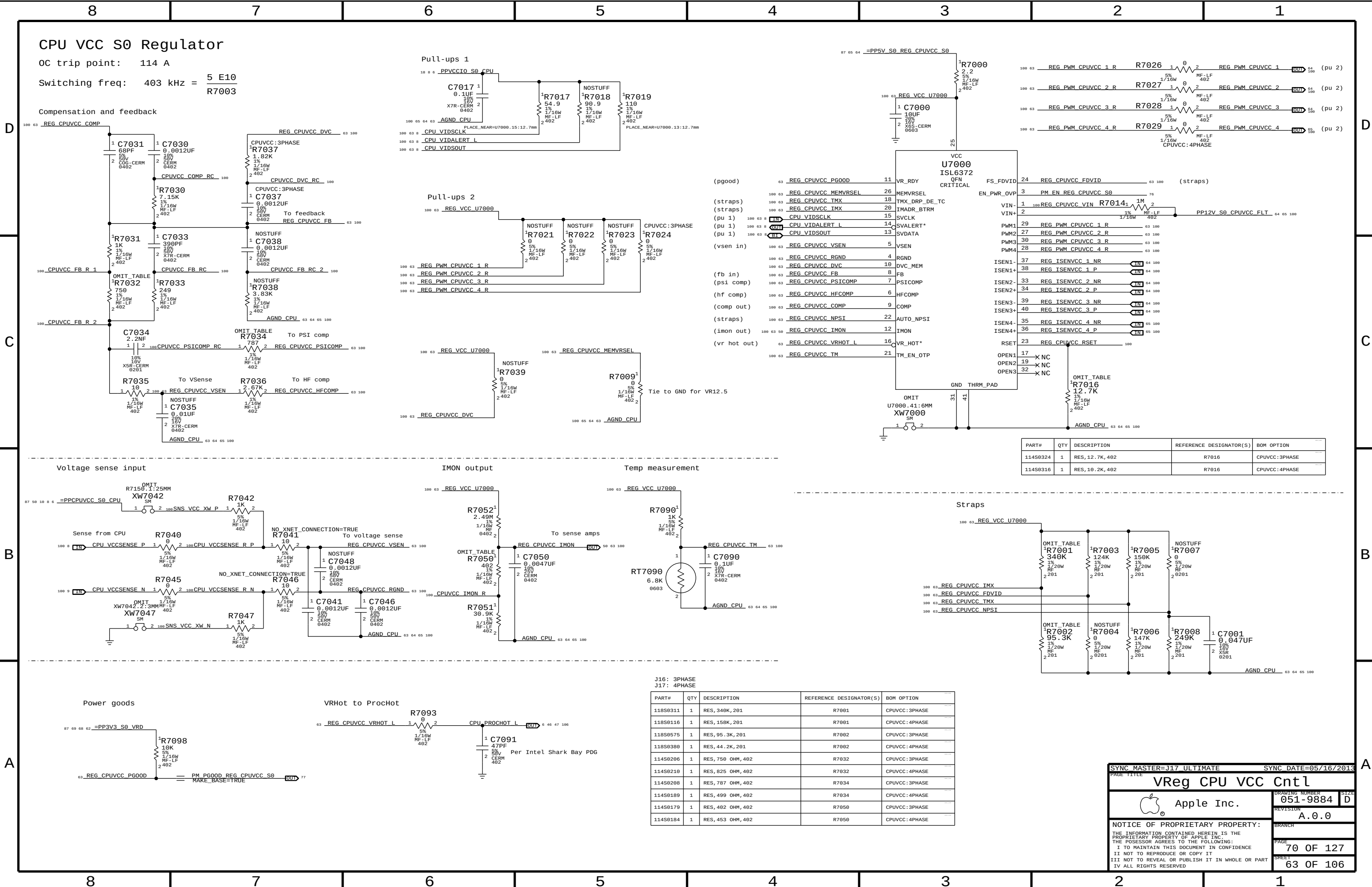
IPHS HS Detect Debounce CKT



Target Display Mode Detect







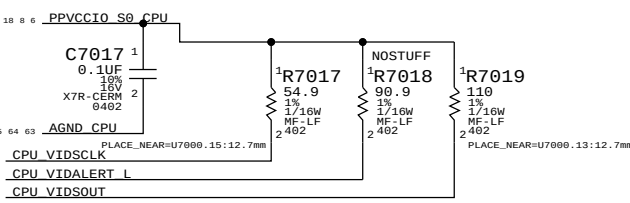
CPU VCC S0 Regulator

OC trip point: 114 A

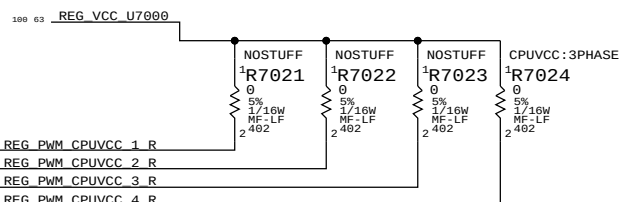
Switching freq: 403 kHz = $\frac{5 \text{ E10}}{R7003}$

Compensation and feedback

Pull-ups 1



Pull-ups 2



(pgood)

(straps)

(straps)

(pu 1)

(pu 1)

(pu 1)

(vsen in)

(fb in)

(psi comp)

(hf comp)

(comp out)

(straps)

(imon out)

(vr hot out)



Table with 5 columns: PART#, QTY, DESCRIPTION, REFERENCE DESIGNATOR(S), BOM OPTION. It lists components R7016 and R7018 with their respective quantities and descriptions.

Voltage sense input

IMON output

Temp measurement

Straps

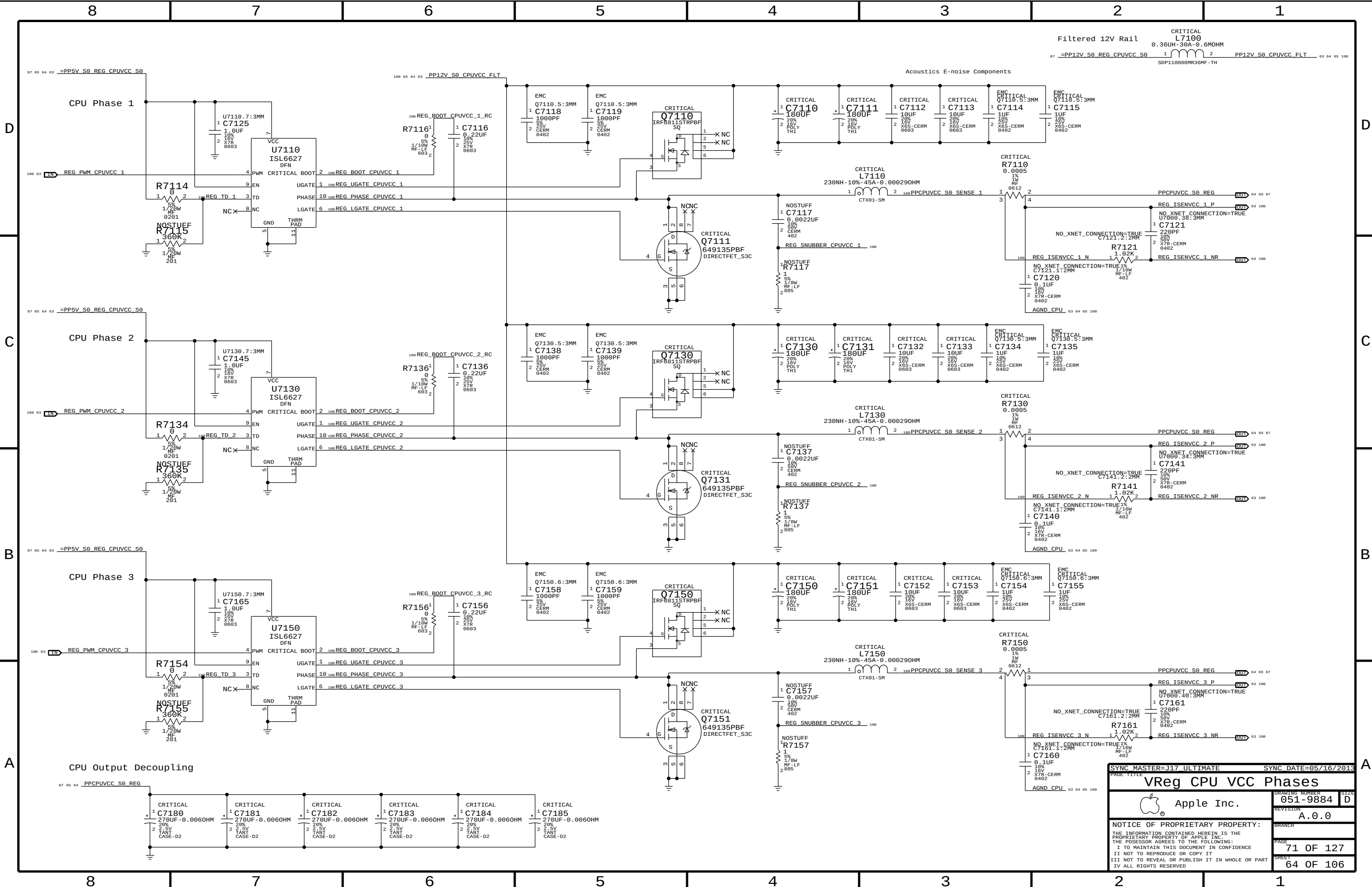
Power goods

VRHot to Prochot

J16: 3PHASE
J17: 4PHASE

Table with 5 columns: PART#, QTY, DESCRIPTION, REFERENCE DESIGNATOR(S), BOM OPTION. It lists various components like resistors and capacitors used in the power goods and VRHot to Prochot sections.

SYNC MASTER=J17 ULTIMATE SYNC DATE=05/16/2013
PAGE TITLE: VReg CPU VCC Cntl
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VReg CPU VCC Phases

Apple Inc.

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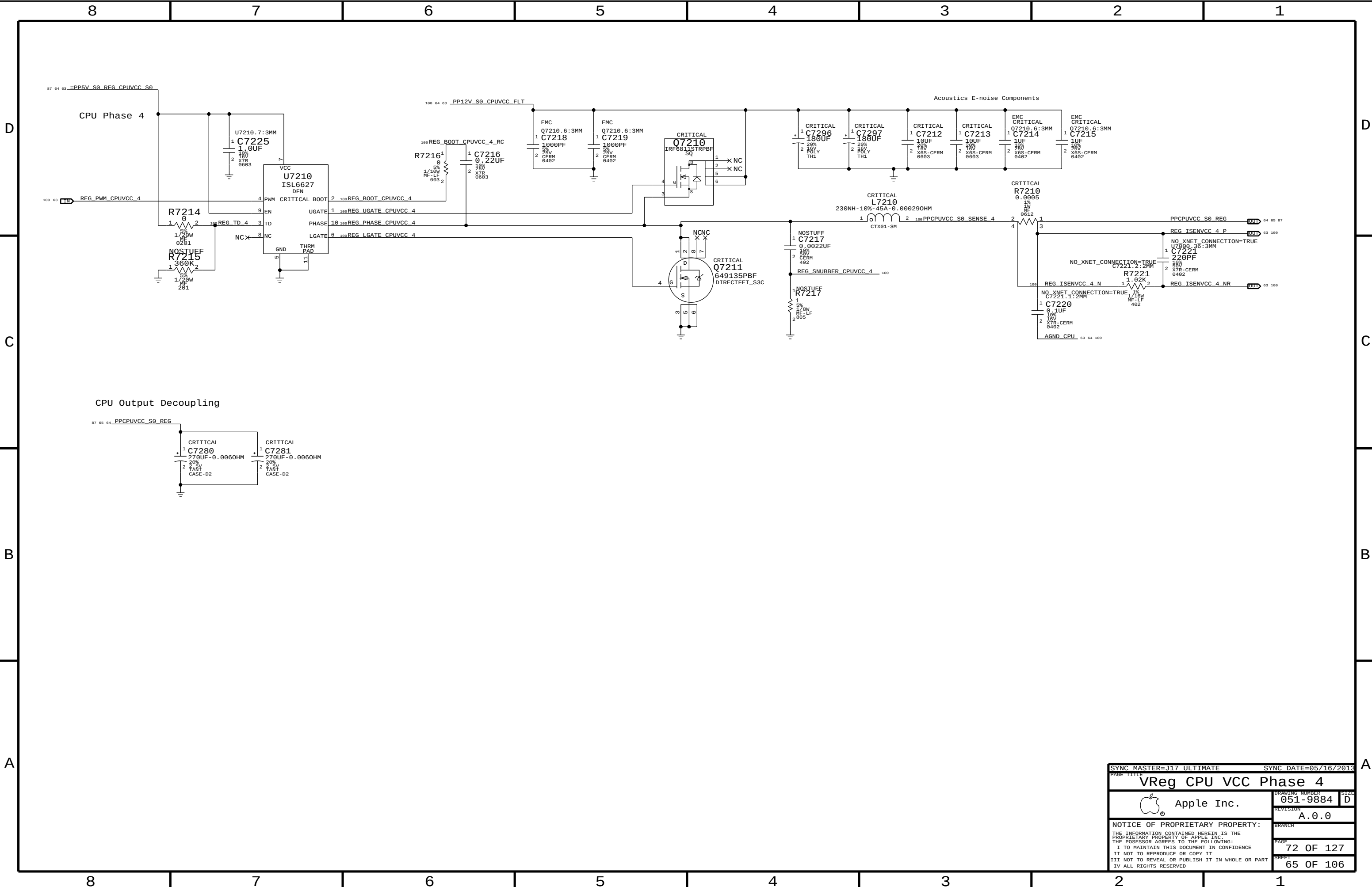
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C

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A



C

B

A

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
VReg VDDQ S3			
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D

C



A

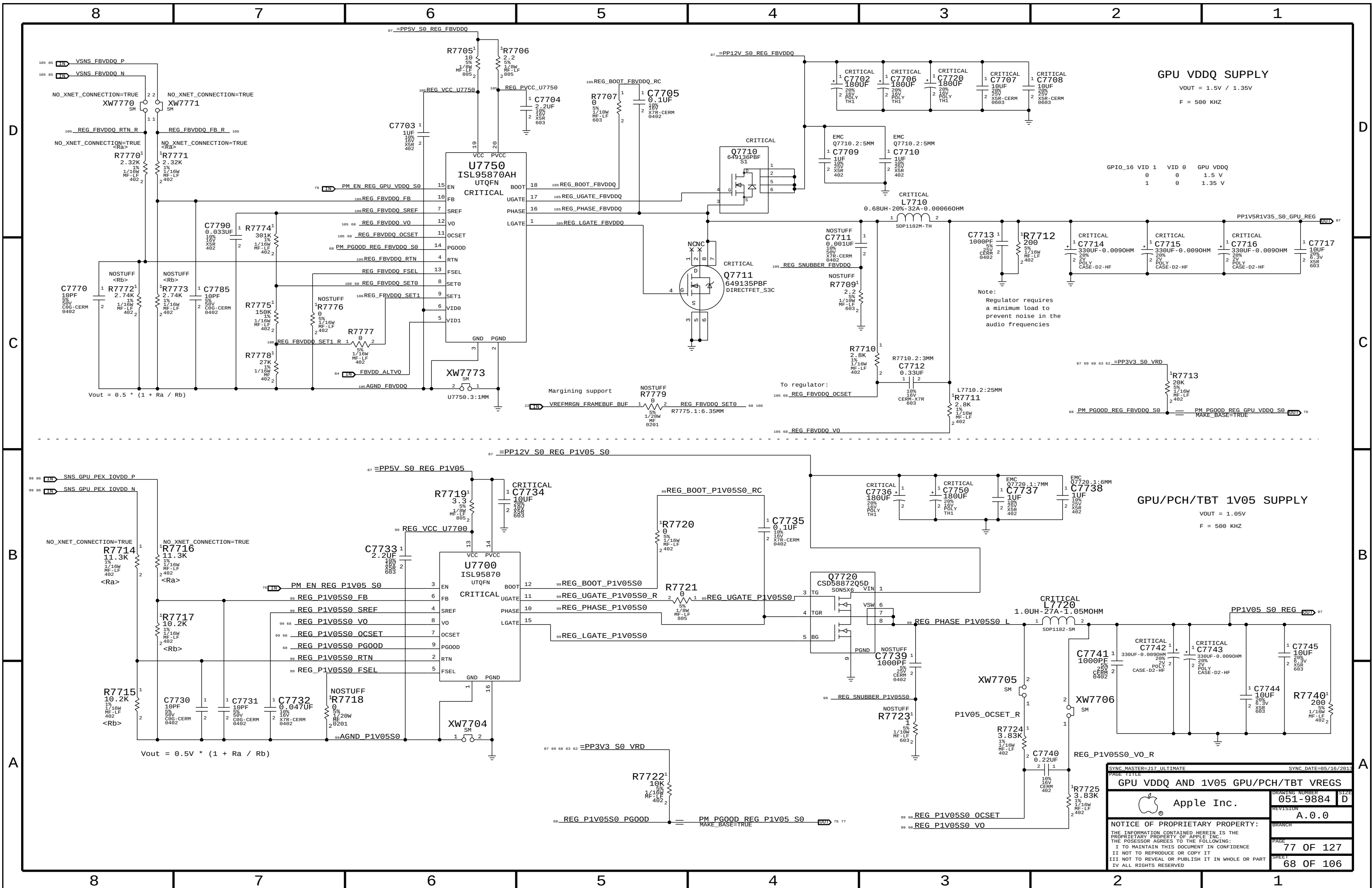
62 43

D

C



A



GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V

F = 500 KHZ

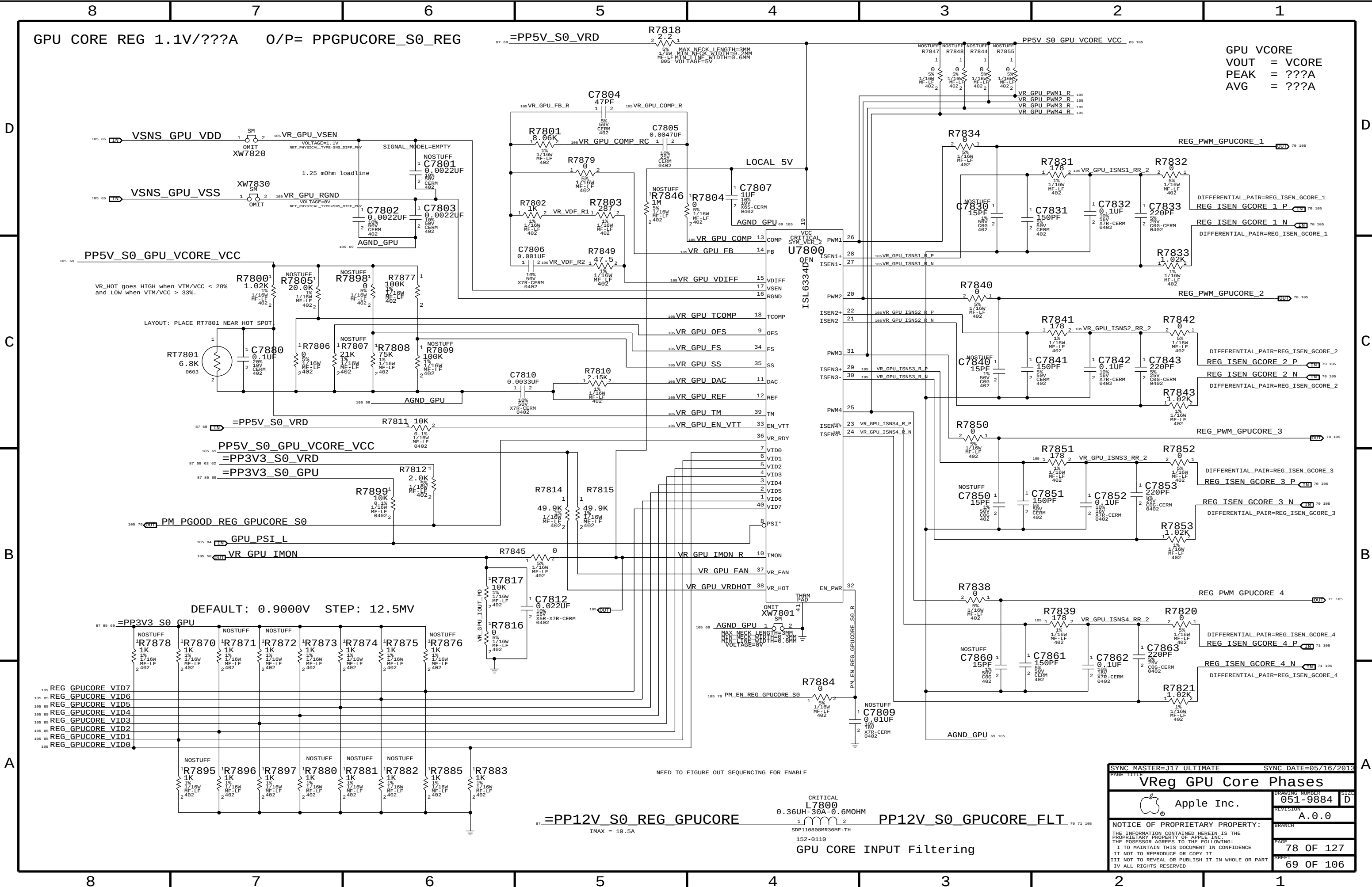
GPIO_16 VID 1 VID 0 GPU VDDQ
0 0 1.5 V
1 0 1.35 V

GPU/PCH/TBT 1V05 SUPPLY

VOUT = 1.05V

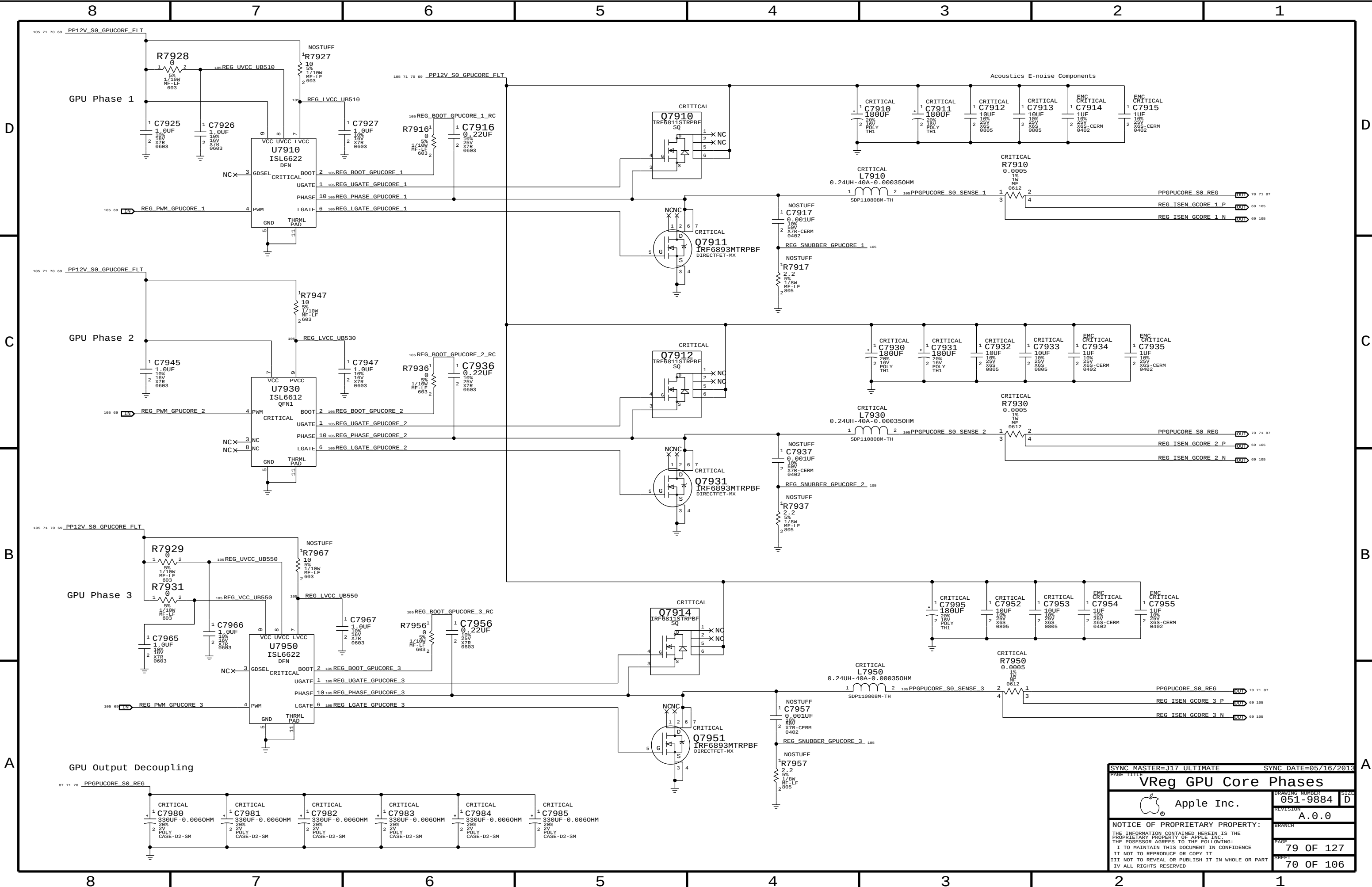
F = 500 KHZ

SYNC MASTER=317 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS	
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		REVISION	A.0.0
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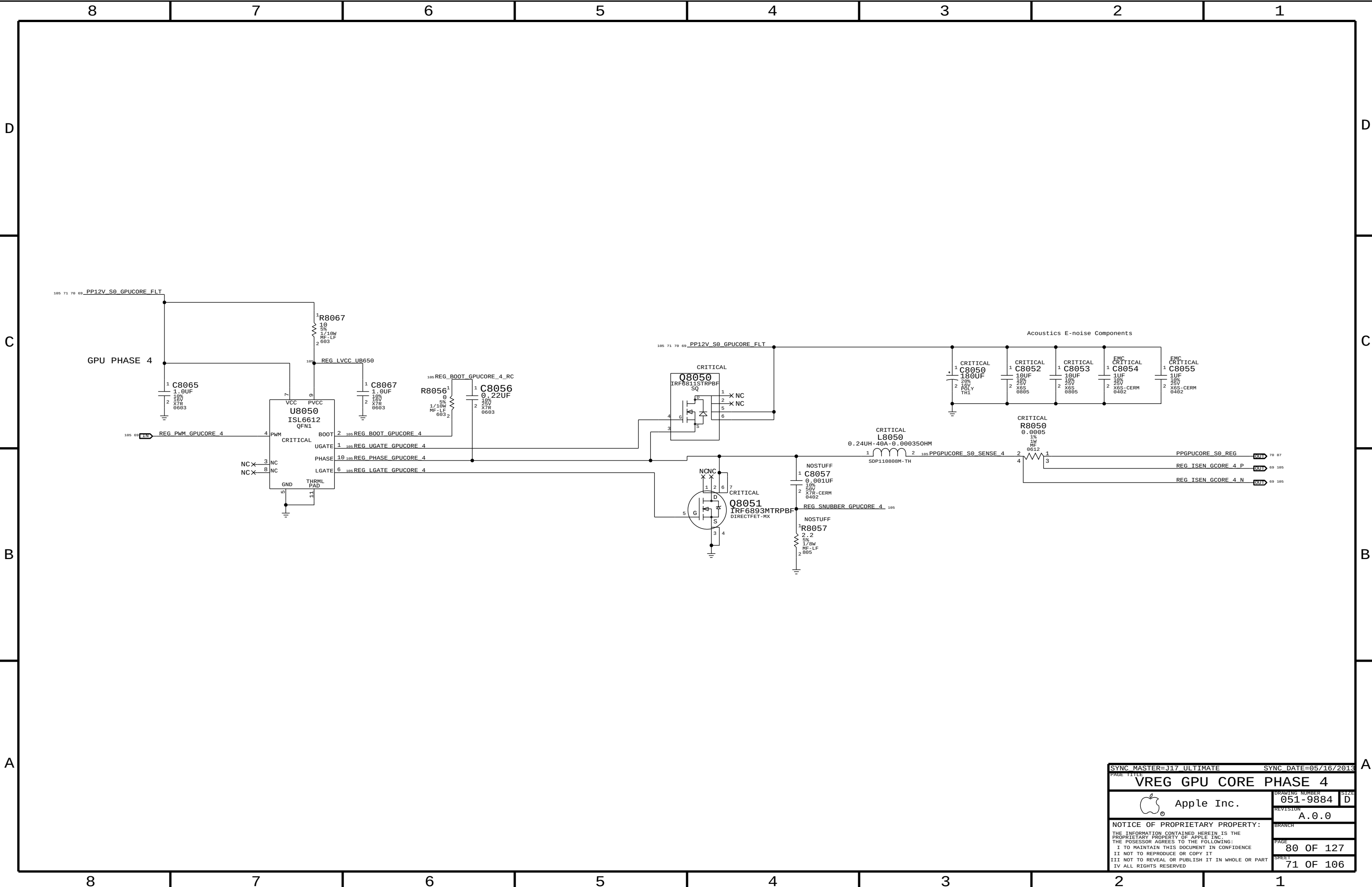


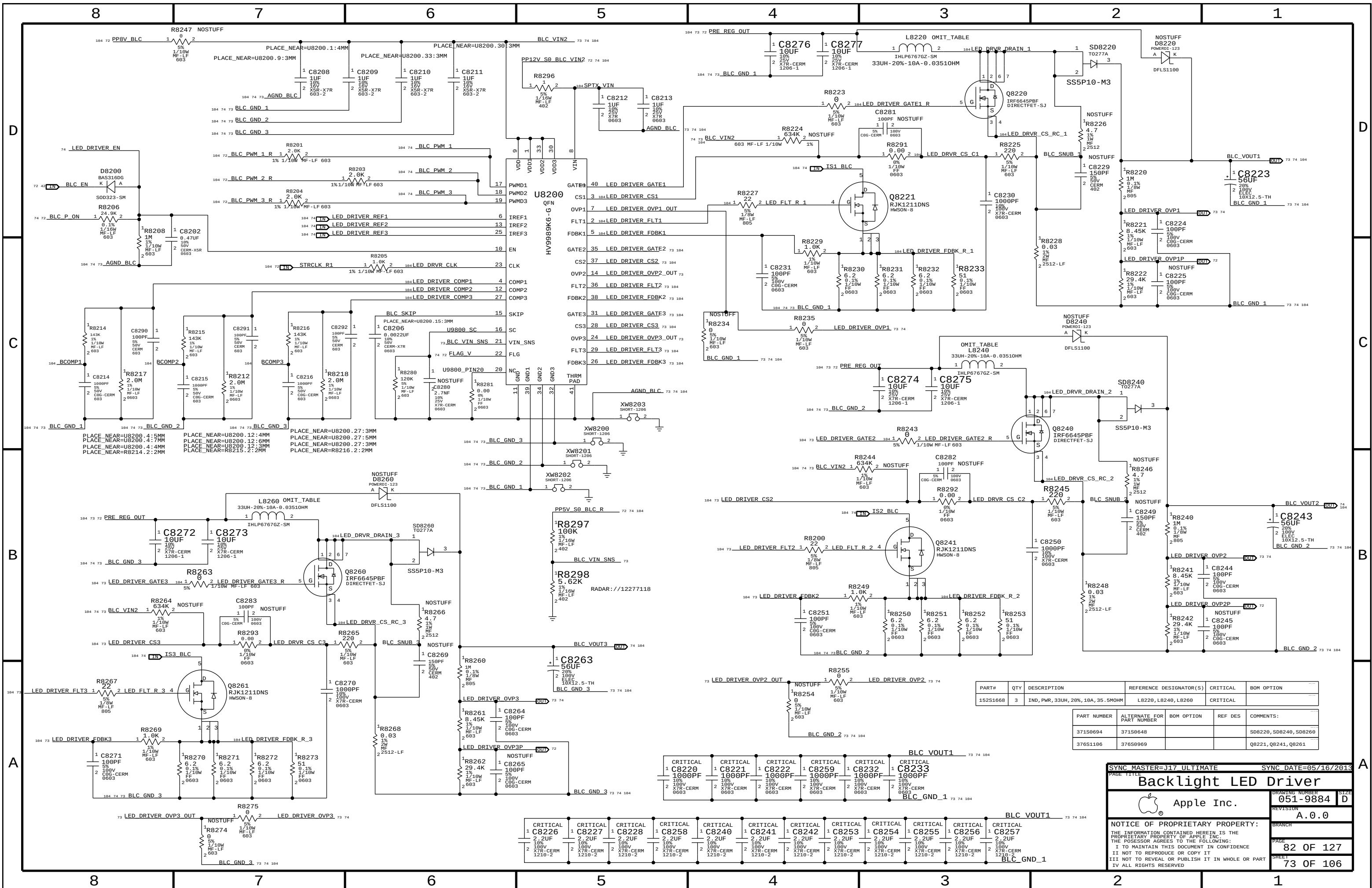
GPU VCORE
VOUT = VCORE
PEAK = ???A
AVG = ???A

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PAGE TITLE		VReg GPU Core Phases	
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	3	IND, PWR, 33UH, 20%, 10A, 35.5MOHM	L8220, L8240, L8260	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0694	371S0648			SD8220, SD8240, SD8260
376S1106	376S0969			Q8221, Q8241, Q8261

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Backlight LED Driver

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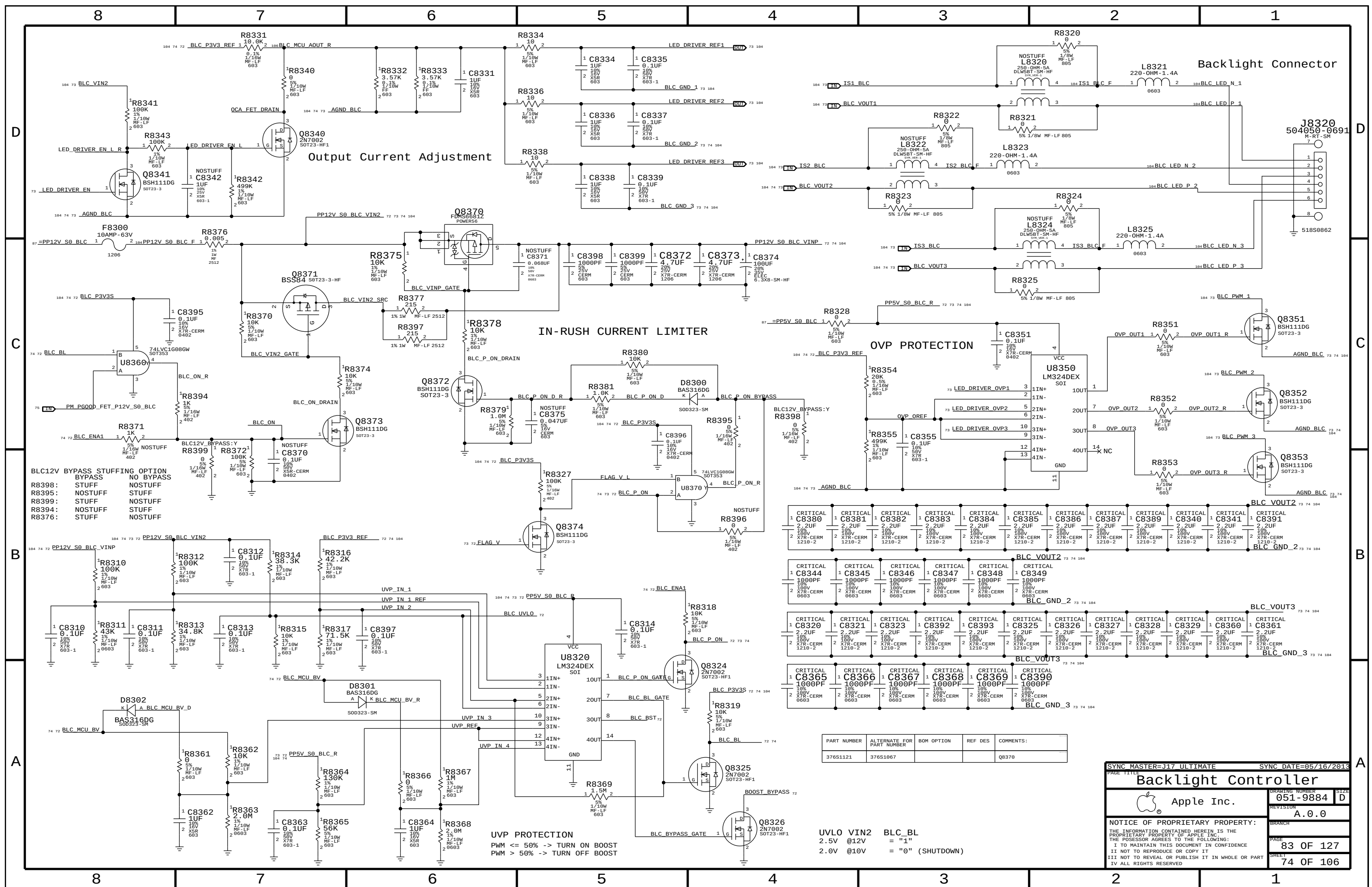
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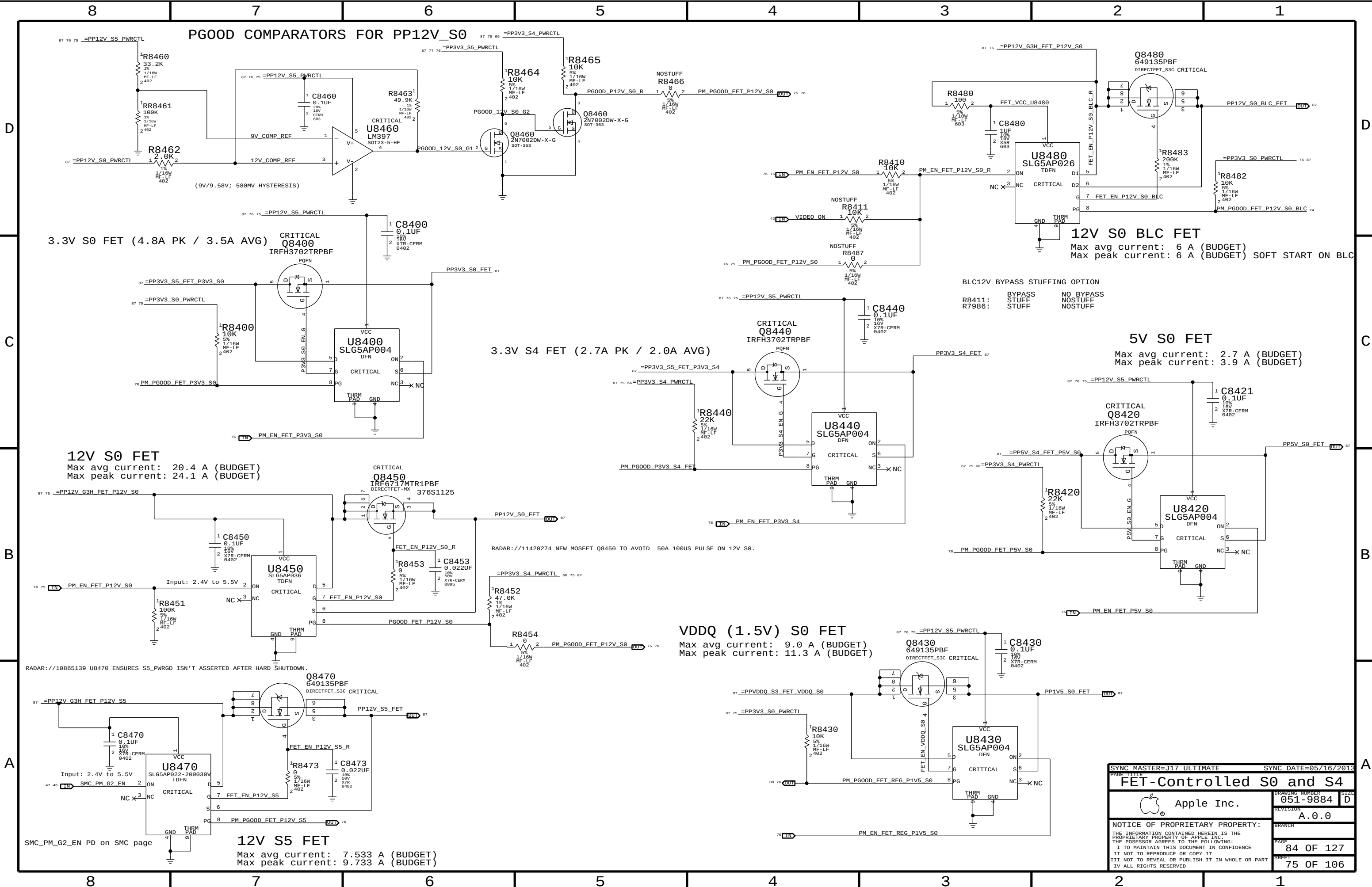
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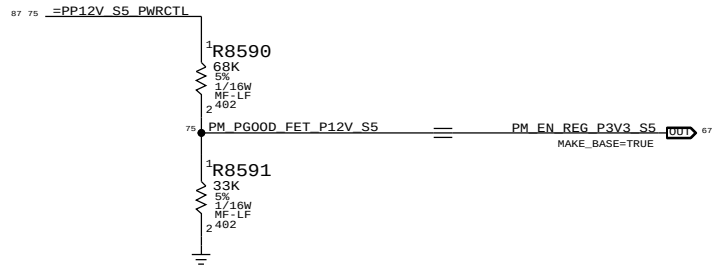
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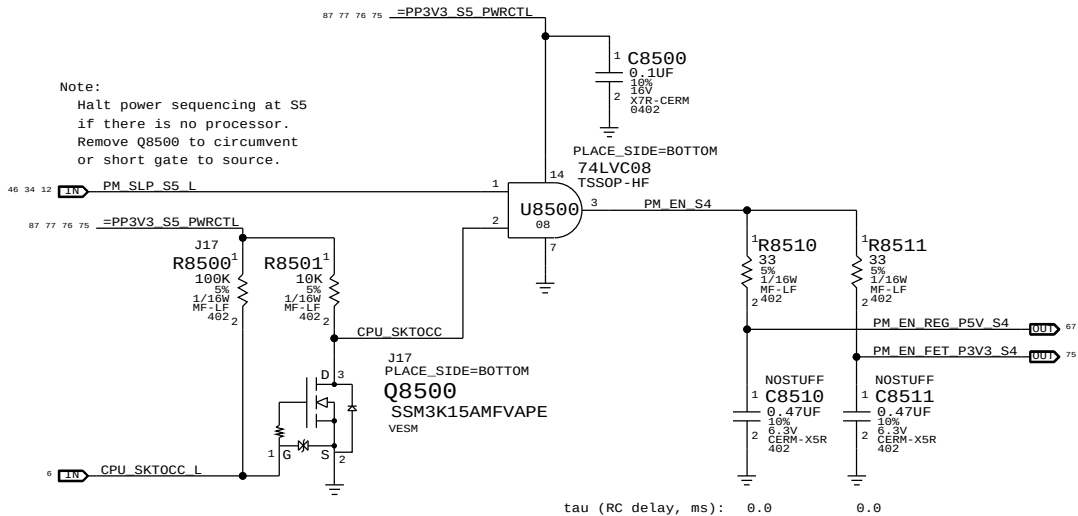




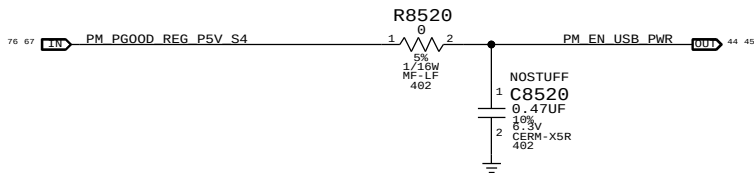
S5 Enable



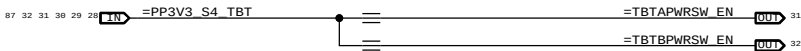
S4 Enables



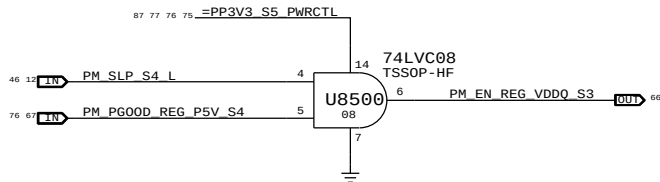
S4 USB Enable



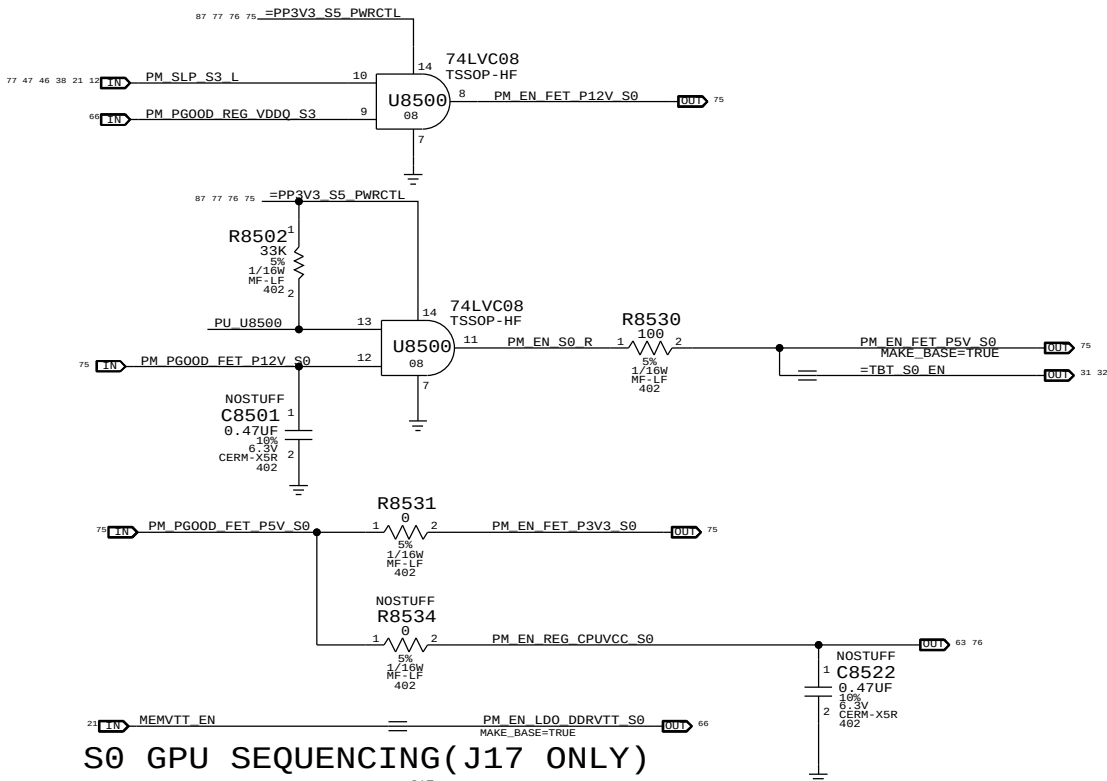
S4 TBT S4 Port Enable



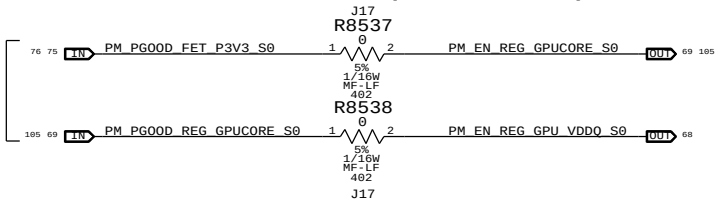
S3 VDDQ Enable



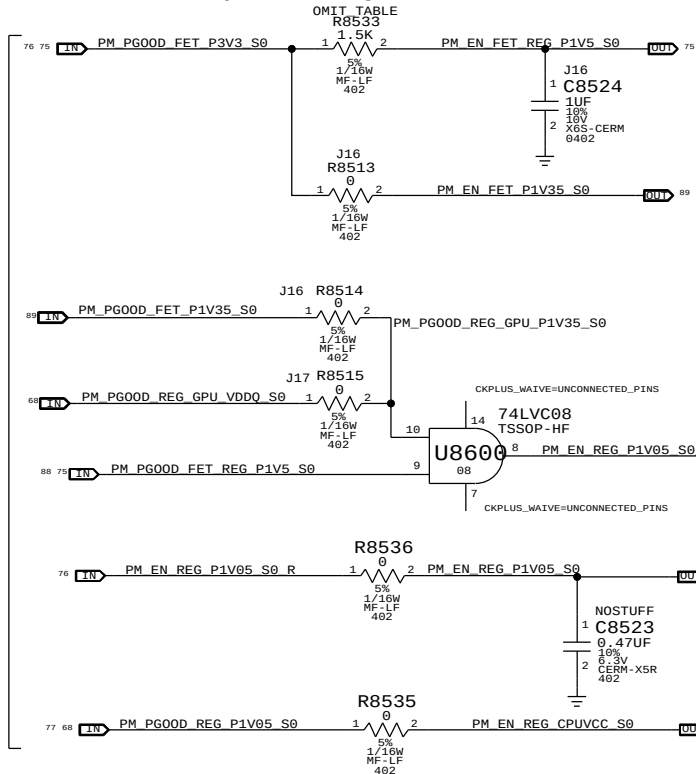
S0 Enables



S0 GPU SEQUENCING(J17 ONLY)



S0 PCH Sequencing



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0070	1	RES,1.5K,0402,5%	R8533	J16
116S0094	1	RES,00HM,0402,5%	R8533	J17

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.5 V, 1.05V for PCH/TBT/GPU)
Uncore: VDDQ

Notes on sequencing requirements

Intel:

1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

NVIDIA:

1. 3V3_S0 must ramp first
2. VDDQ MUST RAMP AFTER GPU_OP-HF
3. PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
4. All rails must reach their target voltages in more than 40 uS

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Page Notes

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- +FP2v2_GFU_V0012

Signal aliases required by this page:
(none)

DOM options provided by this page

(none)

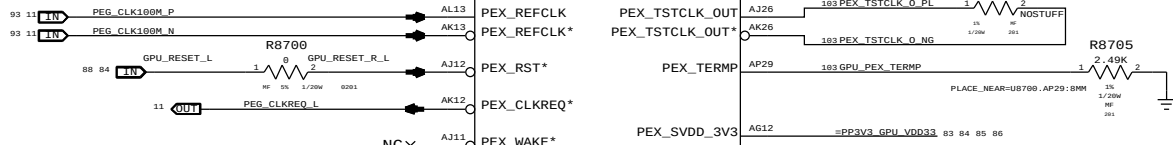
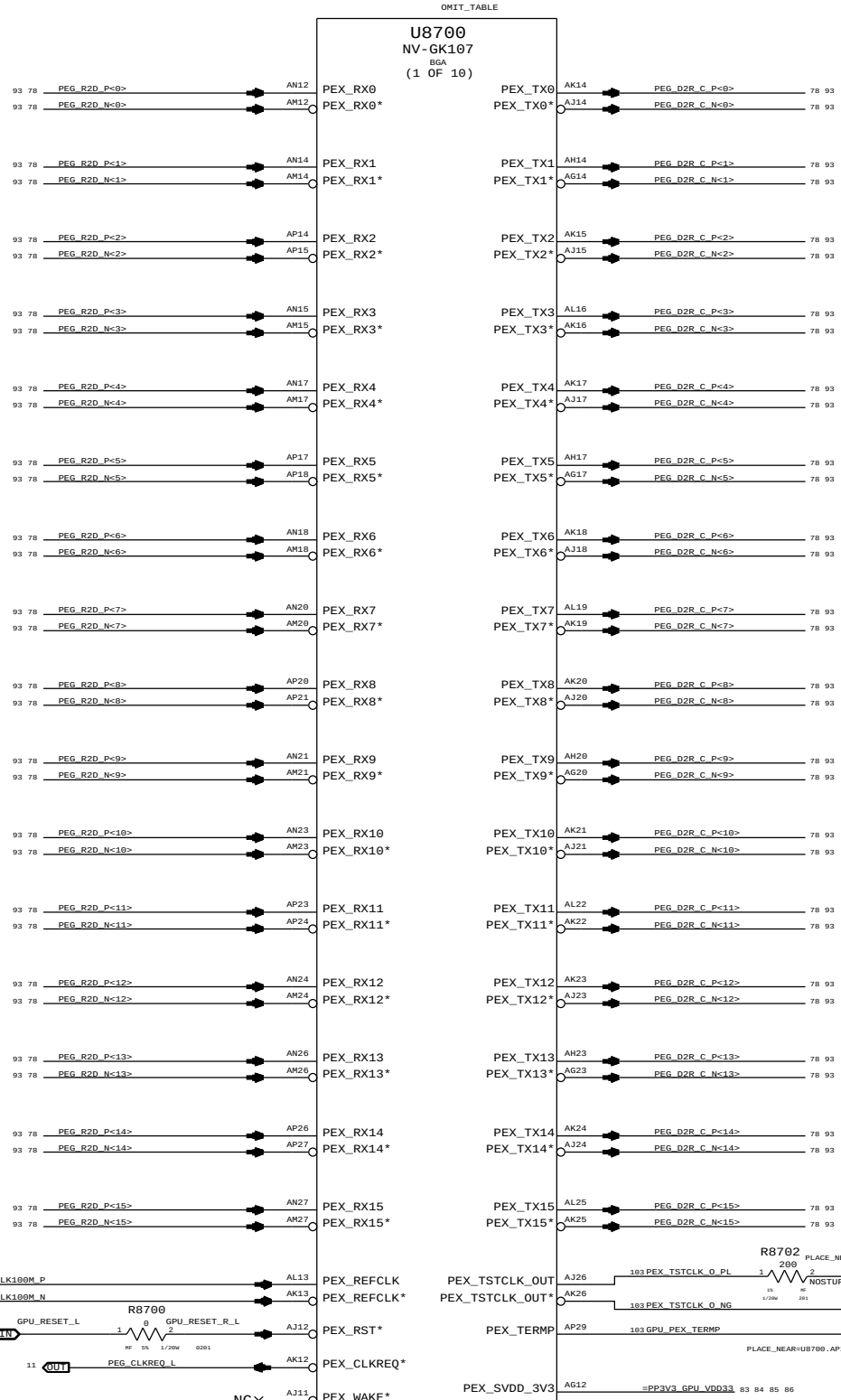
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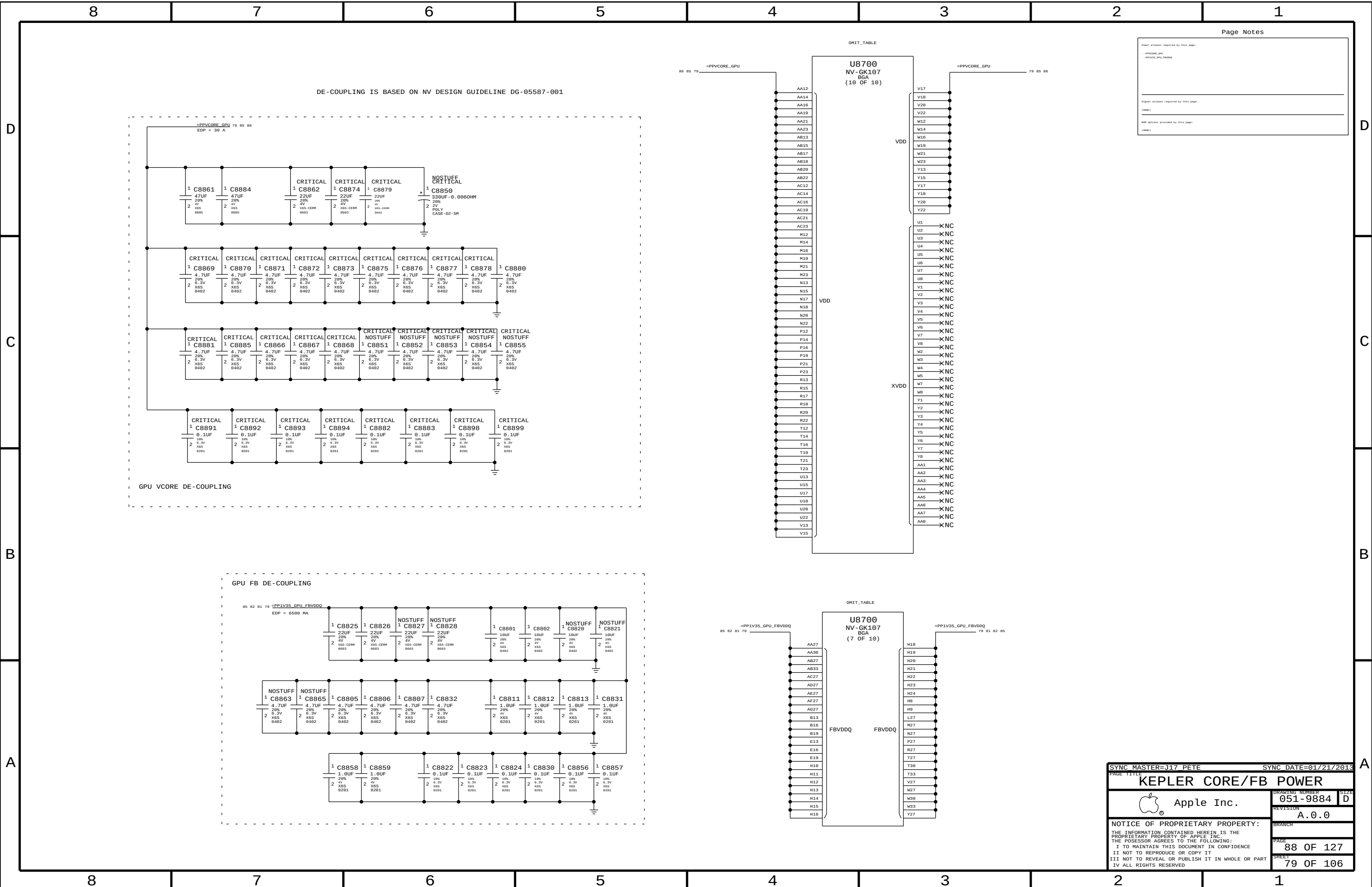
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[illegible]

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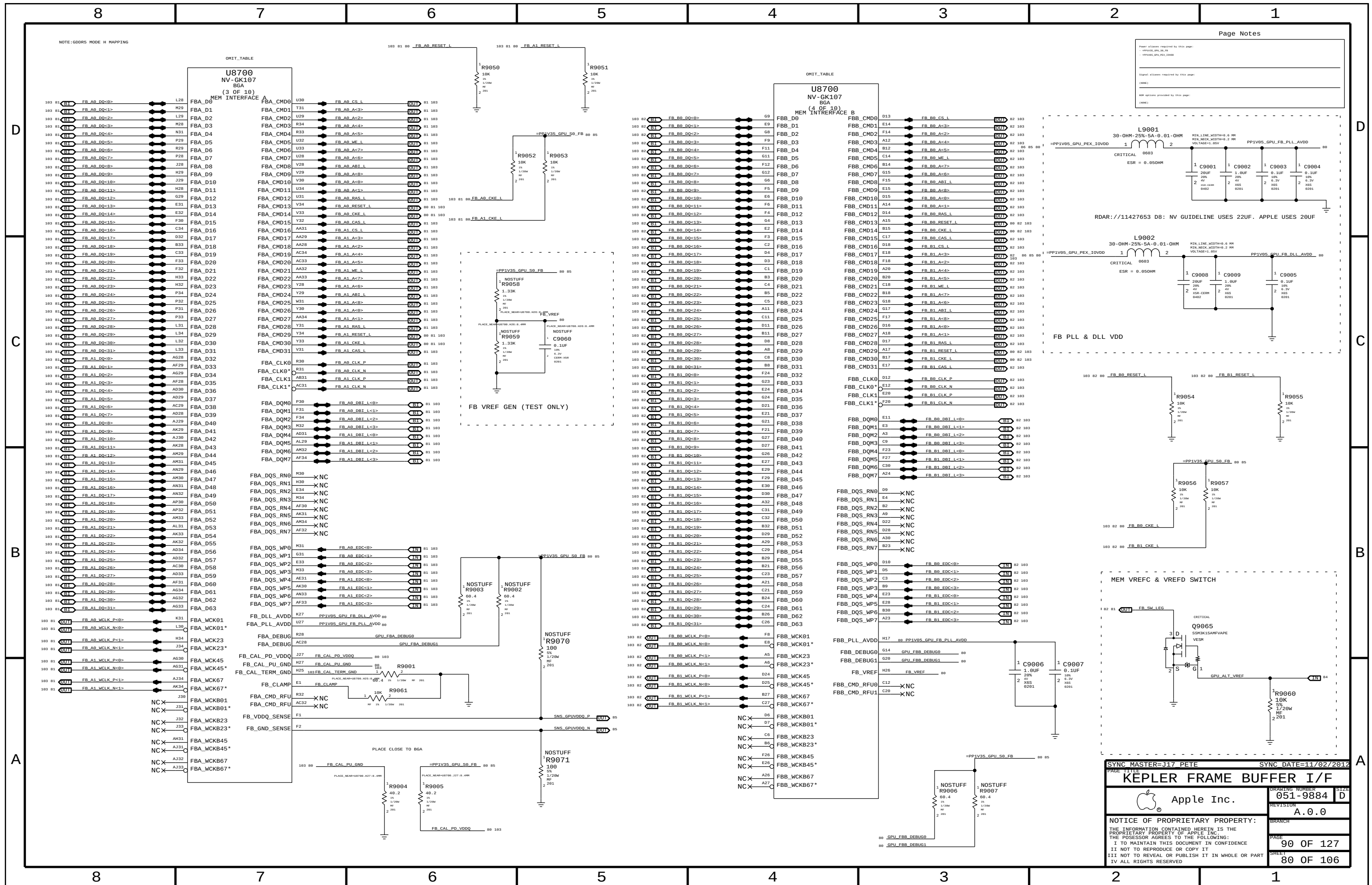
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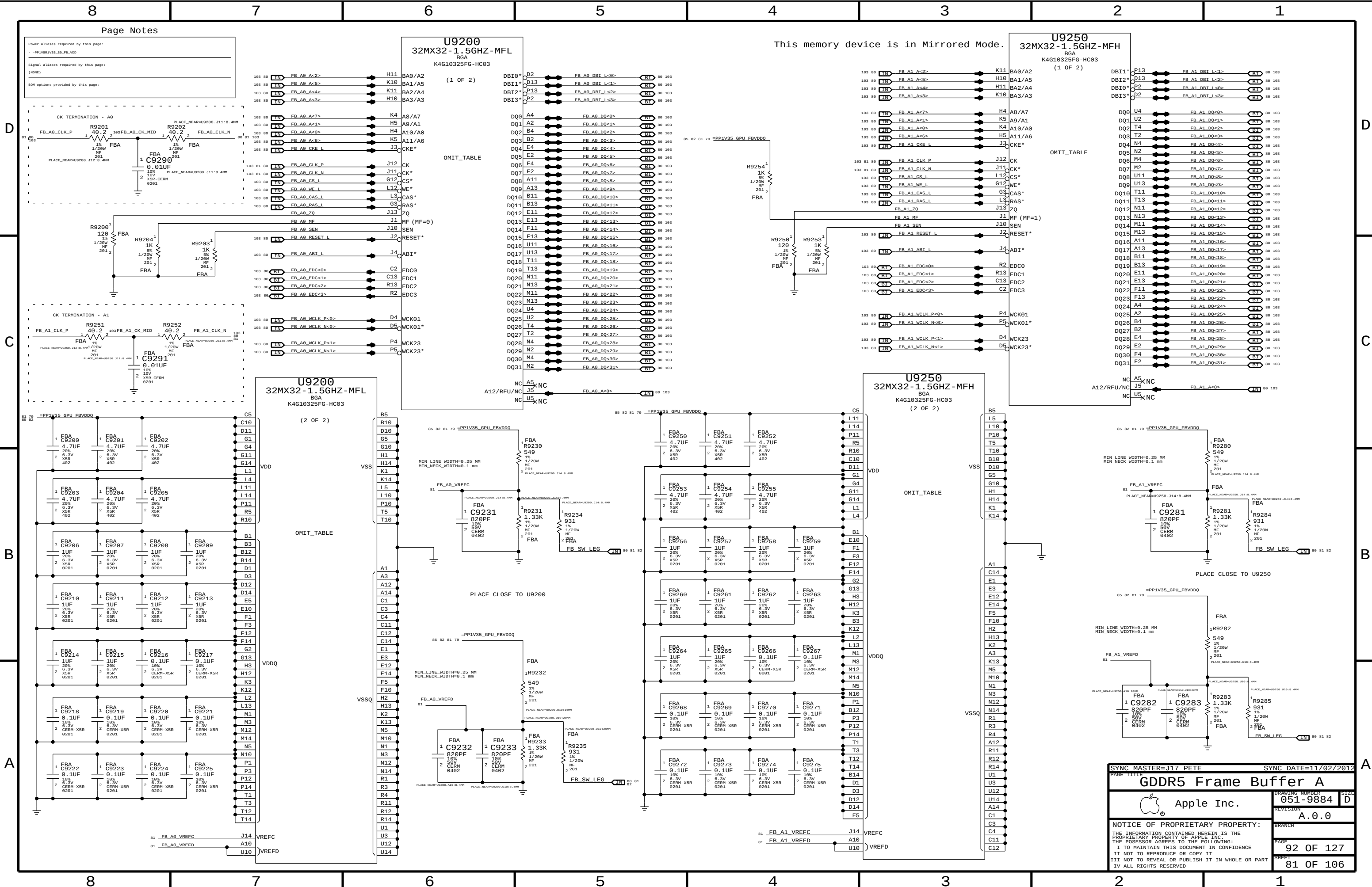
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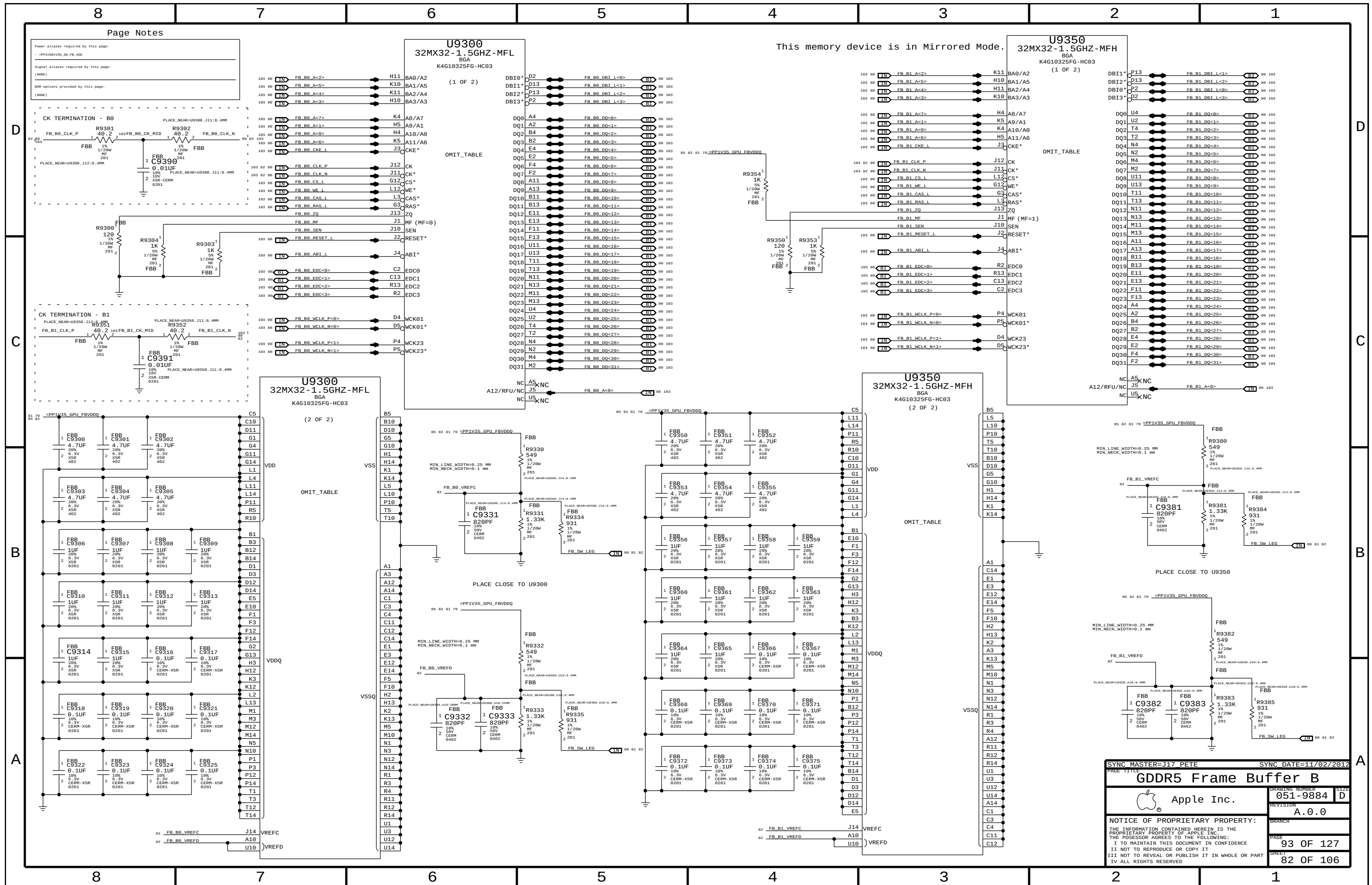
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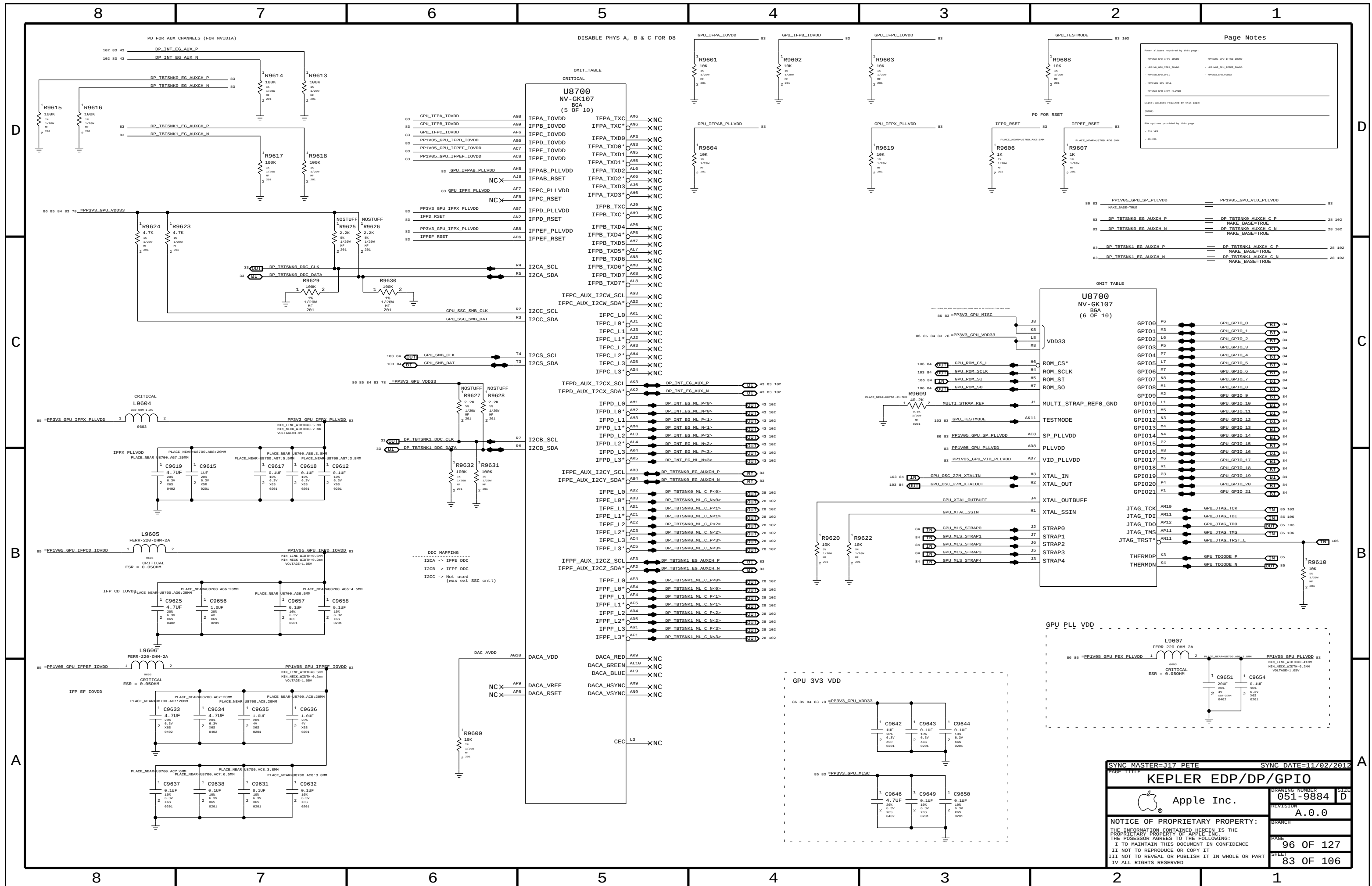
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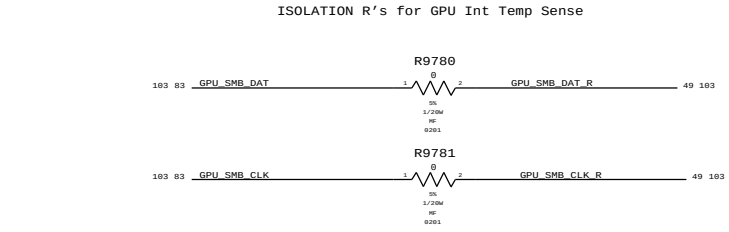
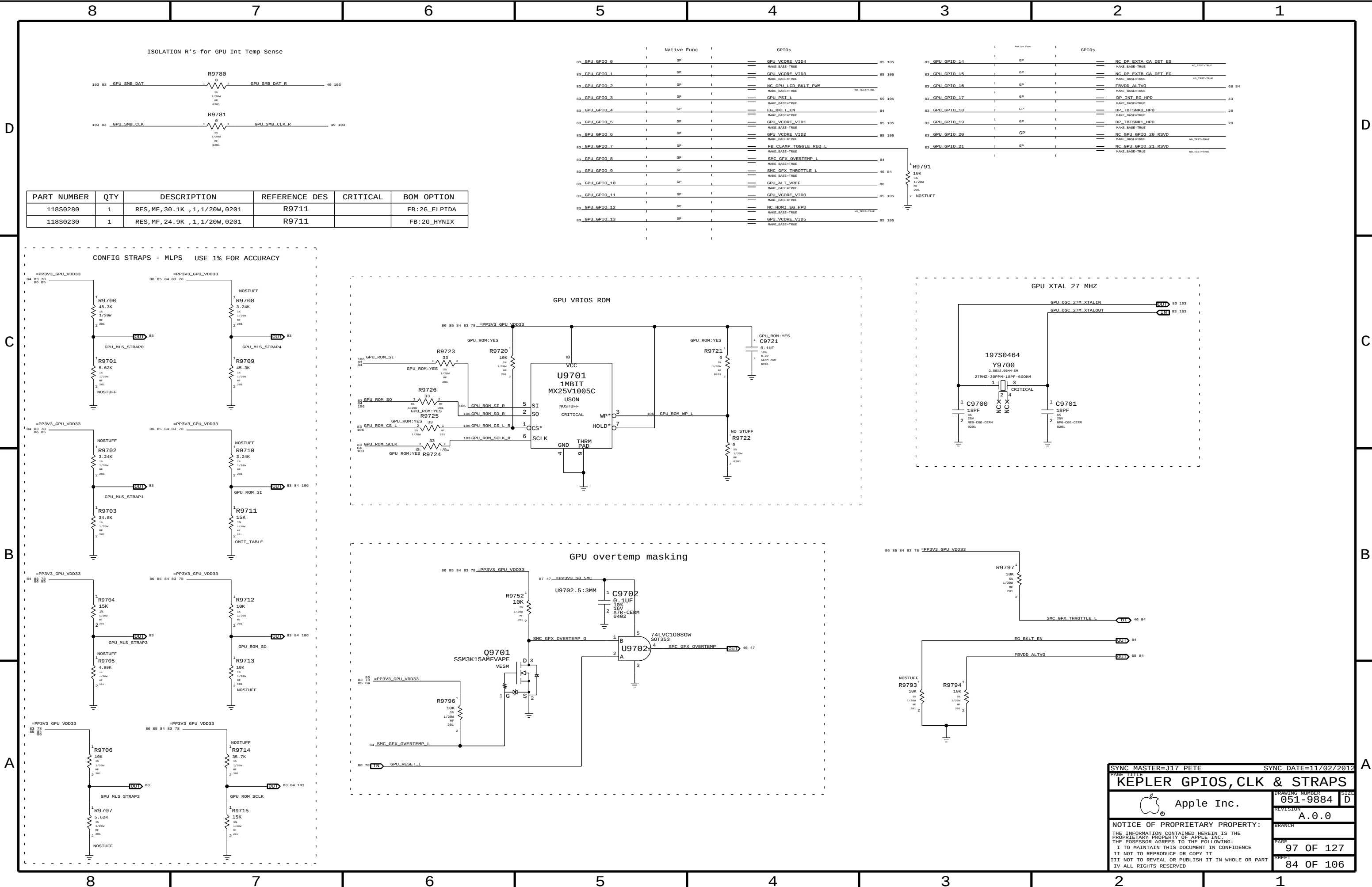




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		REVISION	A.0.0
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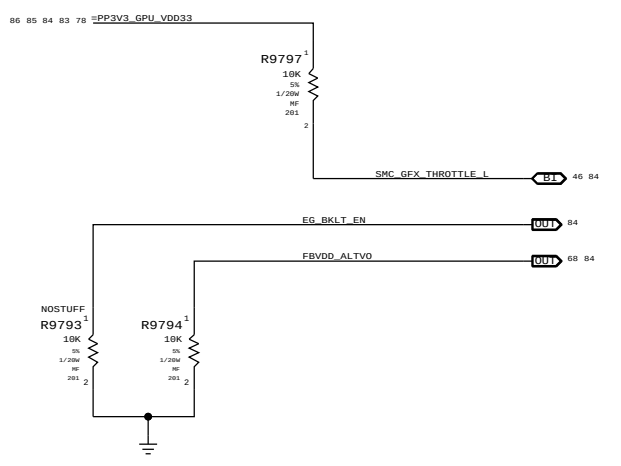
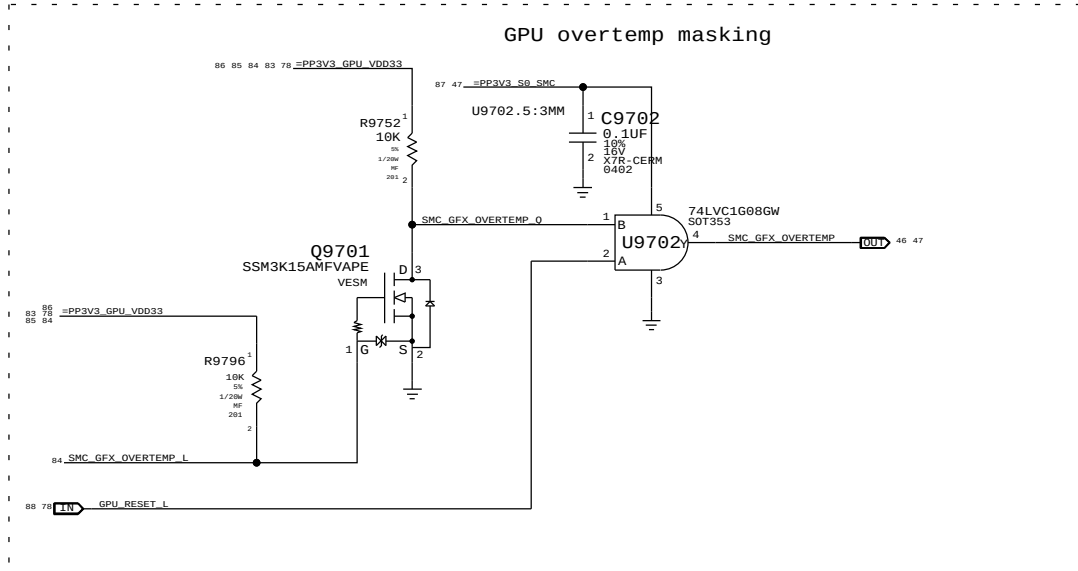
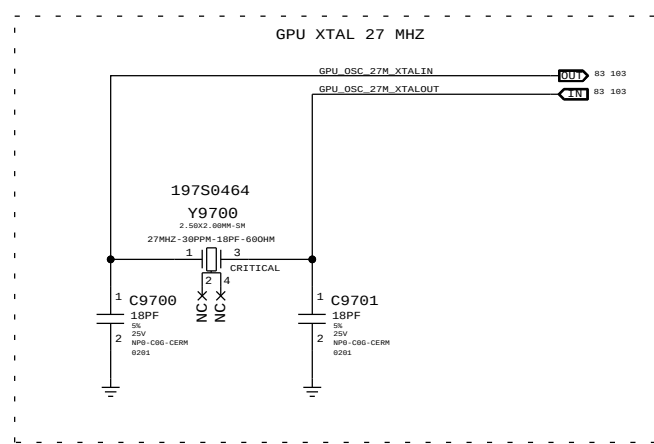
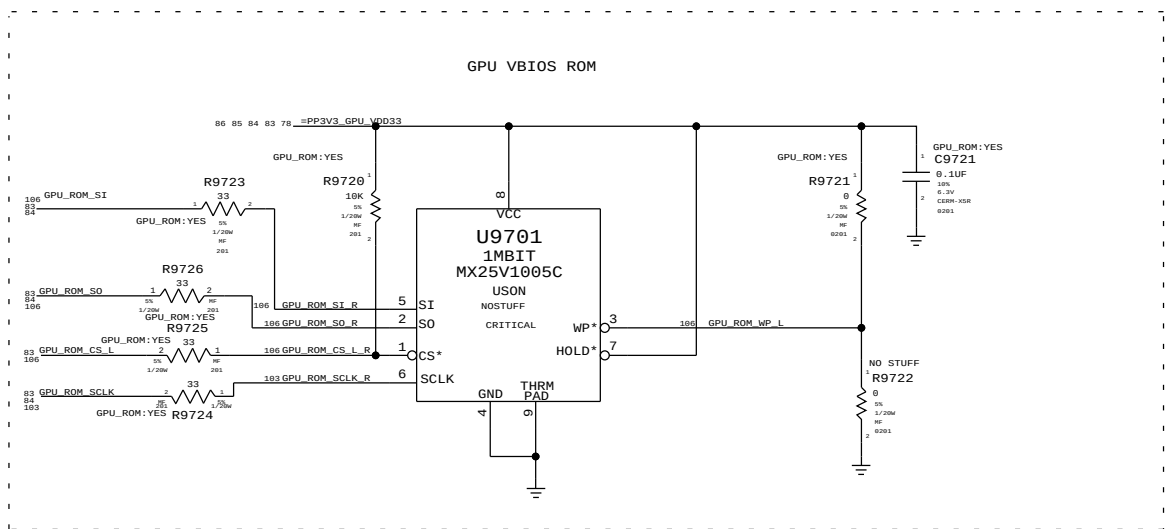
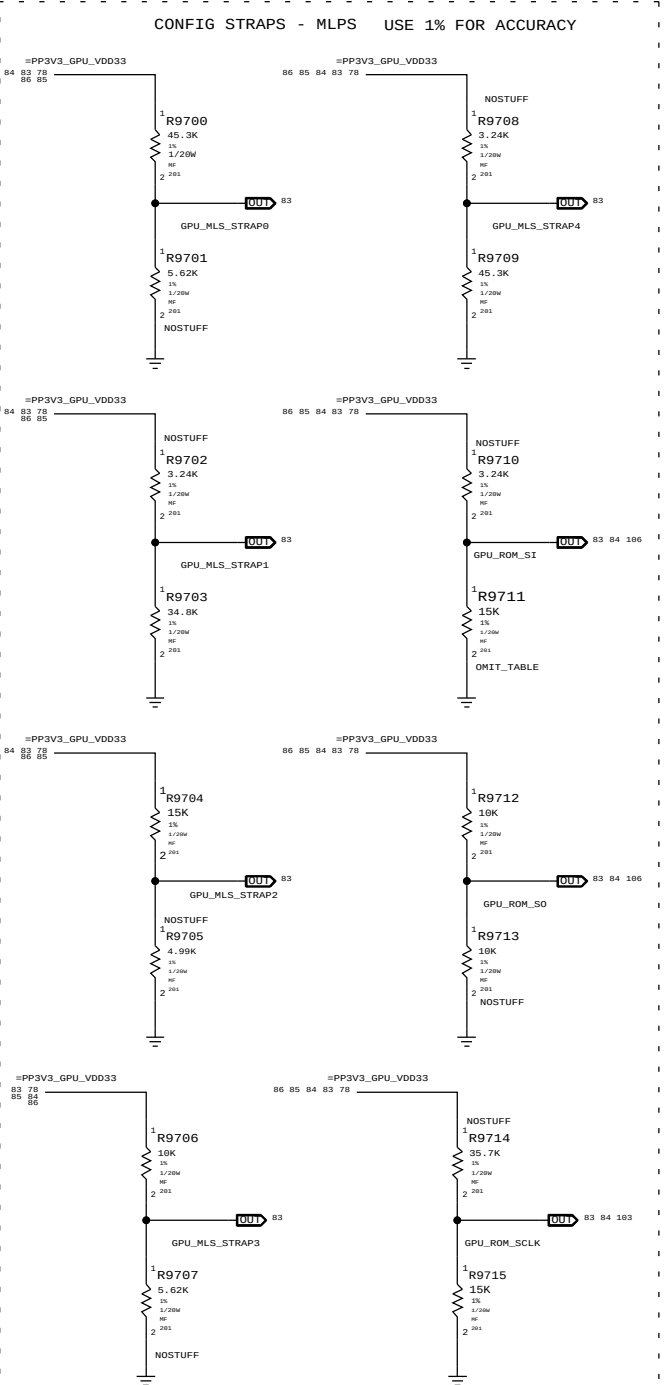




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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118S0230	1	RES,MF,24.9K ,1,1/20W,0201	R9711		FB:2G_HYNIX

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83_GPU_GPIO_2	GP		GPU_VCORE_VID3	85 105
83_GPU_GPIO_3	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_4	GP		NC_GPU_LCD_BKLT_PWM	NO_TEST+TRUE
83_GPU_GPIO_5	GP		GPU_PSI_L	69 105
83_GPU_GPIO_6	GP		EG_BKLT_EN	84
83_GPU_GPIO_7	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_8	GP		GPU_VCORE_VID1	85 105
83_GPU_GPIO_9	GP		GPU_VCORE_VID2	85 105
83_GPU_GPIO_10	GP		FB_CLAMP_TOGGLE_REQ_L	
83_GPU_GPIO_11	GP		SMC_GFX_OVERTEMP_L	84
83_GPU_GPIO_12	GP		SMC_GFX_THROTTLE_L	46 84
83_GPU_GPIO_13	GP		GPU_ALT_VREF	80
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			GPU_VCORE_VID0	85 105
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			GPU_VCORE_VID5	85 105
			MAKE_BASE=TRUE	

	Native Func		GPIOs	
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SYNC DATE=11/02/2012

KEPLER GPIOs,CLK & STRAPS

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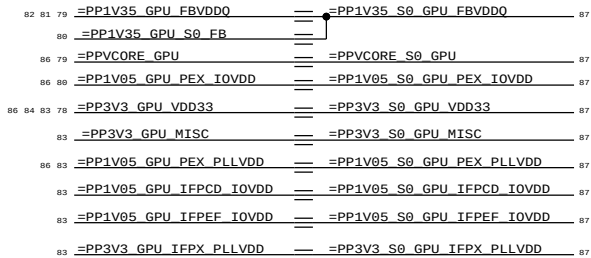
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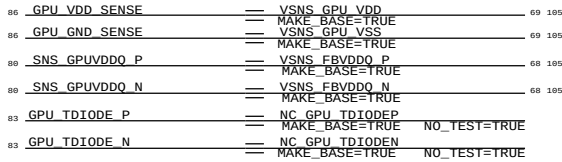
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D

GPU POWER ALIAS

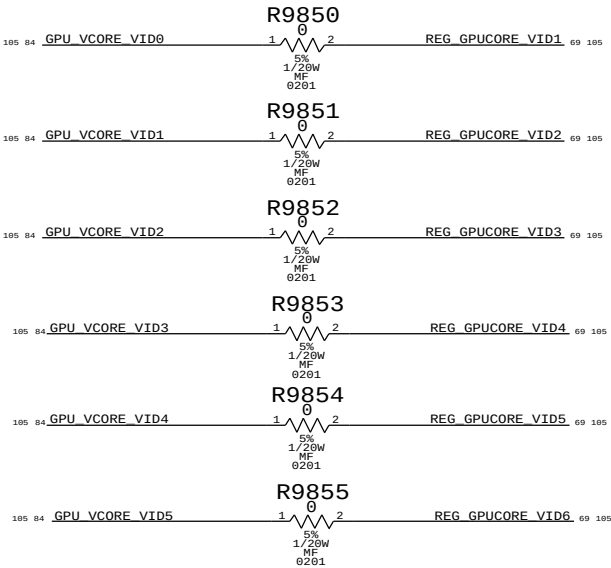


GPU SIGNAL & SENSE ALIAS



C

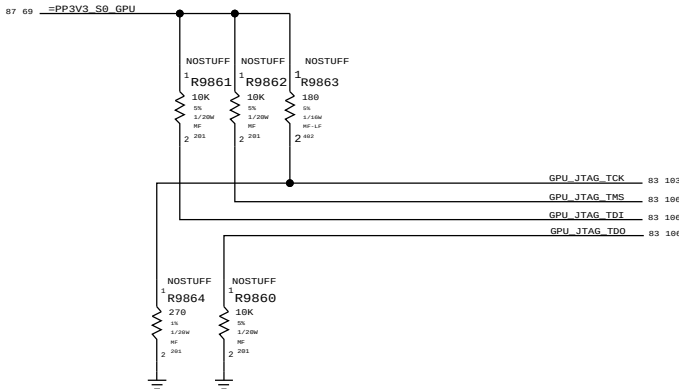
GPU VIDS ALIAS (VR VID0 IS TIED LOW)



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PU/PD IS BASED ON RECOMMENDATION FROM NVIDIA FOR THEIR GPU JTAG DEBUGGER

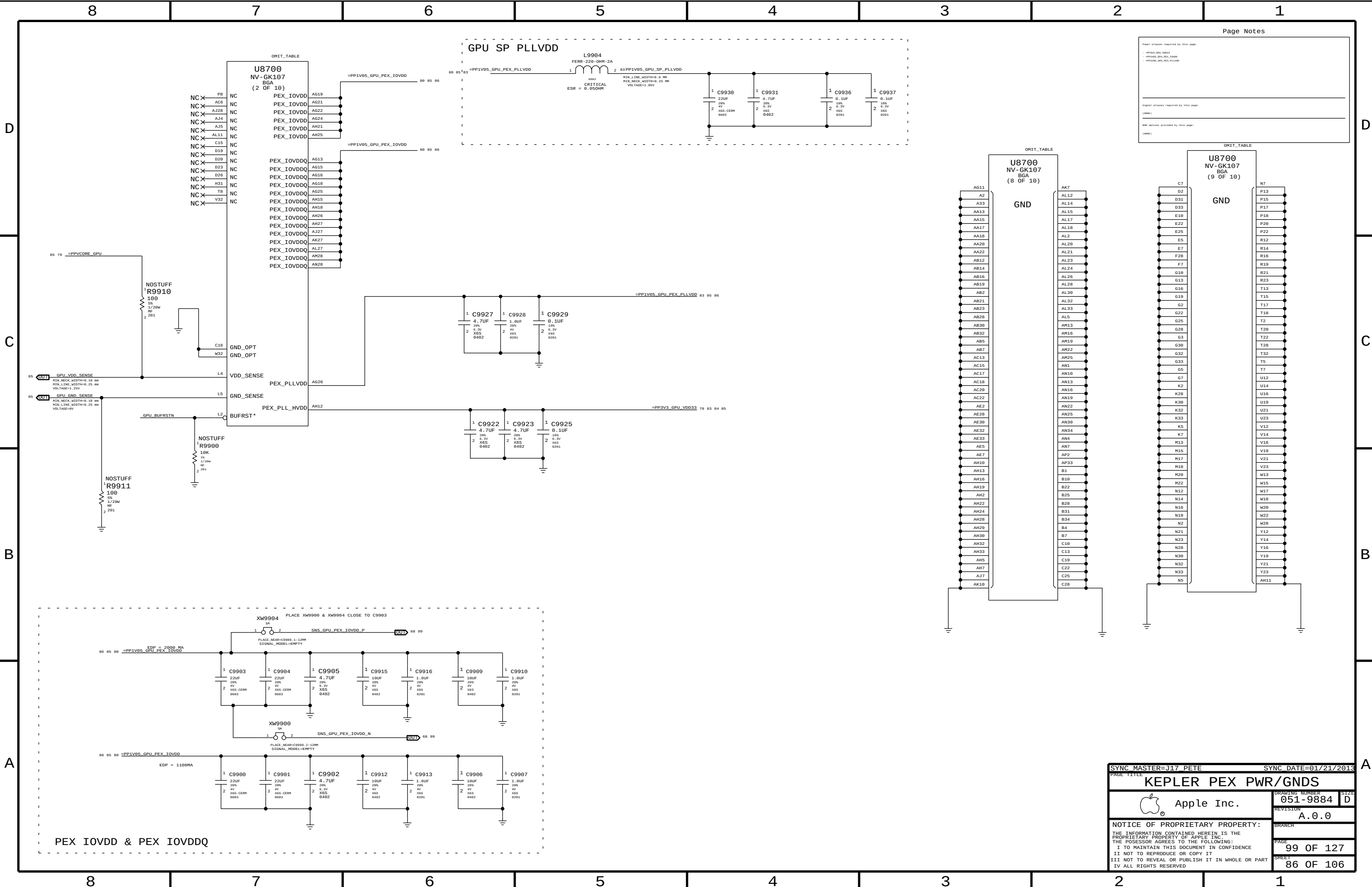


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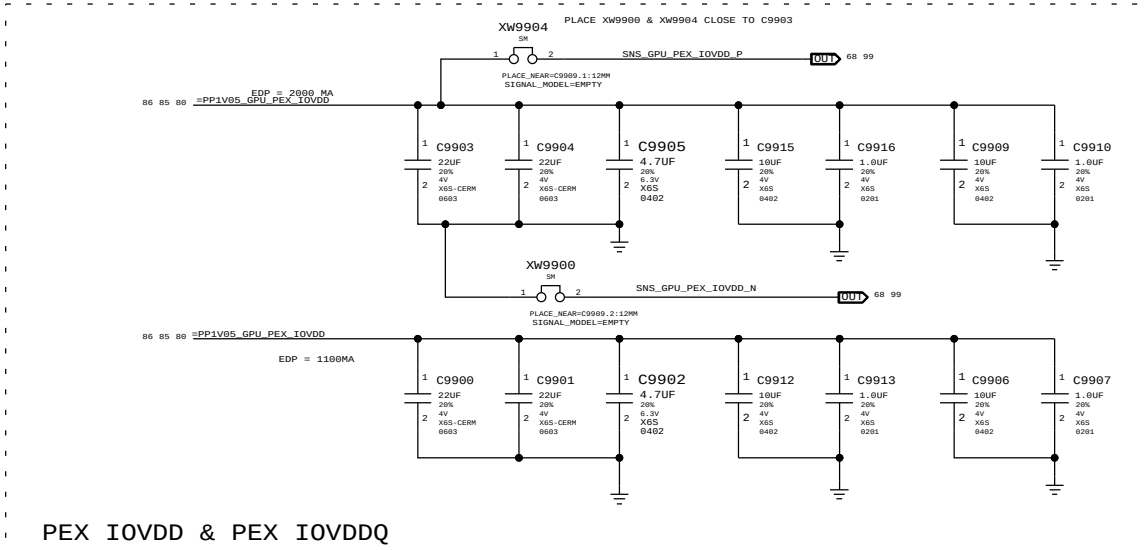
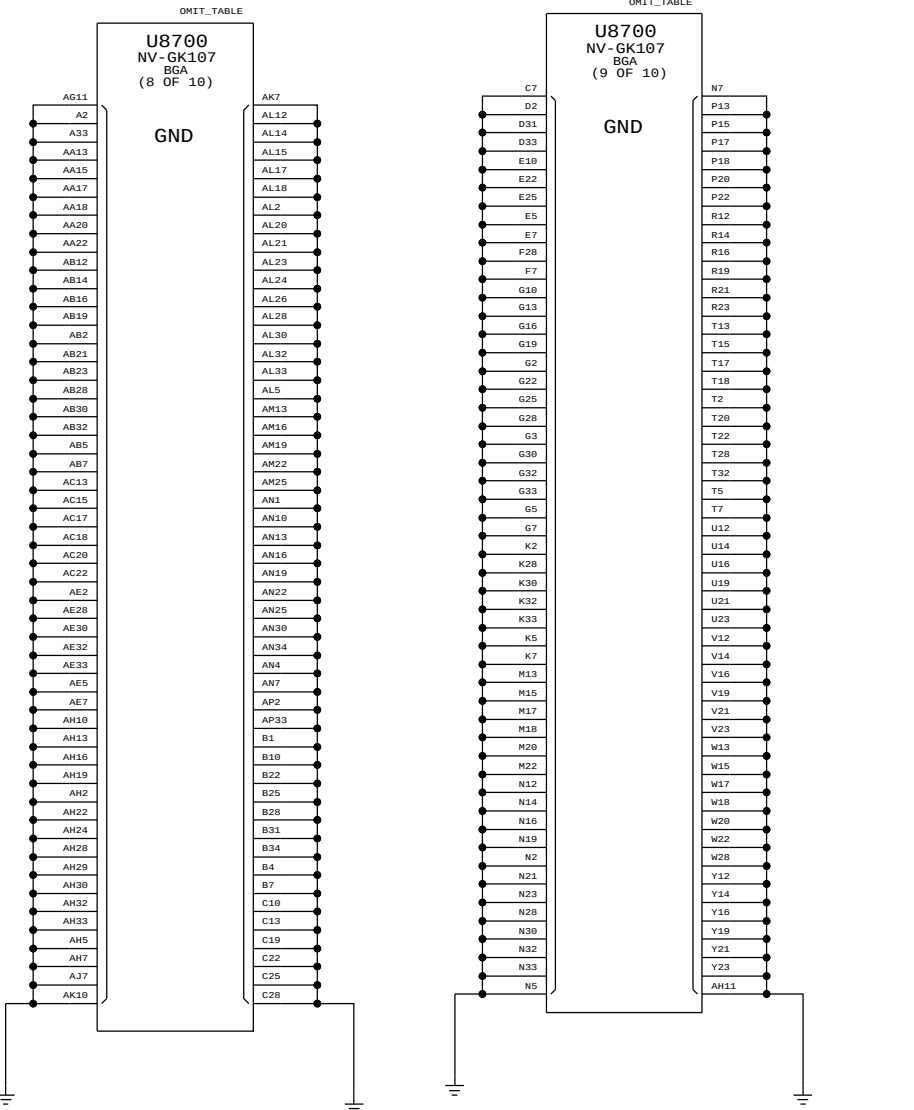
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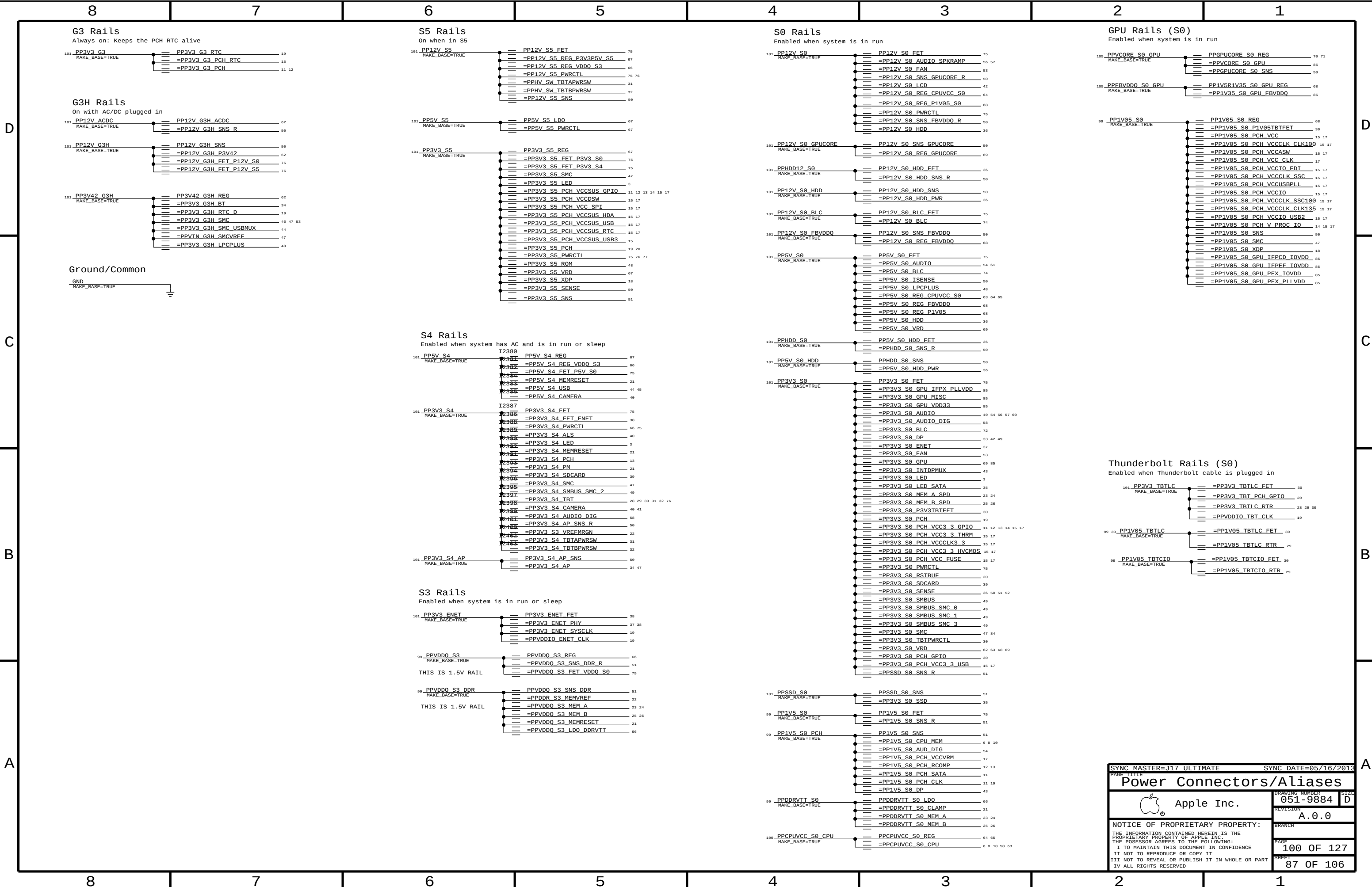
SYNC MASTER=J17 PETE		SYNC DATE=01/21/2013	
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Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_VDDQ1	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_PLLVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



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Power Connectors/Aliases

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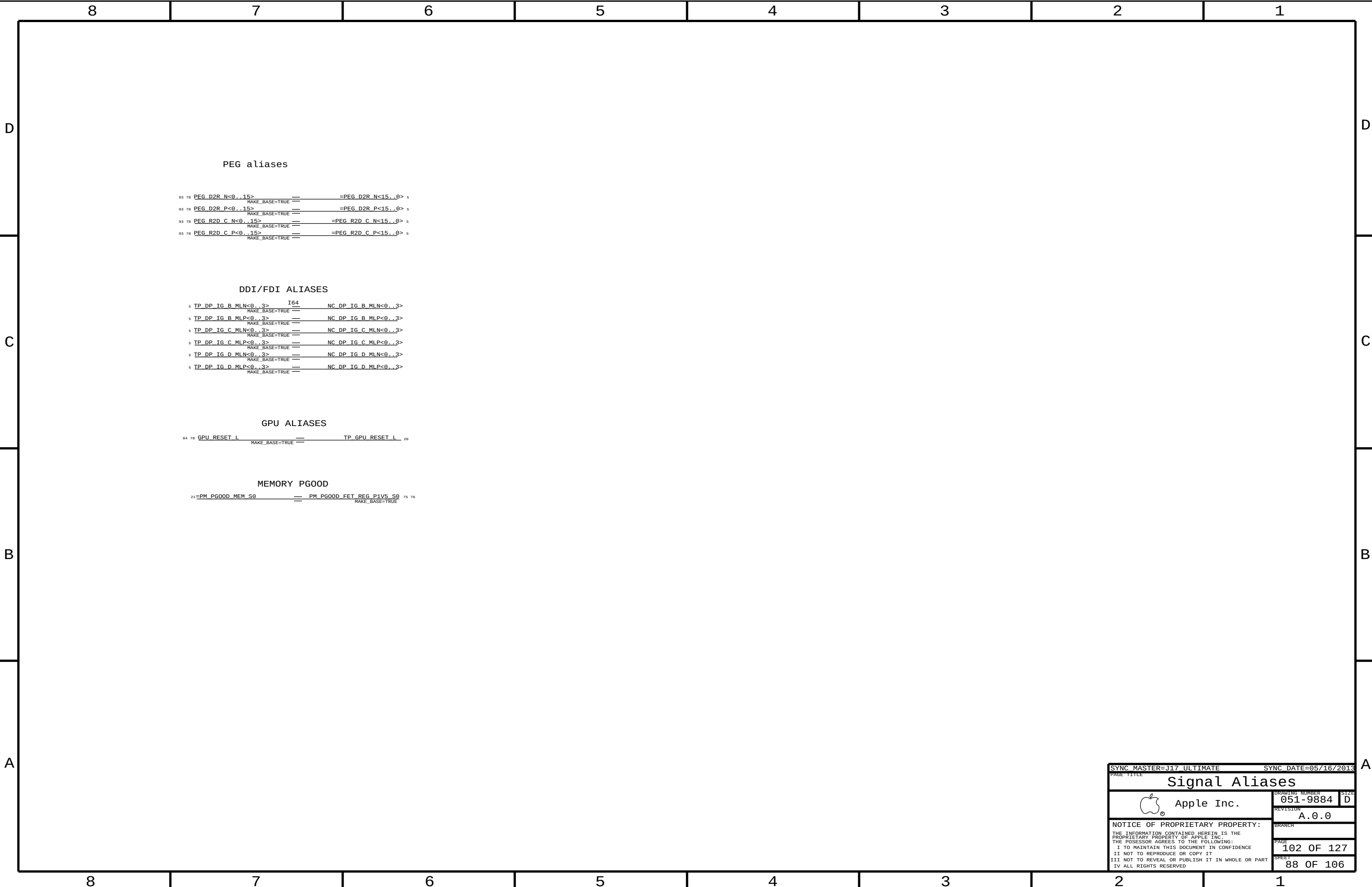
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PEG aliases

93 78 PEG_D2R_N<0..15> == =PEG_D2R_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_D2R_P<0..15> == =PEG_D2R_P<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_N<0..15> == =PEG_R2D_C_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_P<0..15> == =PEG_R2D_C_P<15..0> 5
MAKE_BASE=TRUE ==

DDI/FDI ALIASES

5 TP_DP_IG_B_MLN<0..3> I64 == NC_DP_IG_B_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_B_MLP<0..3> == NC_DP_IG_B_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLN<0..3> == NC_DP_IG_C_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLP<0..3> == NC_DP_IG_C_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLN<0..3> == NC_DP_IG_D_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLP<0..3> == NC_DP_IG_D_MLP<0..3>
MAKE_BASE=TRUE ==

GPU ALIASES

84 78 GPU_RESET_L == TP_GPU_RESET_L 20
MAKE_BASE=TRUE ==

MEMORY PG00D

21 ==PM_PG00D_MEM_S0 == PM_PG00D_FET_REG_P1V5_S0 75 76
MAKE_BASE=TRUE ==

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Signal Aliases

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C							C
B							B
A							A
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Functional / ICT Test

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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_ODR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design	Iso	Design	Comments
4-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

See Note (3)

See Note (1)

See Note (3)

See Note (1)

See Note (2)

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggested 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. D0R3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints $DDR_*_DQ_BYTE*$ to =SAME to win out over $DDR\{A,B\}_DQ_BYTE*$ to $DDR\{A,B\}_DQ_BYTE*$ so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL .

DDR3

Electrical Constraint Set		Physical	Spacing		
Channel A					
H070	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0>	7 23
H071	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0>	7 23
H072	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2>	7 24
H073	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2>	7 24
H100	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0>	7 23
H101	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0>	7 23
H102	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0>	7 23
H103	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2>	7 24
H104	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2>	7 24
H105	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2>	7 24
H110	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0>	7 23 24
H111	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0>	7 23 24
H112	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L	7 23 24
H113	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L	7 23 24
H114	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L	7 23 24
H120	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0>	7 27
H121	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8>	7 27
H122	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16>	7 27
H123	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24>	7 27
H124	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32>	7 27
H125	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40>	7 27
H126	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48>	7 27
H127	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56>	7 27
H130	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0>	7 27
H131	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0>	7 27
H132	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1>	7 27
H133	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1>	7 27
H134	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2>	7 27
H135	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2>	7 27
H136	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3>	7 27
H137	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3>	7 27
H138	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4>	7 27
H139	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4>	7 27
H140	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5>	7 27
H141	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5>	7 27
H142	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6>	7 27
H143	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6>	7 27
H144	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7>	7 27
H145	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7>	7 27
Channel B					
H200	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0>	7 25
H201	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0>	7 25
H202	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2>	7 26
H203	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2>	7 26
H210	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0>	7 25
H211	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0>	7 25
H212	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0>	7 25
H213	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2>	7 26
H214	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2>	7 26
H215	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2>	7 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0>	7 25 26
H217	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0>	7 25 26
H218	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L	7 25 26
H219	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L	7 25 26
H220	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L	7 25 26
H221	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0>	7 27
H222	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8>	7 27
H223	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16>	7 27
H224	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24>	7 27
H225	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32>	7 27
H226	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40>	7 27
H227	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48>	7 27
H228	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56>	7 27
H230	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0>	7 27
H231	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0>	7 27
H232	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1>	7 27
H233	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1>	7 27
H234	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2>	7 27
H235	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2>	7 27
H236	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3>	7 27
H237	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3>	7 27
H238	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4>	7 27
H239	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4>	7 27
H240	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5>	7 27
H241	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5>	7 27
H242	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6>	7 27
H243	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6>	7 27
H244	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7>	7 27
H245	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7>	7 27
Reset					
H246		DDR_50S		MEM_RESET_L	21 23 24
SM COMP					
H049		DDR_COMP_PHY	DDR_COMP	CPU_SM_RCOMP<0..2>	6

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DDR3 Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALL ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing
x16 Graphics		
E6E0 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D
E6E1 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D P<15>
E6E2 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D N<15>
E6E3 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C P<15>
E6E4 PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_R2D C N<15>
E6E5 PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<15>
E6E6 PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<15>
E6E7 PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<15>
E6E8 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_D2R C N<15>
E6E9 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<14>
E6EA PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<14>
E6EB PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<14>
E6EC PCIE GEN3 D2R	PCIE3_PHY	PEG_R2D C N<14>
E6ED PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<14>
E6EE PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<14>
E6EF PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<14>
E6F0 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_D2R C N<14>
E6F1 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<13>
E6F2 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<13>
E6F3 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<13>
E6F4 PCIE GEN3 D2R	PCIE3_PHY	PEG_R2D C N<13>
E6F5 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<13>
E6F6 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<13>
E6F7 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<13>
E6F8 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<13>
E6F9 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<12>
E6FA PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<12>
E6FB PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<12>
E6FC PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_R2D C N<12>
E6FD PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<12>
E6FE PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<12>
E6FF PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<12>
E6G0 PCIE GEN3 R2D	PCIE3_PHY	PEG_D2R C N<12>
E6G1 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D P<11>
E6G2 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D N<11>
E6G3 PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C P<11>
E6G4 PCIE GEN3 D2R	PCIE3_PHY	PEG_R2D C N<11>
E6G5 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<11>
E6G6 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<11>
E6G7 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<11>
E6G8 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<11>
E6G9 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<10>
E6GA PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<10>
E6GB PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<10>
E6GC PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_R2D C N<10>
E6GD PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<10>
E6GE PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<10>
E6GF PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<10>
E6H0 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_D2R C N<10>
E6H1 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<9>
E6H2 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<9>
E6H3 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<9>
E6H4 PCIE GEN3 D2R	PCIE3_PHY	PEG_R2D C N<9>
E6H5 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<9>
E6H6 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<9>
E6H7 PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<9>
E6H8 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<8>
E6H9 PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<8>
E6HA PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<8>
E6HB PCIE GEN3 D2R	PCIE3_PHY	PEG_R2D C N<8>
E6HC PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<8>
E6HD PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<8>
E6HE PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<8>
E6HF PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<8>

PCIe (CPU)

Electrical Constraint Set		Physical	Spacing	
x16 Graphics				
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D P<7> 78
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D N<7> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<7> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<7> 78 88
1E300	PCIe GEN3 D2R RVSD	PCIe3_PHY	PEG_D2R	PEG_D2R P<7> 78 88
1E300	PCIe GEN3 D2R RVSD	PCIe3_PHY	PEG_D2R	PEG_D2R N<7> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<7> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<7> 78 88
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D P<6> 78
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D N<6> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<6> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<6> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<6> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<6> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<6> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<6> 78 88
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D P<5> 78
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D N<5> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<5> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<5> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<5> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<5> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<5> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<5> 78 88
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D P<4> 78
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D N<4> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<4> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<4> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<4> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<4> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<4> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<4> 78 88
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D P<3> 78
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D N<3> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<3> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<3> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<3> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<3> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<3> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<3> 78 88
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D P<2> 78
1E300	PCIe GEN3 R2D	PCIe3_PHY	PEG_R2D	PEG_R2D N<2> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<2> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<2> 78 88
1E300	PCIe GEN3 D2R RVSD	PCIe3_PHY	PEG_D2R	PEG_D2R P<2> 78 88
1E300	PCIe GEN3 D2R RVSD	PCIe3_PHY	PEG_D2R	PEG_D2R N<2> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<2> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<2> 78 88
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D P<1> 78
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D N<1> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<1> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<1> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<1> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<1> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<1> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<1> 78 88
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D P<0> 78
1E300	PCIe GEN3 R2D RVSD	PCIe3_PHY	PEG_R2D	PEG_R2D N<0> 78
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C P<0> 78 88
1E300		PCIe3_PHY	PEG_R2D	PEG_R2D C N<0> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R P<0> 78 88
1E300	PCIe GEN3 D2R	PCIe3_PHY	PEG_D2R	PEG_D2R N<0> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C P<0> 78 88
1E300		PCIe3_PHY	PEG_D2R	PEG_D2R C N<0> 78 88
CPU PCIe Clocks				
1E300	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_P 11 78
1E300	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M_N 11 78
CPU PCIe Compensation				
1E300		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP 5
CPU eDP Compensation				
1E300		COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP 5

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Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIE-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCIE (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCH PCIE Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP

CPU DP REF CLK

Electrical Constraint Set	Physical	Spacing
CPU DP REF CLK		
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_S2N	PCIE_PHY	DMI_S2N
DMI_S2N	PCIE_PHY	DMI_S2N
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_85D
COMP_SATA_PHY	*	SATA_50S
SATA_PHY_90	*	SATA_90D

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

SATA Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
15.2.1	90	95	20	23.62	SATA Gen2, SATA Gen3

SATA Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
15-3	50	50	15	15.75	SATA Gen2, SATA Gen3

FDI Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Imp	Design	Iso	Design	Comments
6-1/6-2	85	85	12	11.81	FDI main length

FDI Compensation Rules (mils)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	55	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

SATA

Electrical Constraint Set		Physical	Spacing	
PCH SATA Port 0 (HDD)				
1E00	SATA_R2D	SATA_PHY_90	SATA	SATA HDD R2D P 35
1E00	SATA_R2D	SATA_PHY_90	SATA	SATA HDD R2D N 35
1E00		SATA_PHY_90	SATA	SATA HDD R2D C P 11 35
1E00		SATA_PHY_90	SATA	SATA HDD R2D C N 11 35
1E00	SATA_D2R	SATA_PHY_90	SATA	SATA HDD D2R P 11 35
1E00	SATA_D2R	SATA_PHY_90	SATA	SATA HDD D2R N 11 35
1E00		SATA_PHY_90	SATA	SATA HDD D2R C P 35
1E00		SATA_PHY_90	SATA	SATA HDD D2R C N 35
PCH SATA Port 1 (SSD)				
1E00	SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D P<0...1> 11 35
1E00	SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D N<0...1> 11 35
1E00	SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D C P<0...1> 35
1E00	SATA_SSD_R2D	SATA_PHY	SATA	SSD R2D C N<0...1> 35
1E00	SATA_SSD_D2R	SATA_PHY	SATA	SSD D2R P<0...1> 11 35
1E00	SATA_SSD_D2R	SATA_PHY	SATA	SSD D2R N<0...1> 11 35
PCH SATA Compensation				
1E00		COMP_SATA_PHY	COMP_SATA	PCH SATA RCOMP 11

FDI

Electrical Constraint Set	Physical	Spacing	
FDI			
1ESD	FDI_SE_PHY	FDI	FDI CSYNC 5 12
1ESB	FDI_SE_PHY	FDI	FDI INT 5 12
FDI Compensation			
1ESD	COMP_FDI_PHY	COMP_FDI	PCH_FDI_RCOMP 12

XDP

Electrical Constraint Set	Physical	Spacing	
CPU XDP			
PC100	XDP_PHY	XDP	XDP BPM L<7..0> 6 18
PC80	XDP_PHY	XDP	CPU CFG<17..0> 6 18 89
PC100 TTP_CLK_CONN	CLK_PCTE_PHY	CLK_PCTE	ITPCPU CLK100M_P
PC100 TTP_CLK_CONN	CLK_PCTE_PHY	CLK_PCTE	ITPCPU CLK100M_N
PC100	CLK_PCTE_PHY	CLK_PCTE	ITPXPDP CLK100M_P 11 89
PC100	CLK_PCTE_PHY	CLK_PCTE	ITPXPDP CLK100M_N 11 89
PC100	CLK_PCTE_PHY	CLK_PCTE	XDP CPU CLK100M_P
PC100	CLK_PCTE_PHY	CLK_PCTE	XDP CPU CLK100M_N
PC100	XDP_PHY	CLK_ITAG	XDP CPU TCK 6 18
PC100	XDP_PHY	XDP	XDP CPU TMS 6 18
PC100	XDP_PHY	XDP	XDP CPU TDI 6 18
PC100	XDP_PHY	XDP	XDP CPU TDO 6 18
PCH XDP			
PC100	XDP_PHY	CLK_ITAG	XDP PCH TCK 11 18
PC100	XDP_PHY	XDP	XDP PCH TMS 11 18
PC100	XDP_PHY	XDP	XDP PCH TDI 11 18
PC100	XDP_PHY	XDP	XDP PCH TDO 11 18

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Constraint Set	Physical	Spacing		
PCI Clock				
	CLK_PCT_55S	CLK_PCT	PCH_CLK33M_PCIIN	11 19
	CLK_PCT_55S	CLK_PCT	PCH_CLK33M_PCIOUT	11 19

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
1633	LPC_55S	LPC	LPC_AD<3..0> 13 46 48
1634	LPC_55S	LPC	LPC_AD_R<3..0> 13
1635	LPC_55S	LPC	LPC_FRAME_L 13 46 48
1637	LPC_55S	LPC	LPC_FRAME_R_L 13
LPC Clocks			
1638	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS 19 48
1640	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R 11 19
1642	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC 19 46
1643	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R 11 19

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
1E40P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB 11 19
1E40P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB_R 11
PCH RTC 32K			
1E40P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 11 19
1E40P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 11 19
1E40P	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 19
SMC 32K			
1E40P	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R 12 47
1E40P	CLK_PCH_55S	CLK_PCH	SMC_CLK32K 46 47

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
H39P	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 19
H40P	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 19
H41P	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 19
25M Reference Clocks			
H39P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET 19 37
H42P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R 19
H43P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT 19 28
H44P	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R 28

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
HDA0	HDA_55S	HDA	HDA BIT_CLK 11 54
HDA1	HDA_55S	HDA	HDA BIT_CLK_R 11
HDA2	HDA_55S	HDA	HDA_RST_L 11 54
HDA3	HDA_55S	HDA	HDA_RST_R_L 11
HDA4	HDA_55S	HDA	HDA_SDOUT 11 54
HDA5	HDA_55S	HDA	HDA_SDOUT_R 11 19
HDA6	HDA_55S	HDA	HDA_SYNC 11 54
HDA7	HDA_55S	HDA	HDA_SYNC_R 11
HDA8	HDA_55S	HDA	HDA_SDIN0 11 54
HDA9	HDA_55S	HDA	AUD_SDI_R 54
SPDIF			
SPD0		HDA	AUD_SPDIF_CHIP 54
SPD1		HDA	AUD_SPDIF_OUT 54 58

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H00P	SPI 50S	SPI	SPI CLK R 13 48
H00Q	SPI 50S	SPI	SPI CLK 48
H00R	SPI 50S	SPI	SPI ALT CLK 48
H00S	SPI 50S	SPI	SPI SMC CLK 46 48
H40P	SPI 50S	SPI	SPI MLB CLK 48
H00T	SPI 50S	SPI	SPI CS0 R L 13 48
H00U	SPI 50S	SPI	SPI CS0 L 48
H00V	SPI 50S	SPI	SPI ALT CS L 48
H40U	SPI 50S	SPI	SPI SMC CS L 46 48
H00Y	SPI 50S	SPI	SPI MLB CS L 48
H00Z	SPI 50S	SPI	SPI MOSI R 13 48
H40Z	SPI 50S	SPI	SPI MOSI 48
H00A	SPI 50S	SPI	SPI ALT MOSI 48
H00B	SPI 50S	SPI	SPI SMC MOSI 46 48
H40A	SPI 50S	SPI	SPI MLB MOSI 48
H40B	SPI 50S	SPI	SPI MISO 13 48
H00C	SPI 50S	SPI	SPI ALT MISO 48
H40C	SPI 50S	SPI	SPI SMC MISO 46 48
H40D	SPI 50S	SPI	SPI MLB MISO 48
H00E	SPI 50S	SPI	SPIROM USE MLB 14 48

SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=1.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMC Generic Control Line Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMC_ISO	*	=1:1_SPACING	?
SMC_3xISO	*	=3:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMC_CTRL	*	*	SMC_ISO
SMC_3XCTRL	*	*	SMC_3xISO

SMC Generic Control Line Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMC_GEN	*	SMC_50S

Current/Voltage Sense

Electrical Constraint Set		Physical	Spacing		
Common					
IE1		SENSE	GND SMC AVSS	46 47 50 51	
12V S5 (System Total)					
IE5	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VG3H_P	50
IE6	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VG3H_N	50
IE8			SENSE	ISNS P12VG3H_R	50 106
IE9			SENSE	ISNS P12VG3H	47 50
IE10			SENSE	VSNS P12VG3H	47 50
12V S0 (GPU Core)					
IE20	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_GPUCORE_P	50
IE21	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_GPUCORE_N	50
IE22			SENSE	ISNS P12VS0_GPUCORE_R	50
IE23			SENSE	ISNS P12VS0_GPUCORE	47 50
IE24			SENSE	VSNS P12VS0_GPUCORE	47 50
HDD					
IE30	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS HDD_P	50
IE31	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS HDD_N	50
IE33			SENSE	ISNS HDDS0_R	50
IE34			SENSE	ISNS HDDS0	47 50
IE35			SENSE	VSNS HDDS0	47 50
IE40	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_HDD_P	50
IE41	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_HDD_N	50
IE42			SENSE	ISNS P12VS0_HDD_R	50 106
IE43			SENSE	ISNS P12VS0_HDD	47 50
SSD					
IE55	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS SSD_P	51
IE57	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS SSD_N	51
IE58			SENSE	ISNS SSDS0_R	51
IE59			SENSE	ISNS SSDS0	47 51
IE60			SENSE	VSNS P3V3S5	47 51
GPU Core (alternate low side sense)					
IE85	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS GPUCORE_P	
IE86	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS GPUCORE_N	
IE88			SENSE	ISNS GPUCORE_ALT	47 50
IE89			SENSE	ISNS GPUCORE_FB	50 106
IE90			SENSE	VSNS GPUCORE_ALT	47 50
VDDQ S3 (DDR)					
IE38	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS VDDQS3_DDR_P	51
IE39	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS VDDQS3_DDR_N	51
IE40			SENSE	ISNS VDDQS3_DDR_R	51
IE41			SENSE	ISNS VDDQS3_DDR	47 51
IE42			SENSE	VSNS VDDQS3_DDR	47 51
VDDQ S0 (GPU FBVDDQ)					
IE87	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_FBVDDQ_P	50
IE88	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P12VS0_FBVDDQ_N	50
IE89			SENSE	ISNS P12VS0_FBVDDQ_R	50 106
IE90			SENSE	ISNS P12VS0_FBVDDQ	47 50
CPU Core					
IE19	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS CPUVCC_P	
IE20	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	ISNS CPUVCC_N	
IE44			SENSE	ISNS CPUVCC_FB	50
IE45			SENSE	ISNS CPUVCC	47 50
IE46			SENSE	VSNS CPUVCC	47 50
PP1V05_S0_PCH					
IE55			SENSE	VSNS P1V05S0_PCH	47 50
PP1V5_S0					
IE55	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P1V5S0_P	51
IE56	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P1V5S0_N	51
IE58			SENSE	ISNS P1V5S0_R	51
IE59			SENSE	ISNS P1V5S0	47 51
IE60			SENSE	VSNS P1V5S0	47 51
Airport					
IE62	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P3V3S4_AP_P	50
IE63	SNS_CURRENT	SNS_DTEFF_PHY	SENSE	SNS P3V3S4_AP_N	50
IE64			SENSE	ISNS P3V3S4_AP_R	50 106
IE65			SENSE	ISNS P3V3S4_AP	47 50

SMC

Electrical Constraint Set	Physical	Spacing	
SMC			
E1	CLK_XTAL	XTAL	SMC XTAL 46 47
E2	CLK_XTAL	XTAL	SMC XTAL 46 47
E31	SMC_GEN	SMC_CTL1	SMC LRESET L 20 46
E30	SMC_GEN	SMC_CTL1	SMC RUNTIME SCI L 14 46
E32	SMC_GEN	SMC_CTL1	SMC WAKE SCI L 14 46
E33	SMC_GEN	SMC_CTL1	SMC FAN_0_CTL 46 53
E34	SMC_GEN	SMC_CTL1	SMC FAN_0 TACH 46 53
E35	CPU1_ASYNC_PHY	CPU1_ASYNC	SMC PECT L 46 47 506
E36	SMC_GEN	SMC_3VCTL1	CPU PECT 0 14 46 47

SMBus

Electrical Constraint Set	Physical	Spacing	
SMC			
E10	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL 46 45
E9	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA 46 45
E8	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL 46 45
E7	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA 46 45
E6	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL 46 45
E5	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA 46 45
E4	SMB_PHY	SMB	SMBUS_SMC_3_SCL 46 45
E3	SMB_PHY	SMB	SMBUS_SMC_3_SDA 46 45
E2	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL 46 45
E1	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA 46 45
PCH			
E48	SMB_PHY	SMB	SMBUS_PCH_CLK 13 45
E37	SMB_PHY	SMB	SMBUS_PCH_DATA 13 45
E36	SMB_PHY	SMB	SML_PCH_0_CLK 13 45
E34	SMB_PHY	SMB	SML_PCH_0_DATA 13 45
Display Tcon			
E50	SMB_PHY	SMB	SMB_DP_TCON_SCL 42 45
E49	SMB_PHY	SMB	SMB_DP_TCON_SDA 42 45

Temperature Sense

Electrical Constraint Set	Physical	Spacing	
EMC1414-1 (Production)			
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC_P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC_N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 3 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 3 N
TMP423 (Development)			
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN_P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN_N
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 P
 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 N
HDD Out-of-Band			
 SNS_TEMP		SENSE	SMC_00B1_D2R_L
 SNS_TEMP		SENSE	SMC_00B1_R2D_L
 SNS_TEMP		SENSE	SMC_00B1_D2R_R
SSD Out-of-Band			
 SNS_TEMP		SENSE	SMC_00B2_R2D_L
 SNS_TEMP		SENSE	SMC_00B2_D2R_L

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		REVISION	A.0.0
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DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	IDIT	NO_TEST	
Input Bus						
E1	POWER	POWER	5V			REG VCC U7700 68
Local Ground						
E3	GND	GND	0V			AGND P1V05S0 68
1.05V S0						
E5	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0 68
E4	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P1V05S0 L 68
E7	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG BOOT P1V05S0 68
E6	VR_DTDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT P1V05S0 RC 68
E8	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0 68
E9	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P1V05S0 R 68
E10	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P1V05S0 68
E12	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P1V05S0 68
E11	VR_CTL_PHY	VR_CTL				REG P1V05S0 OCSET 68
E14	VR_CTL_PHY	VR_CTL				REG P1V05S0 V0 68
E16	VSNS_GPU_P1V05	SNS_DIFF_PHY				SNS GPU PEX IOVDD P 68 86
E17	VSNS_GPU_P1V05	SNS_DIFF_PHY				SNS GPU PEX IOVDD N 68 86
E18		SENSE				REG P1V05S0_FB 68
E19		SENSE				REG P1V05S0 RTN 68
E21	VR_CTL_PHY	VR_CTL				REG P1V05S0 SREF 68
E20	VR_CTL_PHY	VR_CTL				REG P1V05S0 FSEL 68
Output Bus						
E22	POWER	POWER	1.05V			PP1V05_S0 87
FET Switched						
E23	POWER	POWER	1.05V			PP1V05_TBTL_C 87 87
E25	POWER	POWER	1.05V			PP1V05_TBTCIO 87

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	IDT	NO_TEST	
Input Bus					
E43 POWER	POWER	5V			REG_V5IN_U7390
Local Ground					
E44 GND	GND	0V			AGND_VDDQS3
VDDQ S3					
E45 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3
E47 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L
E46 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3
E48 VR_D1DT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC
E49 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3
E50 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R
E51 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3
E51 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3
E53 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS
E55 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF
E56 VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN
E57 VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE
E58 VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP
E59 VR_CTL_PHY	VR_CTL				LDO_DDRVTT0_SNS
E61 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VTTREF
Output Bus					
E60 POWER	POWER	1.5V			PPVDDQ_S3
E62 POWER_DDR	POWER_DDR	0.75V			PPDDRVT S0
FET Switched					
E63 POWER	POWER	1.5V			PP1V5_S0
Sensed					
E64 POWER	POWER	1.5V			PPVDDQ_S3_DDR
E69 POWER	POWER	1.5V			PP1V5_S0_PCH

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VReg Constraints			
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Constraint Set	Physical	Spacing	Voltage	D1D1	NO_TEST	Input Bus						PP12V_S0_CPUVCC_FLT	POWER	POWER	12V			REG_VCC_U7000	POWER	POWER	5V			Local Ground						AGND_CPU	GND	GND	0V			Phase 1						REG_TD_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_1	POWER	POWER	1.8V			REG_ISENVCC_1_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_1_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_1_NR	SNS_D1FF_PHY	SENSE				Phase 2						REG_TD_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_2	POWER	POWER	1.8V			REG_ISENVCC_2_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_2_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_2_NR	SNS_D1FF_PHY	SENSE				Phase 3						REG_TD_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_3	POWER	POWER	1.8V			REG_ISENVCC_3_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_3_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_3_NR	SNS_D1FF_PHY	SENSE				Phase 4						REG_TD_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_4	POWER	POWER	1.8V			REG_ISENVCC_4_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_4_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_4_NR	SNS_D1FF_PHY	SENSE				<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>D1D1</th><th>NO_TEST</th></tr><tr><td colspan="6">ISL6372</td></tr><tr><td>REG_CPUVCC_DVC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_DVC_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_COMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_COMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FB</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_PSICOMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_PSICOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_HFCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>SNS_VCC_XW_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td>1.8V</td><td></td><td></td></tr><tr><td>SNS_VCC_XW_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td>0V</td><td></td><td></td></tr><tr><td>REG_CPUVCC_VSEN</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RGND</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_VIN</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMON</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_IMON_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TM</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_NPSI</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FDVID</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_MEMVSEL</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RSET</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSCLK</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSCLK_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_R_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td colspan="6">Output Bus</td></tr><tr><td>PPCPUVCC_S0_CPU</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	D1D1	NO_TEST	ISL6372						REG_CPUVCC_DVC	VR_CTL_PHY	VR_CTL				CPUVCC_DVC_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	VR_CTL_PHY	VR_CTL				REG_CPUVCC_COMP	VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FB	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_PSICOMP	VR_CTL_PHY	VR_CTL				REG_CPUVCC_HFCOMP	VR_CTL_PHY	VR_CTL				CPU_VCCSENSE_P	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_N	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_R_P	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_R_N	SNS_D1FF_PHY	SENSE				SNS_VCC_XW_P	SNS_D1FF_PHY	SENSE	1.8V			SNS_VCC_XW_N	SNS_D1FF_PHY	SENSE	0V			REG_CPUVCC_VSEN	SNS_D1FF_PHY	SENSE				REG_CPUVCC_RGND	SNS_D1FF_PHY	SENSE				REG_CPUVCC_VIN	SNS_D1FF_PHY	SENSE				REG_CPUVCC_IMON	VR_CTL_PHY	VR_CTL				CPUVCC_IMON_R	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TM	VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_NPSI	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FDVID	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_MEMVSEL	VR_CTL_PHY	VR_CTL				REG_CPUVCC_RSET	VR_CTL_PHY	VR_CTL				CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	VR_VID_PHY	VR_VID				CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	VR_VID_PHY	VR_VID				Output 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CPU_VIDSOUT	VR_VID_PHY	VR_VID																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U624 POWER	POWER	12V			REG VIN_U7600 67
U626 POWER	POWER	5V			REG VCC1_U7600 67
U628 POWER	POWER	5V			REG VCC2_U7600 67
3.3V S5					
U646 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE_P3V3S5 67
U647 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T_P3V3S5 67
U650 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T_P3V3S5_RC 67
U648 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE_P3V3S5 67
U652 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE_P3V3S5 67
U653 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER_P3V3S5 67
U609 VR_CTL_PHY	VR_CTL				REG P3V3S5_ISEN 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_OCSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_FSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT_R 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_FB 67
5V S3					
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE_P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T_P5VS4 67
U618 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T_P5VS4_RC 67
U619 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE_P5VS4 67
U620 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE_P5VS4 67
U617 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER_P5VS4 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_ISEN 67
U618 VR_CTL_PHY	VR_CTL				REG P5VS4_OCSET 67
U619 VR_CTL_PHY	VR_CTL				REG P5VS4_FSET 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT 67
U617 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT_R 67
U618 VR_CTL_PHY	VR_CTL				REG P5VS4_FB 67
Output Bus					
U615 POWER	POWER	5V			PP5V_S5 67
U616 POWER	POWER	5V			PP5V_S4 67
U616 POWER	POWER	3.3V			PP3V3_S5 67
FET Switched					
U626 POWER	POWER	5V			PP5V_S0 67
U624 POWER	POWER	3.3V			PP3V3_S4 67
U626 POWER	POWER	3.3V			PP3V3_S0 67
U625 POWER	POWER	3.3V			PP3V3_S0_SSD 67
U626 POWER	POWER	3.3V			PP3V3_ENET 67
U626 POWER	POWER	3.3V			PP3V3_TBTLIC 67
Sensed					
U625 POWER	POWER	3.3V			PPSSD_S0 67
U627 POWER	POWER	3.3V			PP3V3_S4_AP 67

3.42V G3H

Physical	Spacing	Voltage	DIDT	NO_TEST	
3.42V G3H					
U636 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST 62
U636 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_SW 62
U636 VR_CTL_PHY	VR_CTL				P3V42G3H_FB 62
U636 VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 62
Output Bus					
U636 POWER	POWER	3.425V			PP3V42_G3H 67

3.3V G3

Physical	Spacing	Voltage	DIDT	NO_TEST	
U637 POWER	POWER	3.3V			PP3V3_G3 67

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
FET Switched					
U639 POWER	POWER	5V			PPHDD_S0 67
Sensed					
U640 POWER	POWER	5V			PP5V_S0_HDD 67

12V

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U646 POWER	POWER	12V			PP12V_ACDC 67
FET Switched					
U646 POWER	POWER	12V			PP12V_S5 67
U646 POWER	POWER	12V			PP12V_S0 67
U647 POWER	POWER	12V			PP12V_S0_BLC 67
U648 POWER	POWER	12V			PPHDD12_S0 67
Sensed					
U648 POWER	POWER	12V			PP12V_G3H 67
U649 POWER	POWER	12V			PP12V_G3H_P3V42 62
U648 POWER	POWER	12V			PP12V_S0_GPUCORE 67
U648 POWER	POWER	12V			PP12V_S0_FBVDDQ 67
U647 POWER	POWER	12V			PP12V_S0_HDD 67

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	
Common					
U650 GND	GND	0V			GND

SYNC_MASTER=J17_ULTIMATE

SYNC_DATE=05/16/2013

Platform VReg Constraints

 Apple Inc.

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III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER
051-9884

REVISION
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GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

Trace-to-Trace				Isolation				
Table	Micro	Design	Strip Design	Micro	Design	Strip Design	Comments	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Memory address (MA). Implemented 4.5 Gbps or less rules for K70.	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Address dynamic bus inversion (ADBI)	
5-6/5-7	2:1	2:1	2:1 2:1	5:1 5:1	5:1 5:1	5:1 5:1	Control (CTRL)	
5-6/5-7	5:1	5:1	5:1 5:1	5:1 5:1	5:1 5:1	5:1 5:1	Clock (CLK)	
5-6/5-7	3:1	3:1	3:1 3:1	5:1 5:1	5:1 5:1	5:1 5:1	Data (DQ)	
5-6/5-7	7:1	7:1	7:1 7:1	7:1 7:1	7:1 7:1	7:1 7:1	Error detection pins (EDC). Using larger isolation rules,	
5-6/5-7	3:1	3:1	3:1 3:1	5:1 5:1	5:1 5:1	5:1 5:1	Data dynamic bus inversion (DBI)	
5-6/5-7	5:1	5:1	5:1 5:1	5:1 5:1	5:1 5:1	5:1 5:1	Forwarded clock (WCK)	

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}, y \in \{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR MA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST:TRUE
MEM1	GDDR A1 MA	GDDR MA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST:TRUE
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABI L	NO TEST:TRUE
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABI L	NO TEST:TRUE
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST:TRUE
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST:TRUE
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST:TRUE
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST:TRUE
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST:TRUE
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST:TRUE
FE0A	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST:TRUE
FE0B	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST:TRUE
FE0C	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST:TRUE
FE0D	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST:TRUE
Clock					
FE0E	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST:TRUE
FE0F	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST:TRUE
FE10	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST:TRUE
FE11	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST:TRUE
Data					
FE12	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST:TRUE
FE13	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST:TRUE
FE14	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST:TRUE
FE15	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST:TRUE
FE16	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST:TRUE
FE17	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST:TRUE
FE18	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST:TRUE
FE19	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST:TRUE
Error Detection					
FE20	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST:TRUE
FE21	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST:TRUE
FE22	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST:TRUE
FE23	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST:TRUE
FE24	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST:TRUE
FE25	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST:TRUE
FE26	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST:TRUE
FE27	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST:TRUE
Data Dynamic Bus Inv					
FE28	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST:TRUE
FE29	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<1>	NO TEST:TRUE
FE2A	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<2>	NO TEST:TRUE
FE2B	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<3>	NO TEST:TRUE
FE2C	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<0>	NO TEST:TRUE
FE2D	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<1>	NO TEST:TRUE
FE2E	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<2>	NO TEST:TRUE
FE2F	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBT L<3>	NO TEST:TRUE
Forwarded Clock					
FE30	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST:TRUE
FE31	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST:TRUE
FE32	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST:TRUE
FE33	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST:TRUE
FE34	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST:TRUE
FE35	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST:TRUE
FE36	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST:TRUE
FE37	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<1>	NO TEST:TRUE

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 78
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 78
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 78
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 83 85
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 83 84
SMB			
ES00	SMB_PHY	SMB	GPU_SMB_CLK 83 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT 83 84
ES00	SMB_PHY	SMB	GPU_SMB_CLK_R 49 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT_R 49 84
PCIe Compensation			
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PD_VDDQ 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PU_GND 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_TERM_GND 80

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST-TRUE
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST-TRUE
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST-TRUE
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST-TRUE
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST-TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST-TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST-TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST-TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST-TRUE
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST-TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST-TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST-TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST-TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST-TRUE
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST-TRUE
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST-TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST-TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST-TRUE
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST-TRUE
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST-TRUE
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST-TRUE
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST-TRUE
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST-TRUE
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST-TRUE
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST-TRUE
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST-TRUE
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST-TRUE
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST-TRUE
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST-TRUE
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST-TRUE
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST-TRUE
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST-TRUE
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST-TRUE
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST-TRUE
Data Dynamic Bus Inv					
H009	GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<0>	NO TEST-TRUE
H009	GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<1>	NO TEST-TRUE
H009	GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<2>	NO TEST-TRUE
H009	GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_B_0_DBT	FB_B0_DBT_L<3>	NO TEST-TRUE
H009	GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<0>	NO TEST-TRUE
H009	GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<1>	NO TEST-TRUE
H009	GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<2>	NO TEST-TRUE
H009	GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_B_1_DBT	FB_B1_DBT_L<3>	NO TEST-TRUE
Forwarded Clock					
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST-TRUE
H009	GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST-TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST-TRUE
H009	GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST-TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST-TRUE
H009	GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO TEST-TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO TEST-TRUE
H009	GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO TEST-TRUE

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
H44C	GD0R_A0_RESET	GD0R_56S	GD0R_CTR1	FB_A0_RESET_L
H45C	GD0R_A1_RESET	GD0R_56S	GD0R_CTR1	FB_A1_RESET_L
H46C	GD0R_B0_RESET	GD0R_56S	GD0R_CTR1	FB_B0_RESET_L
H47C	GD0R_B1_RESET	GD0R_56S	GD0R_CTR1	FB_B1_RESET_L
H48C	GD0R_C0_RESET	GD0R_56S	GD0R_CTR1	FB_C0_RESET_L
H49C	GD0R_C1_RESET	GD0R_56S	GD0R_CTR1	FB_C1_RESET_L
H50C	GD0R_D0_RESET	GD0R_56S	GD0R_CTR1	FB_D0_RESET_L
H51C	GD0R_D1_RESET	GD0R_56S	GD0R_CTR1	FB_D1_RESET_L

SYNCH MASTER=D8 AARON		SYNCH DATE=03/13/2012	
PAGE TITLE		PAGE	
GDDR5/GPU Constraints			
 Apple Inc.		DRAWING NUMBER 051-9884	
		SIZE D	
		REVISION A.0.0	
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D

C

B

A

Constraints

BLC High Voltage Output

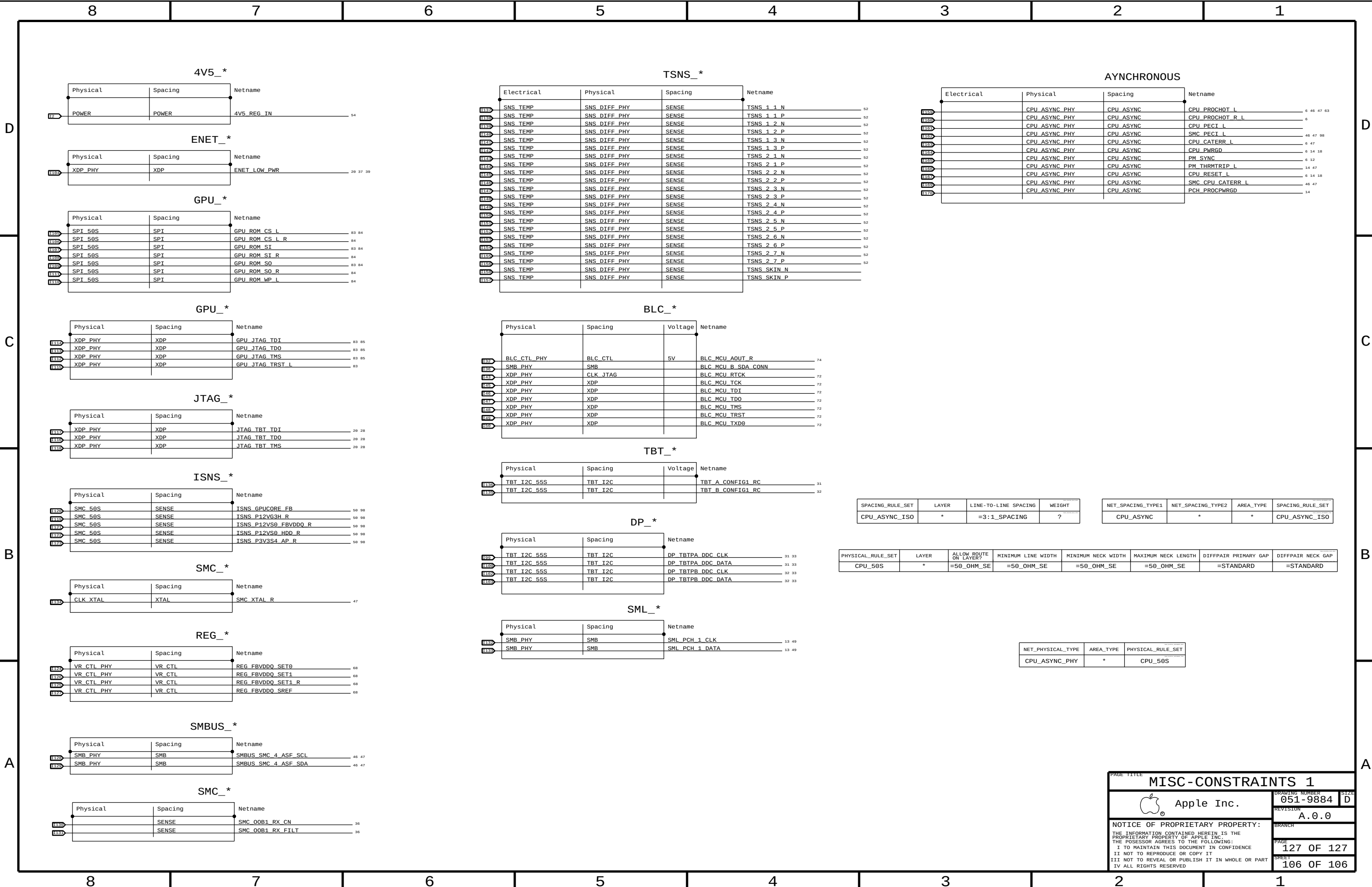
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control			
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Physical		Spacing		Voltage	DIOT	NO_TEST	
Input	Bus						
H460	POWER	POWER	12V			PP12V_S0_BLC_VIN2	72 73 74
H777	POWER	POWER	12V			PP12V_S0_BLC_VINP	72 74
H800	POWER	POWER	14V			PRE_REG_OUT	72 73
H000	POWER	POWER	3_3V			BLC_P3V3S	72 74
H000	POWER	POWER	3_3V			BLC_P3V3_REF	72 74
H000	POWER	POWER	3_3V			BLC_P3V3	
H800	POWER	POWER	3_3V			PP3V3_S0_BLC_R	72
H800	POWER	POWER	8V			SPTX_VIN	73
H000	POWER	POWER	8V			PP8V_BLC	72 73
H000	POWER	POWER	8V			BLC_VIN2	73 74
H000	POWER	POWER	14V			BOOST_FET_DRAIN	72
H800	POWER	POWER	12V			BOOST_VDD	72
H800	POWER	POWER	5V			PP5V_S0_BLC_R	72 73 74
H000	POWER	POWER	12V			PP12V_S0_BLC_F	74
Local Ground							
H790	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_1	73 74
H000	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_2	73 74
H000	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_3	73 74
H000	GND	GND	8V			AGND_BLC	73 74
Backlight							
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1	73
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R	73
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2	73
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R	73
H000	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3	73
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R	73
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_1	73
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_2	73
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_RC_3	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C1	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C2	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRVR_CS_C3	73
H000	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1	73
H000	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2	73
H000	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3	73
H000	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1	73
H000	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2	73
H000	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3	73
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R	72 73
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R	72 73
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R	72 73
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1	73 74
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2	73 74
H000	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3	73 74
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1	73 74
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2	73 74
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3	73 74
H750	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1	73
H460	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2	73
H750	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3	73
H000	BLC_CTL_PHY	BLC_CTL				BCOMP1	73
H000	BLC_CTL_PHY	BLC_CTL				BCOMP2	73
H000	BLC_CTL_PHY	BLC_CTL				BCOMP3	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2	73
H000	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3	73
H000	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1	73
H000	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2	73
H000	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3	73
H000	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R	72
H000	BLC_CTL_PHY	BLC_CTL				BOOST_FB	72
H000	BLC_CTL_PHY	BLC_CTL				BOOST_COMP	72
H000	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C	72
H000	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV	72
H000	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R	72
H000	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS	72
H000	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R	72
H460	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_1	73
H000	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_2	
H000	BLC_CTL_PHY	BLC_HV	80V			LED_DRVR_DRAIN_3	73
OUTPUT BUS							
H460	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F	74
H460	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F	74
H460	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F	74
H460	POWER_BLC_RET	BLC_HV	80V			IS1_BLC	73 74
H460	POWER_BLC_RET	BLC_HV	80V			IS2_BLC	73 74
H460	POWER_BLC_RET	BLC_HV	80V			IS3_BLC	73 74
H000	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1	74
H000	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1	74
H000	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2	74
H460	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2	74
H460	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3	74
H000	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3	74
H000	POWER_BLC	BLC_HV	80V			BLC_VOUT1	73 74
H000	POWER_BLC	BLC_HV	80V			BLC_VOUT2	73 74
H000	POWER_BLC	BLC_HV	80V			BLC_VOUT3	73 74

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SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ASYNC_ISO	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ASYNC	*	*	CPU_ASYNC_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_ASYNC_PHY	*	CPU_50S

MISC-CONSTRAINTS 1

Apple Inc.

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